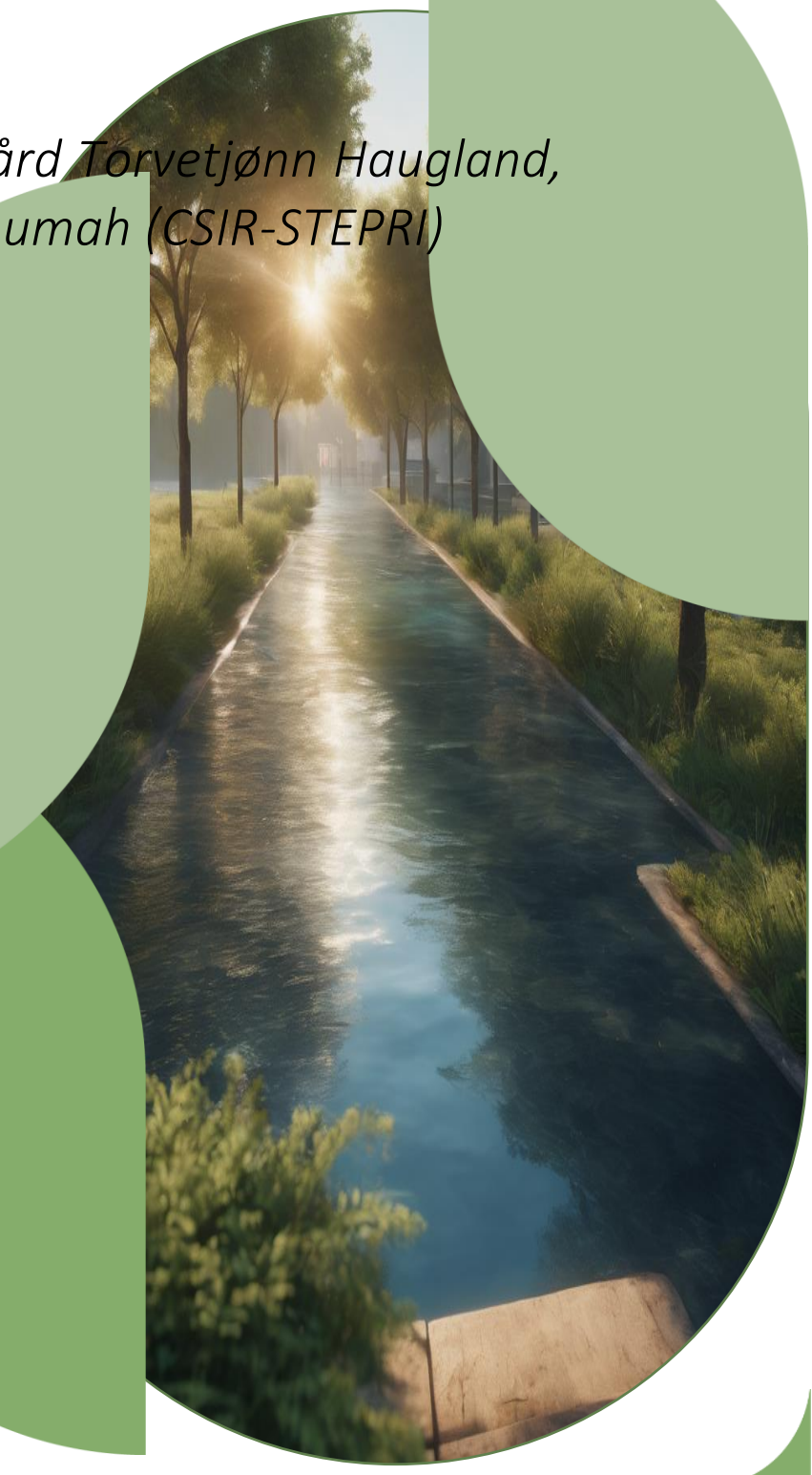
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Policy mixes and pathways towards a circular water economy

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Policy mixes and pathways towards a circular water economy

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ABSTRACT
This report presents the results of an analysis of the policy landscapes and regulatory frameworks surrounding the innovations demonstrated in WIDER UPTAKE, and how these conditions influence the transition pathways associated with the respective solutions. An extended policy mix concept is applied, to shed light on the consistency and coherence of policies and instruments across levels, sectors, and action situations. Regarding consistency, policy alignment is improving within the EU. However, we also identify certain tensions between elements at the EU and national levels, and between objectives and embedded interests in different sector policies. In terms of comprehensiveness, we argue that there is a need to pay more attention to the target users of the reclaimed water and products based on recovered resources, to build acceptance and generate the demand needed to establish viable value chains. Besides case-specific recommendations, we identify three overarching implications: 1) need for increased coordination and collaboration – across levels, sectors, and regions, 2) CE solutions based on water resources must be further incentivised, and 3) future policy must pay more attention to the values in water and how they best can be realised, to maximise economic, social, and environmental sustainability.

KEYWORDS
Policy mix, circular economy, water reuse, resource recovery, transition pathways

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LIST OF ABBREVIATIONS

ADB – African Development Bank

ARERA - Regulatory Authority for Energy, Networks and Environment

CE – Circular Economy

CEAP – Circular Economy Action Plan

CoP – Community of Practice

CWSA - Community Water and Sanitation Agency

EBPR - Enhanced Biological Phosphorus Removal

ECI - Environmental Cost Indicator (ECI)

ECOWAS – Economic Community of West African States

ECREE - ECOWAS Centre for Renewable Energy and Efficiency

EEA – European Economic Area

EEB - European Environmental Bureau

EESC – European Economic and Social Committee

EIB – European Investment Bank

EPR – Extended Producer Responsibility

ERMF – Energy and Resources Factory

FPR – Fertiliser Products Regulation

GHG – Greenhouse gas(es)

GPP – Green Public Procurement

GWCL - Ghana Water Company Limited (GWCL)

IED – Industrial Emissions Directive

ISPRA - Italian Institute for Environmental Protection and Research

INMAP – Integrated Nutrient Management Plan

IWRM – Integrated Water Resources Management

LCA – Life Cycle Assessment

LUP - National Programme for Supplier Development

MESTI – Ministry of Environment, Science, Technology and Innovation

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MMDAs - Metropolitan, Municipal and District Assemblies

MLP – Multi-Level Perspective

N – Nitrogen

NABB - National Approach to Biobased Construction

NCWSS - National Community Water and Sanitation Strategy

NEPAD - New Partnership for Africa's Development

OECD – Organisation for Economic Co-operation and Development

P – Phosphorus

PCR - Product Category Rules

R&D – Research and Development

SDG – Sustainable Development Goal

SME – Small and Medium-sized Enterprises

SNM – Strategic Niche Management

SPP – Sustainable Public Procurement

STI – Science, Technology and Innovation

STOWA – Foundation for Applied Water Research

STRUBIAS - STRUvite, Biochar and ASh-based products

TIS – Technological Innovation Systems

TM – Transition Management

UN – United Nations

UNECE - United Nations Economic Commission for Europe

UNFCCC - UN Framework Convention on Climate Change

UWWTD – Urban Wastewater Treatment Directive

WFD – Water Framework Directive

WiCE - 'Water in the Circular Economy'

WHO – World Health Organisation

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Executive summary

This report presents the results of a policy mix analysis of the policies and policy instruments relevant to the innovations demonstrated in WIDER UPTAKE, and how these conditions influence the transition pathways associated with the respective solutions. An extended policy mix concept is applied, to shed light on the domains and rationales of influence in each case, and the consistency and coherence of policies and instruments between governance levels and across value chains and governance sectors. The analysis is based on a systematic mapping of policies and instruments related to the action situations each innovation involves. The mapping was done by the research team in WP4, in dialogue with the local partners and their Communities of Practice.

The policy mix analyses for the four European demonstration cases show that key policies at the EU level are increasingly aligned. However, there are still tensions, barriers, and gaps. For water reuse (demonstrated in Sicily and the Czech Republic), the key challenge is developing (national level) instruments that make water reuse economically viable for wastewater treatment plants and cost-efficient and practical for end-users.

Considering nutrient recovery, in the form of struvite, innovative mineral-organic fertilisers, and organic soil improvement products demonstrated in Norway, there is a need for more direction, e.g., via the anticipated EU Integrated Nutrient Management Plan. While struvite and some other product categories can achieve end-of-waste status under the Fertiliser Product Regulation, this is not the case for recovered nutrients in biosolids, such as digestate. For products based on these resources, end-of-life criteria, or establishment of standards to ensure and document high quality would be conducive. However, the main challenge identified is a lack of instruments to pull demand, and thereby stimulate further investments.

For material recovery and biocomposite production in the Netherlands, we find a quite consistent and comprehensive policy mix, with specified targets, and a combination of regulatory, economic, and soft instruments, targeting knowledge development, experimentation, and market formation. The challenges identified here relate to costs more than policy and regulations. However, end-of-waste criteria for raw materials, such as reed clippings and recovered cellulose, are still lacking, hindering trade across borders. Furthermore, it is noted that sustainable public procurement (SPP) could be better employed, and assessment methods could still be improved, to encompass wider impacts (e.g., some of the benefits of replacing tropical hardwood with biocomposite materials will not be visible in LCA).

The policy mix identified and analysed for water reuse and biochar in Ghana shows that policies to enable circular water economy are weakly developed as at now. The current regulatory gaps constitute a major barrier for reuse but are less severe for biochar substituting charcoal as a local energy source. The biochar may be classified as a form of biofuel, and there are potential synergies with a national campaign to promote more energy-efficient charcoal stoves. Since

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the solution also appears to be cost-competitive for the users, lack of certification stands out as the main barrier.

In terms of pathways, we see a high degree of consistency across levels for the reuse case in Sicily. As sizeable investments in facilities for water reuse already have been made, and the region is facing a severe water crisis, strong drivers are in place. Thus, this is a pathway with a high transformative potential. However, upscaling may depend on more systemic instruments, addressing how to organise and finance operations on a permanent basis. In the Czech Republic, the case for urban water reuse took a blow when the national authorities decided to postpone the implementation of the Regulation on minimum requirements for water reuse (Reg. (EU) 2020/741). A way to circumvent this barrier for the specific case has been found, but high costs, lack of infrastructure, and uncertainty regarding future demand are other concerns, leading us to associate this pathway with a slower and more gradual shift, amongst other depending on future costs and pressure from the EU, and the pace of urban development, where reuse infrastructure may be planned ahead.

The transition pathway associated with nutrient recovery in Norway is considered to be at a critical branching point, where the implementation of the revised Urban Wastewater Treatment Directive (UWWTD), however debated, constitutes a window of opportunity. Proposals for new national regulations for the production, trade, and use of organic fertilisers, are challenging current sludge management practices, but since the volumes and opportunities for utilising recovered nutrients are variable, other solutions (e.g., pyrolysis) are also seen as relevant. In the Dutch case, the national Transition agenda for the Building and Construction sector (Holland Circular Hotspot, 2018) sees biocomposites as an alternative for targeted design products. Moreover, the National Approach to Biobased Construction (NABB) (Government of the Netherlands, 2023) envisages that more than 30% of new riverbank protection panels shall be made from biocomposites or willow. Thus, national policies define a scenario where the WIDER UPTAKE innovation persists as one of several pathways towards a sustainable, circular economy.

In the Ghana case, reuse for urban farming seems challenging at present mainly due to the absence of infrastructure for the supply of the treated water to farmers who are not compelled (by regulation and enforcement) to use safe water for farming in the cities. However, the local Community of Practice sees reuse as an option in a longer-term perspective, for wastewater treatment plants in rural or developing areas, where potential users are located in the vicinity. The biochar pathway seems more feasible in a shorter-term perspective, as it is expected to meet an existing demand and is synergetic with existing value chains and transition policies. Indeed, the biochar production plant has been linked to a startup distributor that has a network of charcoal consumers that have shown interest to adopt the biochar.

While providing a set of more specific recommendations for each case, the final discussion chapter highlights three implications for policy to enhance the feasibility of the studied pathways: 1) the need for increased coordination and collaboration – across levels, sectors, and

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regions, 2) that CE solutions based on water resources must be further incentivised, and 3) that future policy must pay more attention to the values in water and how they best can be realised, to maximise economic, social, and environmental sustainability.

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1 Introduction

WIDER UPTAKE aims to facilitate industrial symbiosis to increase resource efficiency, limit climate gas emissions and develop sustainable business based on water-smart solutions. As the barriers to uptake of water-smart solutions are not only technological but also of organizational, regulatory, social, and economic character, the project aims to identify and demonstrate means to overcome barriers related to:

1. Monitoring and control of health and quality risks
2. Circular economy (CE) and efficiency potential
3. Governance and business models for industrial symbiosis
4. Measuring water smartness and progress towards achievement of the SDGs.

The project runs demonstrations of wastewater reuse for agriculture and urban greening, phosphorus recycling, biogas, digestate/organic fertilizer and biochar utilisation, and production of biocomposites for materials with resources recovered from the water cycle. Thus, the water-smart solutions in WIDER UPTAKE are essentially CE solutions, based on water resources. The project partnership includes 11 water utilities and industry partners, and seven research institutions, distributed across five countries (Norway, Netherlands, Czechia, Italy and Ghana), as illustrated in Figure 1.

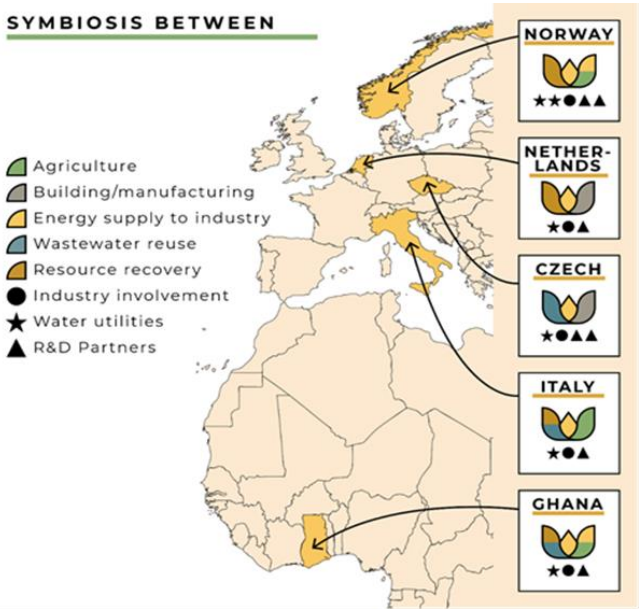


Figure 1: WIDER UPTAKE demonstration cases.

This report presents results from activity in Work Package 4: Governance and business models, carried out between Month 36 and Month 48. Our task was identification and analysis of the relevant policy frameworks for the demonstration cases, and how they may influence the sustainability transition pathways associated with the focal solutions. The results are applied in the final roadmap from of the project (D6.8). Since the contexts of the demonstration cases

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are different, the assessment has been carried out and is presented country by country, before we provide an overarching discussion of the findings and their implications for the transition pathways associated with the WIDER UPTAKE solutions.

The concept of sustainability transition pathways addresses the timing, scope, and scale of change necessary for a region, industry, or economy to get from its current state towards a sustainable future, e.g. a water-smart economy and society. Where strategies are plans that outline how to get from A to B, pathways outline what needs to be considered and what matters when assessing whether a given strategy *can* get us from A to B. In the study of sustainability transitions, three core conceptions of pathways prevail (Rosenbloom, 2017). The first, biophysical conception, understands transition pathways as long-term trajectories describing human-climate interactions over time. The second, techno-economic conception, understand pathways as sequences of techno-economic adjustments linking current sector configurations to desirable future states.

The third conception, which we relate to in this report, is referred to as socio-technical transition pathways and highlights the interconnected nature of technological and social change. As it encompasses the broader political, institutional, cultural, and behavioural dynamics in long-term societal change, it resonates more closely with narrative than quantitative approaches (Geels, 2011). Three study angles have been dominant: (1) providing more knowledge of the complex dynamics of transition processes; (2) focusing on how to stimulate transitions; and (3) efforts to bridge socio-technical and quantitative modelling approaches (Rosenbloom, 2017). By attending to the wider social and material forces that constrain or enable innovation, socio-technical analysis allows assessment of the feasibility of alternative techno-economic pathways.

For such assessment, four key dimensions have been identified (Turnheim and Nykvist, 2019): 1) *The maturity of options* (technological readiness, readiness for market introduction and/or widespread implementation of the focal technological, social or institutional innovation), 2) *system integration and infrastructure* (to what extent the innovations can be implemented within existing processes, practices, and infrastructure systems, or there is the need to change or replace existing infrastructure), 3) *societal acceptability* (what characterizes the perceptions and attitudes towards the focal innovations, to what extent there are controversies, or anxieties regarding the expected use of a new solution, and how this may affect its implementation), and 4) *political feasibility* (likelihood of decisions supporting the implementation of a particular pathway, or addressing obstacles, versus resistance of actors with vested interests).

In a previous deliverable (D4.2), we discussed these dimensions as part of a process to explore new business models for the WIDER UPTAKE innovations. The present report assesses to what extent the existing policy mixes enable uptake of the water-smart solutions in the demonstration cases, as pathways towards CE based on water resources, and eventually towards a water-smart society. To do so, we first map the policy mixes surrounding each

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demonstration case. Our focus is on policies and pathways for the key products and services, which were in focus in the previous business model exploration and have been the main topics in the local Communities of Practice (CoPs). We assess the consistency and comprehensiveness of the respective policy mixes, with a view to how they promote or hinder the emergence of the focal solutions. Then, we discuss the implications of the assessed policy mixes for the other three dimensions identified by Turnheim and Nykvist (2019), and the overall feasibility of the envisaged transition pathways.

While shedding light on context-specific factors that must be considered, our findings highlight overarching challenges and opportunities in terms of coordination, incentivization, and the need to make the principles of circular economy a strategic priority in policymaking.

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2 Objective

The objective of this report is to discuss possible transition pathways for the key CE solutions demonstrated in WIDER UPTAKE. In particular, we focus on interlocking social and technical (socio-technical) patterns of change.

By way of five case-specific policy mix analyses, we assess the influence of current policies and instruments on the uptake of certain CE solutions based on water resources. The results from these analyses serve as a starting point for discussing viable transition pathways in each case. The case-specific discussions are followed by a comparative discussion which seeks to identify some overarching dynamics. The report concludes with some overarching recommendations for further promoting CE solutions based on water resources.

The original title of the deliverable, stated in the description of action, was “Pathways towards a water-smart society”. However, the transition towards a water-smart society (e.g., as defined by Water Europe, 2023) has multiple dimensions, beyond the scope of WIDER UPTAKE and the objective of this report. We have therefore adjusted the title, to one that better fits the objective and content of the deliverable.

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3 Background

Against the backdrop of the global water challenges, Water Europe has launched a vision of a water-smart society. In their conception, a water-smart society is a society where:

> the value of water is recognised and realised to ensure water security, sustainability, and resilience; all available water sources are managed so that water scarcity and pollution are avoided; water and resource loops are largely closed to foster a circular economy and optimal resource efficiency; the water system is resilient against the impact of climate and demographic change; and all relevant stakeholders are engaged in guaranteeing sustainable water governance. (Water Europe, 2023: 4).

Acknowledging that there is an impending water crisis, also in Europe, the President of the European Commission has announced the intention to provide a new Water Resilience Strategy for the EU in the Political Guidelines 2024–2029. Moreover, the European Economic and Social Committee (EESC) and Members of the European Parliament have called for an EU Blue Deal, as a cross-cutting strategic priority.

Circular economy (CE) is central in the vision of a water-smart society, as in the European Green Deal. As noted in the EU Circular Economy Action Plan (COM/2020/98 final), the EU needs to accelerate the transition towards a regenerative growth model that gives back to the planet more than it takes, and therefore strives to reduce its consumption footprint and double its circular material use rate during this decade (European Commission 2020, p. 2). The Circular Economy Action Plan lays down seven key areas as central for achieving a CE. One key area is water and nutrients but, other areas have received more attention. However, the European Commission clearly aims to facilitate water reuse and efficiency, while ensuring more sustainable application of nutrients and stimulating the markets for recovered nutrients (ibid.).

CE can broadly be defined as follows:

> an economic system in which resource input and waste, emission, and energy leakages are minimized by cycling, extending, intensifying, and dematerializing material and energy loops. This can be achieved through digitalization, servitization, sharing solutions, long-lasting product design, maintenance, repair, reuse, remanufacturing, refurbishing, and recycling. (Geissdorfer et al., 2020, p. 3)

CE is most frequently depicted as a combination of so-called “R strategies”. The most well-known is 3R, which encompasses reduce, reuse and recycle activities, with recover activities often included as a fourth R. Further Rs have also been suggested, adding up to 6 Rs (Sihvonen and Ritola, 2015) and even 9Rs (Potting et al., 2017). However, regardless of the exact number

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of Rs, these perspectives share a strong focus on material flow analysis and the product life cycles.

For the water and wastewater sector, a CE model framework with six actions has been proposed: 1) reduction — prevent wastewater generation in the first place, by reducing water usage and pollution at source, 2) reclamation (removal) — removing pollutants from water and wastewater inflow, 3) reuse — reuse of wastewater as an alternative source of water supply (non-potable usage), 4) recycling —recovery of water from wastewater for potable usage, 5) recovery — recovery of resources such as energy and nutrients from water-based waste, and 6) rethink—rethinking how to use resources to create a sustainable economy, without waste and emissions (Smol et al., 2020: p. 682). Several studies highlight the value creation potential associated with reuse, recycling, and resource recovery from wastewater (Wilcox et al., 2016; Ghergel et al., 2019; Guerra-Rodríguez et al., 2020). Besides energy and nutrients, other products such as nitrate, ammonium sulphate, calcium phosphate, adsorbents and heavy metals, as well as cellulose fibres, biopolymers, bioplastics, and proteins, may be extracted and provide added value (van der Hoek et al., 2018; Guerra-Rodríguez et al., 2020). However, the uptake of circular water technologies is still hindered by non-technological factors, such as concerns over economic feasibility (Mallory et al., 2020), and regulatory barriers and uncertainty around the application of policy frameworks (Gherghel et al., 2019; Guerra-Rodríguez et al., 2020; Damman et al., 2022).

The framework proposed by Smol et al. (2020) emphasizes that CE requires more than implementation of specific technologies: as the water cycle must be managed from catchment to consumer, and back to catchment, the transition must address the consumption and production of resources across this entire value chain and create connections and synergies with other sectors. In line with IWA's (2016) call for integrated urban resource management, closer connection with stakeholders beyond traditional boundaries, and innovation in leadership and business models, the CE strategies proposed point to organisational and societal changes, e.g., change of practices to reduce the level of pollutants, and 'rethink' as the basis for systematic changes in the whole value chain for water, across various sectors (Smol et al., 2020). This reflects previous studies on CE, where the need for systemic perspectives is observed (e.g., Geissdorfer et al., 2017, Kirchherr et al., 2017).

Ghisellini and Ulgiati (2020) is one of several research contributions proposing that the transition towards a CE fruitfully may be studied in a socio-technical system perspective. In such a perspective, sectors like the water and wastewater sector can be conceptualised as socio-technical systems, consisting of actor networks and institutions (e.g., norms, regulations, standards of good practice), as well as material artifacts and knowledge, which are closely intertwined and interact to provide specific services for society (Markard et al., 2012, p. 956). A sustainability transition, in turn, is considered as a fundamental shift in socio-technical systems, involving change along multiple dimensions: technological, material, organizational, institutional, political, economic, and socio-cultural. These processes typically run over several decades, where new products, services and business models emerge, compete, and eventually complement or substitute for existing ones (ibid.).

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Socio-technical sustainability transition studies consist of several research strands. The so-called Multi-Level Perspective (MLP) (Geels, 2002) is particularly concerned with the interplay of dynamics at different levels; how exogenous "landscape" pressures such as climate change may put pressure on regimes of existing structures and practices, and open windows of opportunities for radical niche innovations to break through and enable fundamental changes (Markard et al., 2012). This strand forms the basis for the socio-technical conception of transition pathways, starting from typologies of pathways based on variation in the timing and quality of niche-regime-landscape interactions (e.g., Geels and Schot, 2007b). Other research strands commonly highlighted are Strategic Niche Management (SNM), focusing on the creation and support of radical innovation niches (Hoogma et al., 2002; Kemp et al., 1998), Technological Innovation Systems (TIS), which mobilizes ideas from innovation systems theory and industrial economics to shed light on the drivers and barriers faced by radical sustainability innovations (Hekkert et al., 2007; Bergek et al., 2008a), and Transition Management (TM), which aims to provide more prescriptive insights on how policy-makers can shape transitions (Rotmans et al., 2001; Loorbach, 2010).

The last decade has seen a growing interest in how policy mixes promote (or impede) transitions (Kivimaa and Kern 2016; Lindberg et al., 2019). Whereas a policy mix in simple terms can be understood as a combination of several policy instruments, recent definitions typically include the ultimate objective(s) of the policy mix, as well as the interaction between policies, objectives, and instruments in the focal domain(s), and/or their dynamic interaction over time (Rogge and Reichardt, 2016). Within sustainability transitions research, there are calls for an extended policy mix concept, which addresses complexity and the processes whereby "policies emerge, interact and have effects" (Flanagan et al., 2011, p. 702), and pays special attention to their long-term, strategic components. Key studies highlight the need for a system perspective in policy making, e.g., not merely focusing on demand pull or technology push, but covering all concerns (Rogge and Reichardt, 2016), including both policies that aim for the 'creation' of new practices and systems and policies aimed at destabilising the old (Kivimaa and Kern, 2016).

As noted above, CE based on water and wastewater resources faces specific regulatory barriers, making an analysis of relevant public policies particularly relevant. Moreover, CE involves interactions across sectors and policy domains, which fruitfully may be analysed in policy mix framework. Therefore, the analysis in this report builds on frameworks for policy mix analysis that recently have been developed and applied within sustainability transition studies (e.g., Kivimaa and Kern, 2016; Rogge and Reichardt, 2016; Bach and Hansen, 2023). In the following chapter, we provide a more detailed description of the key concepts and methods applied.

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4 Methodological approach

Our policy mix approach builds upon Rogge & Reichardt's (2016) codification of the concept, which has three core building blocks: elements, policy processes, and characteristics. The first, *elements*, refers to two components: policies and instruments. Policies are stated goals and the associated strategies for achieving them, while instruments are "concrete tools to achieve overarching objectives" (Rogge & Reichardt, 2016, p. 1623). Rogge and Reichardt (2016), distinguish between three instrument types (economic, regulatory, and soft) and three primary instrument purposes (technology push, demand pull, and systemic). In their perspective, an ideal instrument mix is characterized by synergies, whereas an adequate instrument mix simply avoids contradictions (Rogge & Reichardt, 2016, p. 1622).

The second building block, *policy processes*, refers to the processes that establish new policies and instruments (Rogge & Reichardt, 2016, p. 1625). This encompasses both policy making and policy implementation. This dual focus serves to highlight that the efficacy of policies and instruments does not depend solely on their design. A poor implementation of a well-designed instrument might limit its performance. The third building block, *characteristics*, is an assessment of the other two. In Rogge & Reichardt's (2016) framework, this includes four aspects:

1. ***Consistency of elements*** refers to the consistency between different policies, the consistency of their related instrument mixes, and the consistency between the overarching policy strategy and the instrument mix. Consistency (i.e., a high degree of alignment between elements), is seen to increase the efficiency of policy mixes, while constant changes or inconsistencies can impede innovation uptake (Reichardt and Rogge, 2016; Uyarra et al., 2016).

2. ***Coherence of processes*** refers to the degree of synergism and systematicity in policy making and implementation, contributing towards the achievement of policy objectives (Rogge and Reichardt, 2016, p. 1626). This includes the ability to coordinate policies across different policy fields and/or policy levels (e.g., transnational, national, and regional levels), as well as the ability to think and act holistically.

3. ***Credibility*** refers to whether a policy mix appears believable or can be trusted to work and/or be implemented as intended. As Rogge and Reichardt (2016, p. 1627) indicate, this term is somewhat elusive. However, political commitment, the level of support offered by policy instruments, and policy mix consistency may all be indicators of credibility.

4. ***Comprehensiveness:*** Whether policy strategies include objectives and principal plans with associated instruments, and how extensive and exhaustive the mixes of instruments are in addressing market, system, and institutional failures. A high degree of comprehensiveness is seen to promote policy mix effectiveness and efficiency,

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leading to increased levels of innovation (Costantini et al., 2017), whereas less comprehensive mixes promote the status quo (Sanz-Hernandez et al., 2020).

In the perspective of Kivimaa and Kern (2016), it is important to also distinguish between policies aiming for the 'creation' of new and for 'destroying' (or withdrawing support for) the old, to identify where there might be elements lacking in existing policy mixes. Therefore, they distinguish between creative elements, supporting the focal sustainability innovation (e.g., elements supporting knowledge development and diffusion, market formation, price-performance improvements, experimentation, resource mobilization, incentives, and legitimation of the novel solutions); and destructive elements, aiming to destabilize the existing unsustainable practices and systems that one aims to change or do away with (e.g., control policies, radical change in economic frame conditions, reduced support for dominant technologies, and/or change in social networks (ibid.).

While Rogge and Reichardt (2016) stress the importance of mapping both instruments, strategies, and processes, it has since been argued that it is challenging to operationalise and measure the four characteristics for both policy elements, processes, and the overall policy mix (Rogge, 2019). Bach and Hansen (2023), therefore, focus on consistency and comprehensiveness. When it comes to comprehensiveness, they move away from the typologies suggested by Rogge and Reichardt (2016). Instead, they bring in policy strategy comprehensiveness, in terms of the breadth of topics covered and objectives' level of concretisation (Bach and Hansen, 2023, p. 3), and instrument mix comprehensiveness, where they apply Kivimaa and Kern's (2016) distinction between creative and destructive processes. For the overall policy mix, they consider to what extent the policy objectives are addressed by a broad set of instruments (Bach and Hansen, 2023, p.4).

Concerning consistency, Bach and Hansen (2023) follow Rogge and Reichardt (2016) in distinguishing between three levels: 1) consistency of the instrument mix (whether the instruments complement, reinforce, or contradict each other), 2) consistency of the policy strategy (synergies versus potential trade-offs), and 3) policy mix consistency, as the level of alignment between the strategy and the instrument mix. In the present policy mix analysis, we largely apply the approach used by Bach and Hansen. However, we also find it important to distinguish between generic instruments, i.e. instruments that are highly relevant for the studied cases, but target CE or innovation at large, rather than relating specifically to the management and use of water-based resources.

Because our analysis is linked to specific cases, the boundaries for the policy mix were set from the viewpoint of the cases in question. That is, the policy mixes were assessed considering the solutions the WIDER UPTAKE partners aim to realise.

The focal solutions in the WIDER UPTAKE demonstration cases may each be considered to consist of a network of distinct, but interdependent action situations, or spaces where actors make decisions and take actions, choosing among available options in light of information

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about the likely actions of others and the benefits and costs of potential outcomes (McGinnis, 2011; Kimmich and Villamayor-Tomás, 2019; Damman et al., 2022) Action situations can be formal and quite strictly regulated, as in the case of wastewater treatment, or more informal, e.g., for urban farmers making joint decisions about irrigation. Each action situation includes a distinct set of resources, actors, and governance tasks, but they may also be linked, through institutions, information flows, common participants, or biophysical conditions, such as the resource flow associated with water cycle services. Thus, each action situation may be closely connected with a specific policy domain, but also influenced by others, associated with linked action situations (Kimmich and Villamayor-Tomás, 2019).

For the policy mix analysis at hand, we therefore started with a mapping of the relevant action situations and policy domains for the key solution in each demonstration case. In dialogue with representatives of the cases, the key policies and instruments of influence for each action situation were identified and classified in Excel, according to level (transnational, national, regional), type (regulative, financial, or soft; specific or generic), and key objectives, with a brief extract of relevant content included.

Thereafter, we took a more detailed look into the policy mix for each case. This was an iterative process, where the core team in WP4 started the assessment, based on closer reading of the policy documents, asked questions, and got feedback from the local partners, then updated the assessment, and had another round of consultation with the local partners. The process was challenging, since many of the documents at national and regional level are available only in local language. The key concepts of consistency and comprehensiveness were operationalised more or less the same way as by Bach and Hansen (2023), as shown in Table 1.

Table 1: Operationalisation of key concepts (based on Bach and Hansen, 2023, p.4).

Dimension	Concepts	Operationalisation
Consistency	Policy strategy (alignment between policy objectives)	Degree of synergies and conflicts
	Instrument mix (alignment between policy instruments)	Degree of synergies and conflicts
	Overall policy mix (alignment between policy objectives and instruments)	Degree to which policy instruments contribute to achievement of the objectives
Comp-rehen-siveness	Policy strategy (breadth of topics covered + level of concretisation)	Coverage of relevant issues in objectives Degree of concretisation
	Instrument mix (regulatory, economic and soft instruments + targeting creative and destructive processes)	Division between regulatory, economic and soft instruments Identification of instruments targeting creative and destructive processes
	Overall policy mix (comprehensiveness across topics)	Comparison of breadth of policy objectives and degree of concretisation with characteristics of instruments

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The distinction between instruments targeting creative and destructive processes was operationalised as follows (Table 2):

Table 2: Operationalization of distinction between 'creative' and 'destructive' policy elements (based on Bach and Hansen, 2023; Kivimaa and Kern, 2016).

Policies targeting 'creation'	Policies targeting 'destruction'
• Knowledge development and diffusion (e.g., innovation platforms, networks, training schemes, guidelines) • Establishing market niches/market formation (e.g. market-based policy instruments such as labelling) • Entrepreneurial experimentation (e.g., relaxed regulatory conditions for experimenting, goal-based regulation) • Resource mobilisation (e.g., financial: R&D funding Human: educational policies, labour-market policies, secondment of expertise • Legitimation/support from powerful groups (e.g., innovation platforms, foresight exercises, etc.) • Influence on the direction of search (e.g., goals set and framing in strategies, targeted R&D funding schemes, goal-based regulations, tax incentives, voluntary agreements, technical performance standards, etc.)	• Control policies (e.g., regulation and taxes, stricter discharge requirements, etc.) • Significant changes in regime rules (e.g., structural reforms, new overarching laws) • Reduced support for dominant regime technologies (such as huge, centralized plants, chemical treatment) • Changes in social networks (e.g., involving not only incumbents in policy advisory councils, formation of new organisations and networks)

Preliminary results from the assessment were presented at the final project consortium meeting in Prague in late April 2024 and have since been updated and refined through online meetings with the leaders of the local CoPs.

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5 Policy mix for reuse in agriculture (Sicily)

5.1 National context

Italy is the European country with the highest per capita consumption of drinking water and the second highest consumption in agriculture. More than 30 billion cubic metres of water are withdrawn annually, with approximately half being used for agricultural purposes (Eurispes, 2023). While Italy ranks third in Europe in terms of water availability, numerous studies have shown a steady reduction in the amount of renewable water in Italy (e.g., ISPRA, 2023), and parts of Italy have already experienced water scarcity due to droughts. In 2024, Sicilian authorities declared a state of crisis and implemented water rationing (EU Observer, 2024). The challenge of water scarcity is further exacerbated by a poorly maintained water distribution network. In 2020, approximately 42.2% of the water that entered Italy's water infrastructures was lost due to leakages. In Sicily, the number is even higher, with approximately 52.5% of the water being lost (Eurispes, 2023). Hence, neither Italy nor Sicily can assume that water will be readily available in the future.

Responsibilities in Italy's water sector are divided between the national and regional levels. The central government is responsible for the national water legislation, including the transposition of EU directives into Italian law (Division of Powers, n.d.). The Ministry of Environment, Land and Sea is responsible for setting environmental standards, including water pollution, and the Ministry's work is supported by the Italian Institute for Environmental Protection and Research (ISPRA) which provides scientific and technical support relating to environmental protection. The Italian Regulatory Authority for Energy, Networks and Environment (ARERA) is responsible for overseeing water services. Water tariffs are proposed by local regulators (Ente di Governo d'Ambiente, EGA) and subsequently approved by ARERA (EurEau, 2020: 40–41).

Regional authorities have a number of responsibilities regarding water management, including the development of river basin management and flood risk management plans and overseeing water service areas (Division of Powers, n.d.). Each region has a Regional Agency for Environmental Protection (ARPA), which, among other duties, is responsible for ensuring that wastewater treatment is compliant with legislation at the EU and national levels. Municipalities are tasked with choosing how to manage water services, and though ownership structures vary across the country, approximately half of Italy's population is served through delegated public management (EurEau, 2020: 40).

Italy's water resources face challenges relating to climate change and under-investments in an already aging infrastructure. As climate change is expected to reduce water availability already in the short term (ISPRA, 2023), measures are needed to address these challenges. One possible measure is to reuse treated wastewater for agricultural irrigation in order to reduce the pressure on Sicily's dwindling water resources.

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5.2 Solution in focus

The focal solution demonstrated in Sicily is reuse of reclaimed wastewater for agricultural irrigation. This means that domestic and/or industrial wastewater will be collected, treated, and then distributed to be reused for irrigation. Sicily has fourteen facilities for wastewater reuse, but only four of these have been formally approved by the authorities. Regardless, the overarching concept is that water from these treatment plants is distributed to local farmers. This requires establishing a set of strategies and regulations that can realise this concept, including a sustainable business model where local farmers are enlisted as customers.

The solution in the Sicily case can be divided into four steps: wastewater collection, wastewater treatment, additional treatment to achieve the minimum quality required for reuse, and distribution and reuse. These steps can also be considered as a network of distinct but inter-related action situations, where decisions are made and different technologies, actors, institutions, and policies and institutions are involved (Figure 2). To better understand the drivers and barriers to realisation of the full chain of activities, it is necessary to explore the mix of policies and regulations associated with the different steps and how they interact.

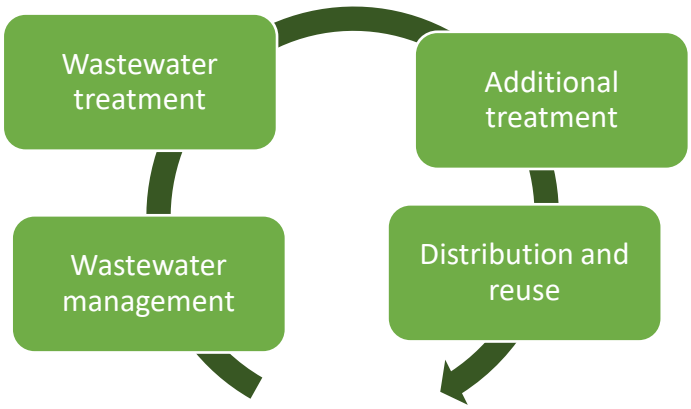


Figure 2: Circularity and action situations considered for the case in Sicily.

5.3 Policy documents considered

To assess the policy mix for the demonstration case in Sicily, we identified a series of relevant policy documents and instruments in collaboration with the project partners at UNIPA. Further documents were identified through strategic searches, and subsequently double-checked with UNIPA. The documents that have been considered in the Sicily case are listed in Table 3. Instruments are denoted by soft colours while policies and strategic documents are denoted by strong colours. The columns relate to the different action situations in question. The rows are ordered according to administrative level (transnational (EU), national, regional). Furthermore, they visualise which policy elements that only target one or few of the linked action situations, and which ones that are more generic, by the length of the coloured row and the number of columns it cuts across.

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Table 3: Policy documents assessed for the demonstration case in Sicily.

	Wastewater collection	Wastewater treatment	Additional treatment	Distribution and reuse
Transnational	A Blueprint to Safeguard Europe's Water Resources			
	Water Framework Directive			
	Urban Wastewater Treatment Directive			
			Regulation (EU) 2020/741	
			Guidelines to support the application of Regulation 2020/741	
				Farm to Fork Strategy
	Industrial Emissions Directive			
	European Regional Development Fund			
				Agricultural Fund for Rural Development
				Common Agricultural Policy
	Rural Development Programme for Italy			
	Rural Development Programme for Sicily			
	EU Circular Economy Action Plan			
	EU Zero Pollution Action Plan			
	EU Taxonomy for Sustainable Finance			
National		Technical standards for the reuse of wastewater		
	National water tariff			
	Ministerial Decree 185/2003			
	Decree Law No. 39/2023			
		Code on the Environment		
Local/regional	Regional Law no. 4 of March 22, 2022			
				Regional plan against the drought
				Temporary funding for distributing reclaimed water
	Resolution No. 3/2014 (regional water tariff)			
	Smart Specialization Strategy Sicily 2021–2027			

5.4 Mix of policy domains and rationales

In the Sicilian case, each step is quite clearly delimited. There are two main rationales: pollution (which includes health and food safety) and water security. *Step 1* relates exclusively to water management: whether the Water Framework Directive or the national and regional water tariffs, these are instruments that in some way relate to water services. *Step 2* implicates more policy domains. Whereas the water sector remains the focus, the quality standard placed upon reclaimed water relate to its use in agriculture and, by extension, to food safety. *Step 3* is quite narrow: it concerns the distribution of wastewater, which in the Sicilian case is carried out by trucks. Whereas this implicates both transport and climate policy by introducing a new source

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of CO_2 emissions into the system, this distribution solution is considered temporary. Finally, *Step 4*, there are a series of strategies and instruments that apply to reuse. These overlap considerably with those identified in Step 2. For example, the performance standards relate to environmental pollution, but also seek to ensure human health and food safety (as per Regulation 2020/741 and the accompanying guidelines).

The distribution of treated wastewater destined for reuse has been emphasised the least. Currently, the plan is to temporarily distribute the reclaimed water using trucks, as there is currently no dedicated pipe network for transporting reclaimed water. This distribution strategy is temporary: the municipality of Corleone has committed to funding truck-based distribution for one year. In the medium to long-term, repairing the existing but currently decommissioned pipe network appears more feasible. However, a closer look at the instruments in Step 4 also show that these do not really *promote* reuse; rather, they specify the necessary conditions for it to be allowed. The policies and instruments in Step 4 primarily focus on health and safety (for food and for workers), though some policies (e.g., Farm to Fork) emphasise the importance of reducing the quantity of drinking-quality water being used for irrigation – for example, by using reclaimed water.

5.5 Policy development at different levels

5.5.1 Transnational (EU) level

A key document in EU's water policy is the Urban Wastewater Treatment Directive. This was first published in 1991, and a revised version was adopted by the European Parliament in 2024. This regulation specifies the legal framework for collecting, treating, and discharging wastewater. As such, the Directive primarily deals with wastewater as pollution, with the goal of preserving both human health and the environment more generally. The Water Framework Directive (2000) is the main legislation for water protection within the EU. Upon establishment, a core objective of the Water Framework Directive was for Europe's water bodies to achieve good status by 2015. In 2012, the EU published A Blueprint to Safeguard Europe's Water Resources (COM/2012/0673). The Blueprint concludes that the EU member states are set to fail the objective set forth in the Water Framework Directive. The Blueprint goes on to present a development path for the EU's water policies. Maximisation of water reuse is listed as one of the objectives. In terms of funding, the Blueprint lists the European Investment Bank and the EU Cohesion Fund as possible sources. In terms of regulation, the Blueprint refers to "possible regulation in 2015" (Blueprint, 2012, p. 22).

The EU has since developed regulation geared towards reuse, but rather than being finished by 2015, the regulation was first implemented in 2020 through Regulation 2020/741 Minimum requirements for water reuse. This regulation specifies the minimum quality requirements for specific kinds of agricultural reuse. It also specifies how to monitor water quality and how to carry out a risk assessment before reusing treated wastewater. Regulation 2020/741 has since been supplemented with a set of Guidelines to support the application of Regulation 2020/741,

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an official document that further guides EU member states in implementing Regulation 2020/741.

5.5.2 National level

In Italy, the main regulations of relevance for water reuse are Legislative Decree 152/2006 and Ministerial Decree 185/2003. The 152/2006 is the overarching environmental protection legislation in Italy and regulates emissions to water, as well as to air and soil. Ministerial Decree 185/2003 regulates the use of reclaimed wastewater and specifies the maximum levels of different pollutants. As opposed to the EU's Regulation 2020/741, which only concerns agricultural reuse, the Ministerial Decree encompasses urban, agricultural, and industrial reuse of treated wastewater. The regulations follow a fit-for-all approach, meaning that they state that reusable water should hold the same high standard regardless of area of application. Maximum pollutant concentration is defined along 55 parameters, and in many cases the levels are as strict as for drinking water.

In 2023, Italy effected Legislative Decree 39/2023, a regulation that was developed to combat the country's water crisis. In practice, Legislative Decree 39/2023 excised agricultural reuse from Ministerial Decree 185/2003 and simultaneously implemented the risk assessment required by the EU's Regulation 2020/741. In practice, this served to harmonise the national regulations with the ones introduced by Regulation 2020/741. However, beyond this harmonisation, the national level policies and instruments offer little in terms of support, economic or otherwise, for ensuring compliance with new regulations. This also applies to water reuse practices: there are no specific instruments geared towards generating customer-side demand for reclaimed water for agricultural irrigation. Italy has got a National Strategy for the Circular Economy, adopted by the Ministry of Ecological Transition (2022) within the Recovery and Resilience Plan for Italy. The plan includes multiple generic measures to foster CE (e.g., a CE innovation platform and mandatory Green Public Procurement), as well as an Italian Phosphorus Platform (IPP), but water and wastewater are not defined as focus areas (EEA, 2022).

5.5.3 Regional level

In Sicily, wastewater treatment was previously regulated through Regional Law 27/1986. The Regional Law placed stricter limits upon phosphorous and nitrogen content than what would later be imposed through Ministerial Decree 185/2003. This discrepancy has since been rectified with Regional Law 4/2022, which (a) is the regional transposition of Italy's Legislative Decree 39/2023 and (b) aligns the regional regulations for phosphorous and nitrogen content with those listed in the national Legislative Decree 152/2006. Regional Law 4/2022 was set in effect through Implementing Degree 6/2024.

5.6 Consistency

The policy mix for the Sicilian case is consistent in terms of treatment requirements. Simultaneously, the policy mix is stronger on treatment than reuse. In terms of water

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treatment, the current policy mix is well-aligned. Barring some minor inconsistencies that have been rectified, such as the variegated national and regional maximum levels for phosphorous and nitrogen content, there are no obvious contradictions in the policy mix relating to wastewater treatment. The main inconsistency was between the EU's Regulation 2020/741 and Italy's 185/2003, where Italy's 185/2003 established stricter constraints on the use of treated wastewater than what is present in the EU's Regulation 2020/741. This has since been rectified with Italy's Decree 39/2023, which brought the two regulations in compliance. Further, with Sicily's Implementing Decree 6/2024, the regional maximum pollutant levels were harmonized with those of EU and Italy. The consistency between these three core policy elements is illustrated as green dotted arrows in Figure 3.

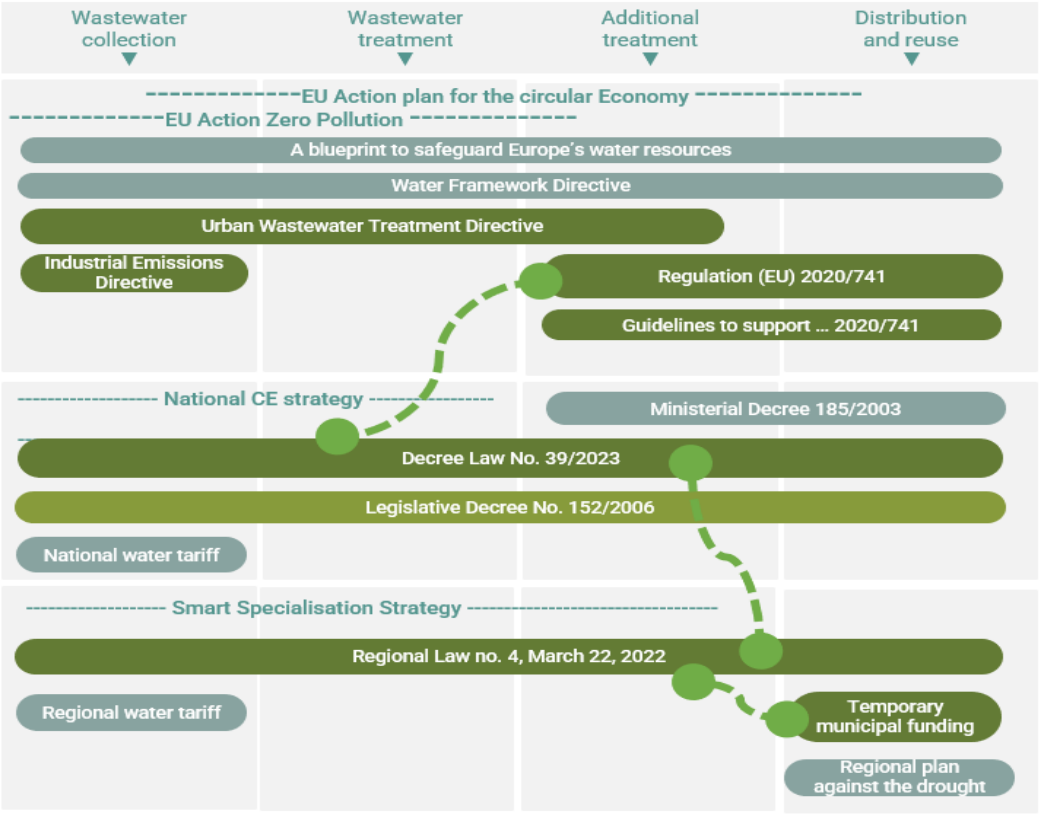


Figure 3: Key policy elements and relationships identified for the Sicily case.

In the figure, green boxes illustrate elements promoting the circular water solutions in question (the darker colour the more active promotion), the grey boxes signify elements that are relevant, without actively promoting, and orange boxes illustrate elements that are critical or central to the case. Green dotted lines connect element that strongly support each other, grey dotted lines illustrate tension, and red arrows show a conflicting relationship (not all illustrations are used in all figures).

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The policy mix in the Sicilian case emphasises the collection and proper treatment of wastewater. Barring Corleone Municipality's temporary funding for distributing reclaimed water, the existing instruments specify the conditions that must be fulfilled for reuse to happen, without necessarily promoting the *practice* of reusing reclaimed water. Whereas there is a considerable number of pull instruments (e.g., performance-based regulations which promote the appropriate treatment of wastewater), there are few economic instruments of this type (e.g., economic incentives for farmers to use reclaimed water rather than groundwater). Further, there are few system-type regulations aimed at promoting reuse (e.g., market design, (permanent) infrastructure provision). Hence, despite being *consistent*, framework conditions for promoting the reuse of reclaimed water remain underdeveloped in Sicily.

The policy mix is consistent and promotes the reuse of water within specific parameters. Whereas there have been some tensions (if not contradictions) in the policy mix previously, these have recently been straightened out. Currently, trans-national, national, and regional policies regarding water treatment are consistent in the sense that they require the same performance from treatment facilities. The more stringent pollutant levels introduced with the revised Urban Wastewater Treatment Directive represent a further challenge. The increased stringency may incur higher operating costs, both due to monitoring and because further treatment is necessary. Hence, the implementation of the Urban Wastewater Treatment Directive might also impact the uptake of solutions for reuse.

5.7 Comprehensiveness

In terms of comprehensiveness, the policy mix shows some notable *gaps,* as opposed to the *inconsistencies* observed, e.g., in the Czech case. In particular, this applies to the goal of promoting agricultural reuse of treated wastewater. In this regard, one might argue that the policy mix is not sufficiently comprehensive to promote treatment *and* reuse. Whereas there are policies and instruments that address every step in the process, instrument types and purposes are limited. A key challenge appears to be the market for reclaimed wastewater. The treated wastewater must either be transported using vehicles or by repairing a currently decommissioned pipe network. However, reclaimed water requires treatment, which raises the question of who will pay. If agricultural customers were to pay for reclaimed water, economic logic dictates that it cannot be more expensive than groundwater. Simultaneously, reclaimed water production remains an underdeveloped practice as water companies' revenues do not significantly improve when wastewater is reused instead of just being discharged. Hence, it is necessary to establish a funding scheme that specifies the distribution of additional costs.

Furthermore, a permanent distribution infrastructure is necessary for the reuse of treated wastewater to continue after the regional funding for truck transport runs out. Who will repair the decommissioned distribution network for reclaimed water, and who will bear the costs? While the existing components of the policy mix are consistent, these questions suggest that the policy must be made more comprehensive for reuse to become viable. Specifically, an important gap is instruments for establishing a market for reclaimed wastewater and

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instruments for (permanent) infrastructure provision. For example, approximately half the water extracted in Italy per year is used for agricultural irrigation, and whereas policies such as Farm to Fork emphasise the importance of establishing sustainable agricultural practices, water reuse is not listed as a possible measure for reducing the quantity of potable water being used for irrigation purposes.

5.8 Summary characterisation of policy mix

The policy mix relating to the Sicily case encapsulates two main objectives. The first objective relates to pollution: collected wastewater should be treated in specific ways and the treated wastewater should fulfil specific quality requirements with regards to various kinds of pollutants. This objective also reflects concerns regarding health and safety: the treated wastewater should be safe for workers who will come in contact with the water and for those who will consume products that have been irrigated with the treated wastewater. The other objective relates to water scarcity: by using treated wastewater rather than groundwater for agricultural irrigation, the solution in question would reduce the use of groundwater and thus counteract the projected water scarcity in Sicily. To achieve this objective, one must establish a (permanent) distribution network for treated wastewater, as well as a customer base – in this case, farmers.

In the policy mix relating to the Sicily case, economic instruments are the most common. However, these also tend towards the generic: this includes a considerable number of EU funding mechanisms, instruments that provide funding for a wide variety of purposes rather than specifically for water treatment and/or reuse. A key exception among the economic instruments is the funding provided by Corleone Municipality, which represents highly specific – though temporary – funding for infrastructure provision. In terms of instrument design features (Rogge & Reichardt, 2016: 1624), this funding represents a high degree of support, although with limited – though predictable – duration. However, this predictability simultaneously introduces a lack of predictability that has yet to be addressed: whether the decommissioned pipe network will be repaired and thus be able to reliably supply the farmers with reclaimed water. Farm practices are addressed in a series of policies and instruments (e.g., the EU's rural development programmes and agricultural funds), but there are no specific instruments that seek to promote the use of reclaimed water. Finally, it is worth noting that the EU has channelled significant funding to Sicily in an effort to make the region's wastewater plants operational.

As opposed to the generic nature of most of the economic instruments, the regulatory instruments are specific. (A minor exception would be Italy's Environmental Code (152/2006), which covers air, soil, and water pollution; still, its treatment of water is quite specific.) The regulatory instruments are performance-based: rather than specify *how* wastewater should be treated to be reused, they specify threshold values to be reached (e.g., in terms of bacteria, nutrients, or salinity per volume). As such, they leave the question of the exact treatment up to the agencies and/or organisations that carry out the treatment (i.e., they are flexible

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instruments; on instrument design features, see Rogge & Reichardt, 2016: 1624). This applies to trans-national regulations as well as national and regional ones.

The instruments in the policy mix also include information. Here, a key instrument is the Guidelines associated with the EU's 2020/741, which specify how to carry out the risk assessment required by the 2020/741. The mandatory risk assessment became part of Italian legislation with Italy's Decree 39/2023, and the Guidelines thus promote the fulfilment of the newly established EU regulation. Further, Horizon 2020 is an example of a systemic instrument that emphasizes information: for example, participants in WIDER UPTAKE share knowledge about water treatment and reuse, as to facilitate the uptake of solutions that promote – for example – the reuse of treated wastewater.

In terms of instrument *purposes*, the policy mix contains push instruments (mainly various kinds of EU funding), systemic instruments (information provision and sharing, (temporary) infrastructure provision), and pull instruments (regulation). Pull instruments are most prevalent, as exemplified by the regulations setting requirements for water treatment. These regulations emphasise wastewater treatment: by instating stricter standards, these will conceivably create a demand pull for the technologies needed for appropriately treating wastewater. However, this does not promote reuse, which points to the need for different kinds of instruments. For example, an obvious gap relates to systemic regulation, specifically market design. As of writing, it is not obvious how the operation and maintenance costs related to the additional treatment of wastewater for reuse will be distributed, especially if the costs associated with treatment mean that the price of reclaimed wastewater will exceed that of pumping irrigation water from other sources, including groundwater.

The policy mix consists of both creative and destructive instruments (Bach & Hansen, 2023; Kivimaa & Kern, 2016), with creative instruments being most common. Key examples of creative instruments include knowledge creation/diffusion (R&D funding) and experimentation (funding for temporary distribution infrastructure). Control policies – the aforementioned regulations – represent the sole destructive instrument. Control policies are often put in place to challenge the existing regime (for elaboration, see Kivimaa & Kern, 2016). However, in this specific case, the performance-based regulations help bolster a regime that emphasizes water treatment. This is not a problem per se, as reuse requires treatment. However, there is a notable gap pertaining to distribution and reuse of reclaimed water. Whereas a series of policies and instruments *promote* the use of reclaimed wastewater, including in the agricultural sector, less attention has been paid to the distribution of this water. Again, by considering gaps in the policy mix rather than looking for inconsistencies, it becomes clear that creative instruments that seek to establish a market for reclaimed water are missing – this includes market design as well as the physical infrastructures necessary for such a market to function.

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6 Policy mix for reuse and biochar (Ghana)

6.1 National context

Since achieving independence in 1957, Ghana has undergone notable demographic transformations, marked by swift population growth and the rapid pace of urbanization. The 2021 census reported the country's population at approximately 30.8 million, a sharp rise from figures recorded in past decades, showcasing a significant trend toward the expansion of urban areas which now host the majority of the Ghanaian populace. Indeed, reports indicate that over half of Ghanaians reside in urban centres (GSS, 2014). Projections suggest that by 2050, urban inhabitants might comprise as much as 70% of Ghana's population (Tuholske et al., 2020, p. 420). This shift is vividly illustrated by the urbanization of areas like Greater Accra, which had an urbanization rate of 90.5% in 2010, making it the most urbanized region in the nation.

In Accra, where more than 90% of the population lives in urban areas, the urban sprawl since the 1990s has resulted in notable deficiencies in services and infrastructure, underscoring the urgent need for developmental planning and action (UN Habitat, 2015). Urbanization's impacts span various aspects, affecting food and nutrition security, availability of agricultural land, freshwater extraction and use, and environmental quality, including water body pollution. The health implications tied to urban living conditions are linked to wastewater management, touching on both its quality and quantity, and the potential for human contact (Van Rooijen & Drechsel, 2008). These elements highlight the intricate relationship between urban growth and sustainable development, stressing the necessity for cohesive strategies in urban planning and environmental stewardship.

Despite the rise in urbanization, discussions on water reuse solutions for the nation and specific urban locales experiencing chronic water shortages have not been priority. In major urban centres, including Accra, urban farming serves as a livelihood strategy. Urban farmers, situated along major drains and streams, draw untreated water from these sources to irrigate their crops directly. Research by Fianko & Korankye (2020, pp. 140–141) found that water utilized for such informal irrigation in the Greater Accra Region contained substantial microbial loads, indicating that its usage should be restricted.

Often, wastewater is discharged with little to no treatment. Still, urban farmers are sometimes forced to use this wastewater for irrigation, at risk of negative health effects due to pollution (Qadir et al., 2010). Meanwhile, wastewater is inherently rich in nutrients. Using treated wastewater in agriculture may lessen dependency on synthetic fertilizers, reduce farmers' financial burdens. It can not only boost agricultural yields, but also further sustainable resource use by incorporating nutrient cycles into urban ecosystems (Scott et al., 2004). Therefore, optimising current water usage through the adoption of water-efficient practices, including water reuse, is essential.

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6.2 Solutions in focus

As urbanization increases and land use rapidly evolves, the availability of arable land within the city diminishes. Moreover, the transformation of streams and creeks into drainage systems has reduced the access to clean water necessary for cultivating high-quality vegetables in urban centres, particularly during the dry season. Simultaneously, Ghana's reliance on wood-based charcoal as a primary energy source persists.

Against this background, the Ghana demonstration case, located in Accra, is centred on the innovative practices of reusing treated wastewater for urban agriculture, and transforming sludge from wastewater treatment into biochar. Sewerage Systems Ghana Limited (SSGL), tasked with wastewater treatment in Accra, is collaborating with a group of urban farmers and the Council for Scientific and Industrial Research (CSIR) to reclaim water from treated wastewater for irrigating vegetables in selected areas in Accra's inner-city areas. Additionally, the case includes converting sludge into biochar, a biofuel alternative to traditional wood-based charcoal, which is being tested in collaboration with local SMEs. The activities involved thus encompass different sectors and action situations, including wastewater treatment, production of reuse water (adding extra treatment, monitoring and documentation of safe water quality), and application of the reuse water in informal urban farming, as well as biochar production and eventual use of biochar as a substitute for regular charcoal. This implies that multiple policy domains are involved, and there is a need to understand how the mix of relevant policy elements interact to promote and/or hinder the development and diffusion of the focal solution.

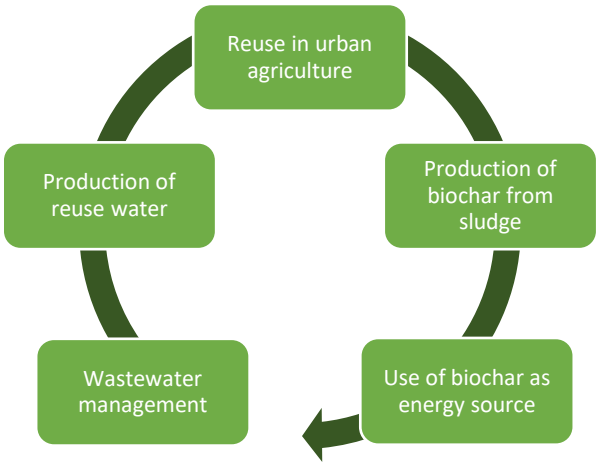


Figure 4: Circularity and action situations in focus in the Ghana case.

6.3 Policy documents considered

To assess the policy mix, a series of relevant policy documents and instruments was identified in collaboration with the project partners at CSIR-STEPRI. Further documents were identified through strategic searches, and subsequently double-checked with CSIR and the other core members of the local CoP. The documents considered are listed below (Table 4). Instruments are denoted by soft colours while policies and strategic documents are denoted by strong colours. The columns relate to the different action situations in question. The rows are ordered

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according to administrative level (transnational (Africa, ECOWAS), national, regional). Furthermore, they visualise which policy elements that only target one or few of the linked action situations, and which ones that are more generic, by the length of the coloured row and the number of columns they cut across.

Table 4: Policy documents considered for the Ghana case.

	Wastewater management	Production of reuse water	Reuse in urban agriculture	Biochar production	Biochar use
Transnational	Africa Water Vision 2025				
	West Africa Water Resources Policy				
	Guidelines on the Right to Water in Africa				
		WHO wastewater reuse guidelines			
				ECOWAS Bioenergy Policy (EBEP)	
				ECOWAS Centre for Renewable Energy and Efficiency (ECREEE)	
				ECOWAS White Paper on Access to Energy Services	
National	Ghana Science and Technology Policy				
	Circular Economy Roadmap				
	National Integrated Water Resources Plan (2012)				
				Ghana country action plan for clean cooking	
	The Environmental Sanitation Policy (Revised, 2009)				
	Ghana WASH Sector Development Programme 2021-2030				
	IWRM Policy, 2007,				
	Water Resources Commission Act, 1996 (Act 522)				
	National Water Policy, 2007				
	Environmental Assessment Regulations 1999				
	Environmental and Social Management Framework under Ghana Economic Transformation Project				
				The Renewable Energy Masterplan	
	Sanitation Levy				
				Energy Sector Strategy and Development Plan	
	Ghana Innovation and Research Centre				
	Water Sector Strategic Development Plan (2012-2025).				
				Biofuel production licence	
			Wastewater Use for Agriculture in Ghana		
					Ghana country action plan for clean cooking
	L.I. 1692 Water Use Regulations, 2001				

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		Mining law of Ghana			
				National Energy Policy	
	Ghana Building Code				
Local	National Community Water and Sanitation Strategy (NCWSS)				
	Metropolitan, Municipal and District Assemblies				

The development of the energy and water sector policies are linked to issues of environment, sanitation, health, productive sectors, and socio-economic development plans. These policies are cross-cutting and multi-sectoral interconnected, aligned to National Development Planning Policy Frameworks, as well as regional and international level development trajectories and commitments such as the Sustainable Development Goals, ECOWAS Bioenergy Policy, ECOWAS Centre for Renewable Energy and Efficiency (ECREEE), the New Partnership for Africa's Development (NEPAD), UN Framework Convention on Climate Change (UNFCCC), Paris Agreement and it associated Nationally Determined Contribution and the Africa Water Vision 2025 adopted by the African Union and Economic Commission for Africa.

The key policies and instruments in the water/sanitation sector considered in this study at the national and local levels include Ghana's National Water Policy, the Environmental Sanitation Policy (Revised, 2009), Water Sector Strategic Development Plan (2012-2025), L.I. 1692 WATER USE REGULATIONS, 2001, National Community Water and Sanitation Strategy, Water Resources Commission Act, 1996 (Act 522), among others. At the transnational level, the Guidelines on the Right to Water in Africa, WHO Wastewater Reuse Guidelines, The Africa Water Vision 2025 and the West Africa Water Resources Policy were the main policy mixes identified for the water sector. It suffices to say that most of the polices and instrument mixes for the water sector at the transnational (Africa) level focused on water use and management and not much on wastewater treatment and reuse as seen in Table 4. The key national level energy sector policies on the other hand include the National Energy Policy, Energy Sector Strategy and Development Planning (2010), Bioenergy Policy, Renewable Energy Master Plan (2019-2030), Renewable Energy Act, 2011 (Act 832) and Ghana's Energy Transition Framework 2022 to 2070. At the transnational level, we have the ECOWAS Bioenergy Policy adopted by the ECOWAS Member States in December 2016.

6.4 Mix of policy domains and rationales

The governance of water and its related systems in Ghana are in multiple but interdependent levels from transnational to local administrative level. It is structured through various institutional and regulatory frameworks at both the national and local levels. It must be added that some of these regulatory frameworks are also informed by regional frameworks that Ghana is singed up to, such as the Africa Union and the ECOWAS region. In terms of institutional structures, there is the Ministry of Sanitation and Water Resources, which is the main policy institution in the sector; the Water Resources Commission, Community Water and Sanitation

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Agency (CWSA), Ghana Water Company Limited (GWCL) and Local Government Authorities. Whilst the GWCL is responsible for urban water management and use, the CWSA is solely in charge of peri-urban and rural water needs. There are also the Local Government Authorities who play an oversight role at the local water governance level. For the purposes of the second recovered product, biochar, the Ministry of Energy with its allied agencies such as the Energy commission are the main institutional structures supporting the energy sector.

6.5 Policy developments at different levels

6.5.1 Transnational level (African Union, ECOWAS)

For this policy mix studies, we find that Ghana adopts its policies and guidelines for water use and management as well as sanitation instruments internally but also learns from best practices and aligns with some of the regional policies. Nation States under the Africa Union and ECOWAS are to adhere to the various policies enacted at that level, including the Africa Water Vision, West Africa Water Resources Policy and the Guidelines on the Right to Water in Africa as indicated in Table 4. For the energy sector, there are the ECOWAS Bioenergy Policy and ECOWAS Centre for Renewable Energy and Efficiency (ECREEE) as an institution, where nation states such as Ghana aligns with its frameworks. These policies are mainly guidelines for the management of water and energy resources across the continent, including transboundary water management. What we however find is that these continental policies do not particularly target standards as it is expected that nation states set their own standards. Where there are no standards at the national level and regional level, globally acceptable standards are adopted.

In the case of Ghana, the Ghana Standards Authority has its set standards for safe drinking water and charcoal (not biochar) but what we do not find is the quality standards for safe treated wastewater. Therefore, the main level of governance when it comes to such standards is the adoption of the World Health Organisation (WHO) Wastewater Reuse Guidelines. The WHO guideline uses the Hazard Analysis and Critical Control Points (HACCP) concept, which is a variety of entry points where health risks occur or can best be mitigated before the food is consumed. Hence, the guidelines for safe wastewater reuse ensures that the health of consumers are duly protected. Ghana also adopts her main policies from regional policies and frameworks such as the Africa Water Vision, Guidelines on the Right to Water in Africa and the ECOWAS Bioenergy Policy. Ghana is signed up to the UN Framework Convention on Climate Change (UNFCCC), Paris Agreement and the Nationally Determined Contribution and the National Voluntary Reporting System for the Sustainable Development Goals (SDGs), in the context of WIDER UPTAKE, SDGs 6 and 7.

6.5.2 National and local levels

At the national level, there are a number of policies and instruments that are designed to regulate the entire lifecycle of the solution, from water management through to treatment and the production of alternative energy such as the biochar. One can observe from Table 4 that there are more national level policies and instruments in the water and energy sector that there are at the Africa regional level that member states are required to adopt. There is

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interconnectedness between the various policy frameworks. For example, all policy frameworks in the water sector are based on the mother National Water Policy and the Environmental Sanitation Policy. Though different government ministries and agencies may handle the each of the policies, there is a synergistic relationship between all the frameworks. The Water Sector Strategic Development Plan provides a framework for implementing the vision, policy objectives and targets for the water and sanitation sector under the National Water Policy. During the WIDER UPTAKE Policy dialogue held in January 2023, it came to light that more interministerial coordination to ensure proper alignment of policies is required. However, there are efforts across government ministries to bridge that gap as the Ministry of Sanitation and Water Resources directly depends on coordination with other ministries in the governance of the water sector. Generally, there is very little policy coordination between the Ministry of Sanitation and Water Resources and the Ministry of Energy.

The local assemblies are responsible for the governance of the water sector in Ghana and the National Community Water and Sanitation Strategy (NCWSS) addresses water governance issues at the local level. There are no specific policies and instruments that are focused on local governance in the energy sector however as they all draw on the national level of governance. The Metropolitan, Municipal and District Assemblies (MMDAs) also play a key role in policies and actions regarding waste management at the local level.

6.6 Consistency

Transition to CE pivots strongly on the key national policies and the consistency in articulating the importance of circularity in sustainable development. As far as Ghana's case in the Wider Uptake goes, at the centre of Ghana's policy mix are the national development framework, the National STI Policy, the Environmental Policy, the Water Policy and the Energy Sector Strategy and Development Plan. Across these specific policy documents, the concept of resource recovery or CE consistently gets mentioned to various degrees. Even though there may not be extensive elaboration of CE in the given policy document, there are expressions of the practice. Thus, the National STI Policy underscores the importance of "the inclusion of the Circular Economy (CE) concept and practice and makes emphasis on promoting CE to address the environmental sanitation challenges." (MESTI, 2023, p.6). This is an explicit and emphatic commitment to CE. However, the National Energy Policy touches on CE implicitly and elaborates extensively on the mechanisms for waste-to-energy in Ghana's energy mix. It states the intention "to convert most of the wastes generated in municipal activities, industrial operations and agricultural operations to energy" (Ministry of Energy, 2010, p.23). However, the question is what is the depth of consistency in these policy documents and by extension consistency in the policy instruments? This is where there are inadequacies, which need to be addressed. The policy documents need further elaboration of the CE approaches to give muscle to the policy statements on CE.

In terms of alignment of policies, regulations and instruments, there have been efforts in this direction. For example, MESTI published the Ghana Energy Transition and Investment Plan to

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guide the role of public and private actors in the energy sector towards achieving the country's climate goal of net-zero emissions by 2060. It is commendable that MESTI from its perspective as the oversight establishment for Environment, Science, Technology and Innovation, asserted its responsibility in publishing a transition and investment plan for energy, which conventionally is in the domain of the Ministry of Energy. It is a document that clearly details insights of the energy sector, the socio-economic factors constraining or facilitating the transition and the levels of investment that are required as well as the specific technologies necessary for the transition. Yet, this document does not significantly mention the waste-to-energy technologies that contribute to achieving the net-zero emissions. Rather, in discussing the six main decarbonization technologies, it mentions clean cooking technologies highlighting improved biomass and electric cookstoves. This closely aligns with the Ministry of Energy's action plan for clean cooking. WIDER UPTAKE has demonstrated the possibilities in processing municipal waste to provide veritable options in improved biomass technologies such as biochar, which should attract investment.

The international policy environment of CE spans regional and international policy documents produced by the relevant organisations. For example, WHO's Guidelines for the Safe Use of Wastewater, Excreta and Greywater is a comprehensive publication guiding countries' approach to the multi-purpose use of wastewater, excreta, and greywater with the objectives of addressing core health and safety issues. It elaborates on the risks and trade-offs in the use of these resources, the measures for mitigating risks and the instruments for addressing risks. Safety issues are also the subject of the related African regional documents in the water sector. The African Union, UN Economic Commission of Africa, and the African Development Bank (ADB) produced the Africa Water Vision for 2025: Equitable and Sustainable Use of Water for Socioeconomic Development. In breaking down the vision for 2025, the document includes the "sustainable access to safe and adequate water supply and sanitation to meet the basic needs of all"; that, there must be sufficient water for food and energy security. Thus, there is some concordance in policy-related documents at the international and regional levels. Figure 5 shows the linkages between policies across the water and energy sectors. There are a number of mixes in the various water and sanitation policies as well as between energy related policies. However, we do not find many linkages between water and energy policies directly.

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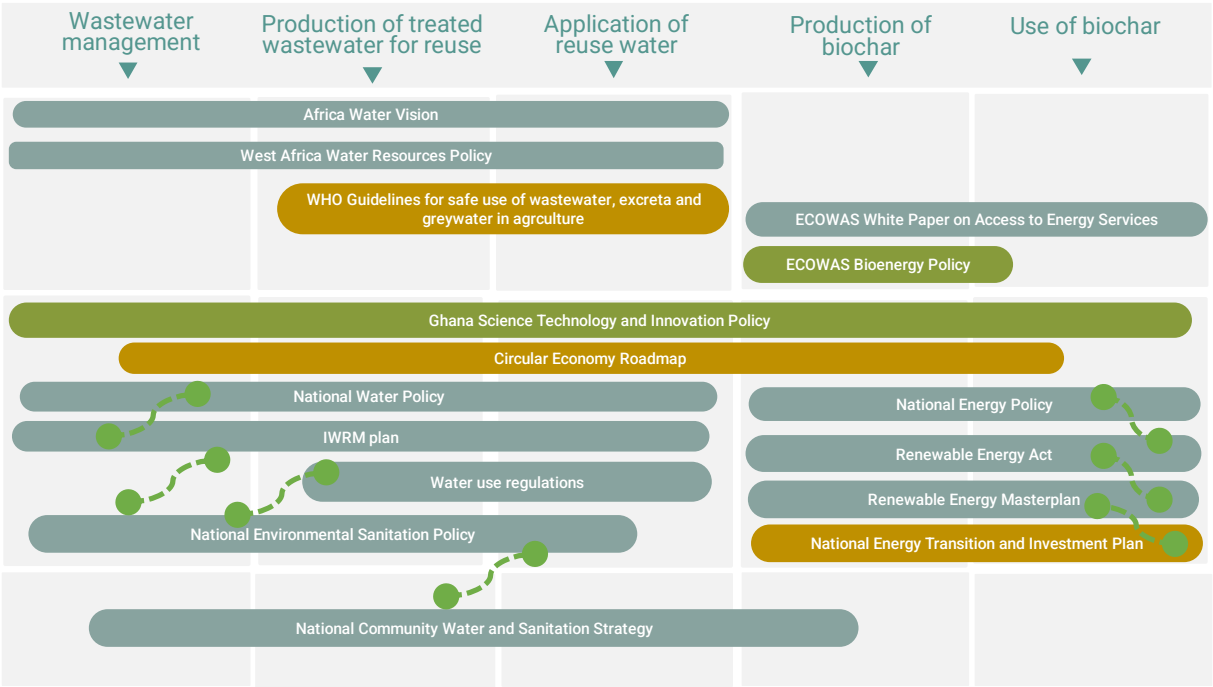


Figure 5: Key policy elements and relationships identified for the Ghana case.

6.7 Comprehensiveness

CE has only come into the forefront of development discourse in recent years and therefore, understandably, some policies, legislations and policy instruments did not take account of CE. For example, Ghana's legislative instrument L.I. 1692 Water Use Regulation 2001 focuses only on abstracted water from the existing water resources of the country. Recycled and treated water is not covered in the legislative instrument. And why not? This is a piece of legislation that provides for the management of the process for water access, distribution, and use. In that sense, it brings into the national system for management of all types of water at the disposal of all stakeholders in the water sector. The inclusion of municipal treated wastewater for urban agriculture will even address the taboo issues surrounding such recycled water.

Furthermore, there are gaps in the assemblage of policy instruments to foster achieving the overall goals of CE in Ghana's cases of biochar and treated municipal water for urban agriculture. For example, whereas there is a drafted document on Standards for Charcoal production, transportation and distribution, there is none for biochar. This has to be addressed.

Nevertheless, Ghana may still want to enact CE-focused legislations and regulations to ensure comprehensiveness and concreteness. Taking the case of wastewater for urban agriculture, the legislation can delve deeper into the process of assuring quality and safety of the treated water. More incentives must be created in policy instruments for the relevant stakeholders to be willing to invest resources into recycling of wastewater. Similar legislation and regulation can be provided also for the energy sector.

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Indeed, the policy engagements at the respective ministries including Ministry of Energy, Ministry of Water and Sanitation and Ministry of Trade and Industry, have proven that there is understanding of the CE issues in promoting the use of treated wastewater and producing energy products from the sludge of wastewater. Advocacy, awareness creation and sensitization of the stakeholders are important activities that must be intensified in Ghana for the sustainability of the outcomes of the Wider Uptake project.

There are also a number of transnational policies that have some level of linkages with water policies in Ghana. Amongst these are the Africa Water Vision and the Guidelines on the Right to Water in Africa. It must however be noted that National states are required to align with the various transnational policies. There is some level of adoption of transnational policies into the development of national policies related to water and energy. The challenge has always been the policy instruments to ensure that the policy objectives are achieved. They are either lacking or flawed especially when it comes to recycled water, which situates in the CE concept.

6.8 Summary characterisation of policy mix

The consultative processes followed during the formulation of these policies consider adaptation of national approaches to local sociocultural, environmental, and economic circumstances without compromising on sustainability of water and energy resources. The rational and policy relevance although varied in focus but have commonalities in terms of structure and policy coherence. Issues of inadequate funding for effective policy implementation, weak institutional coordination and governance systems, data challenges and ineffective enforcement of existing regulations cut across policies reviewed. From the policies reviewed Environmental Impact Assessment is mandatory for the construction of wastewater treatment plant, Ghana's Economic Transformation Project has an Environmental & Social Management Framework 2019 with a component on solid and liquid waste management. Embedded in the Integrated Water Resources Management Plan for Ghana the principle of circularity and the use of treated wastewater for irrigated agriculture particularly among populations with special needs for water, pressures from climate change, increasing population growth and rapidly growing water and energy demands.

The Environmental Sanitation Policy (2009) by the Ministry of Local Government and Rural Development clearly demonstrates the role of communities, individuals, households and traditional authorities and local assemblies in solid and liquid waste management especially in terms of advocacy, sensitization, better enforcement and compliance of environmental sanitation and health related laws and regulations.

Ghana's National Energy Policy (2010) goal of converting most of the waste generated in the municipal activities and industrial operations to energy, the Ghana Renewable master Plan (2020 -2030) with targets for increasing the renewable energy in the national energy generation mix, the ECOWAS Bioenergy Policy that seeks to promote modern, sustainable and vibrant

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bioenergy sector in the region and unlocking potential of the private sector players by removing institutional, legal, financial, social environmental and capacity gaps and barriers are all in sync with the objectives of the Wider-Uptake Project. Central to Ghana's Energy Transition Framework 2022 to 2070 is decarbonisation using cross-cutting approach, across all relevant sectors and decentralization of the energy transition processes. Embedded in the Ghana's Industrial Policy is also the need to build technological capacity of SMEs to enhance economic growth and competitiveness.

Lastly, there is growing recognition for international cooperation for sustainable development of water resources. WHO Guidelines for Safe Use of Wastewater consider socio-cultural aspects and perceptions as well as behavioural change over time that could impact on the circularity principles depending on how these issues are managed.

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7 Policy mix for nutrient recovery (Norway)

7.1 National context

Norway has abundant annual freshwater resources, of as much as 377 billion cubic meters (Norwegian Water, 2021). As climate change is expected to lead to more unpredictable climatic conditions, the country faces increasing challenges in terms of stormwater management and water quality. At the same time, there is a backlog in infrastructure maintenance and development. Recent assessments suggest that investments of NOK 320 billion are needed between now and 2040 (Norwegian Water, 2023). This is due to many factors, including a challenging geography and widely dispersed population, as well as the current structure of the water sector.

Some water utilities are organised as intermunicipal companies, but the majority operate as either technical departments of their owning municipalities, or as municipal companies. By 2022, there were 2 754 wastewater treatment plants with a capacity of 50 person equivalents (pe) or more.[1] The total discharge (excluding overflows), was around 1 480 tons of phosphorus and 19 500 tons of nitrogen, and approx. 133 000 tons of dry matter from sludge was removed and used for different purposes.[1]

Although Norway is not a Member State, the rules and regulations for its wastewater sector are strongly influenced by EU policies, most notably the urban wastewater treatment directive (UWWTD, Directive 1991 (91/271/EEC)). Presently, 10-15% of the population are not connected to the public sewage system. Within areas defined as sensitive, 90% of the population are connected to public facilities with secondary treatment. The corresponding percentages for normal and less sensitive areas are 75% and 29%. In all, around 2/3 of Norway's wastewater treatment facilities do not comply with the current requirements. This is partly due to legal adjustments made when the directive was implemented in Norwegian legislation, and partly due to slow follow-up by the utilities.[2] However, the local and national authorities have recently signalled a more restrictive practice, where temporary exemptions no longer will be granted.

Thus, multiple factors, including foreseen environmental challenges, infrastructure backlogs, an unused resource potential, and increasing pressure to comply with legal requirements, make the time right for development, testing, and implementation of innovative circular water solutions.

7.2 Solutions in focus

1 Statistics Norway (2023). https://www.ssb.no/natur-og-miljo/vann-og-avlop/artikler/kommunale-avlop-2022.ressursinnsats-gebyrer-utslipp-rensing-og-slamdisp/_/attachment/inline/55f4defd-3ce9-4971-a61e-5aa0b19ada33:c2bb61b653fbd48d67930abcbb3c9e29f6bc1ccf/RAPP2023-50.pdf

2 https://www.regjeringen.no/no/sub/eos-notatbasen/notatene/2021/des/revisjon-av-avlopsdirektivet/id2966230/

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Against the background noted above, recovery of nutrients is at the core of the Norwegian demonstration case in WIDER UPTAKE. Energy recovery in the form of biogas production is already implemented at most of the larger utilities, and reuse of treated wastewater is less relevant than in water-stressed countries and regions. Fertilisers and soil improvement products based on waste and wastewater resource recovery are therefore an important focus area for the partners in the two Norwegian demonstration cases in WIDER UPTAKE: the water utility Hias, technology provider Hias How2O, and waste management and recycling company Sirkula in the county of Innlandet, and in Rogaland county; IVAR as the regional utility for management of water and waste, and the consultancy and trade company Terramarine, which is designing and distributing mineral-organic fertilisers produced from digestate at IVAR.

The circularity and value chain for these products includes the management of wastewater, e.g. in terms of regulating the amounts and types of discharge that enter the grid and wastewater treatment facility; innovative wastewater treatment (Enhanced Biological Phosphorus Removal (EBPR) and optimised pre-treatment); sludge management, in terms of adding resource recovery (production of struvite and digestate prepared for use as soil improvement); production of fertilisers and soil improvement; and finally, the use of these products, in agriculture, horticulture, and/or for urban greens, before the resources return to the natural resource cycle.

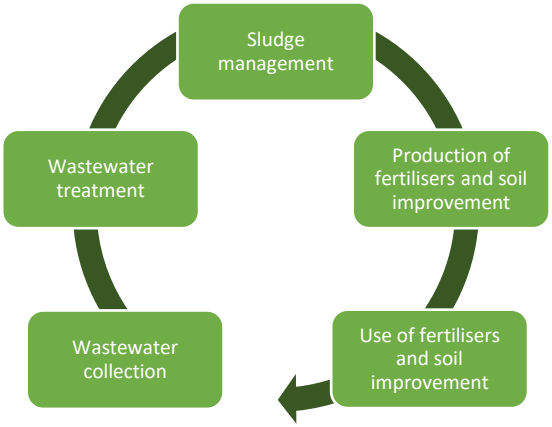


Figure 6: Circularity and action situations in focus for the Norwegian case.

As noted above, this set of activities can be understood as a network of action situations, each associated with a set of key actors, resources, and institutions of governance. Our analysis therefore covers the key policies and policy instruments associated with each step. The following section provides an overview of the main policy documents considered, and the action situations where they mainly exert their influence, before we proceed with the next steps in the analysis.

7.3 Policy documents considered

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The policies and instruments identified as most relevant for the Norwegian case are listed below (Table 5). We have linked each policy and instrument to the action situations where it is of strongest influence, by way of colouring and merging cells in the table. The different hues of colour signify different levels (transnational, national, regional/local), and shades are used to distinguish between policies (darker shade) and key instruments (lighter shade).

Table 5: Overview of policies and instruments considered for the Norwegian case.

	Wastewater collection	Wastewater treatment	Sludge management	Production of fertilisers and soil improvement	Use of fertilisers and soil improvement
	WHO/UNECE Protocol on Water and Health				
Transnational	Blueprint to safeguard Europe's water resources				
	Water Framework Directive (WFD)				
		UWWTD			
			Sewage sludge Directive		
				EU General Food Law	
	EU Action Plan for the Circular Economy				
				Farm to Fork Strategy	
			EU Fertilising Products Regulation (FPR)		
			EU Org. Farming Regulations		
	EU Action Plan for Zero Pollution				
			Waste Framework Directive		
	EU Industrial Emissions Directive (IED)				
	EU Taxonomy for Sustainable Finance				
National	National Water Regulation				
	Act on municipal water and wastewater utilities				
	Norwegian Pollution Regulation				
		Law on public procurement			
		Regulation of public procurement			
				Act on food production and food safety	
			National fertiliser regulation		
			National regulation of organic production		
			National Waste Regulation		
	National strategy for circular economy				
			National bioeconomy strategy		
		Environmental Technology Scheme			
					Regulation of support for agricultural production
Regional/local		Bioeconomy strategy for Innlandet			
		Bioeconomy strategy for Rogaland			
	County-specific discharge requirements				
	Regional plans for climate, energy and environment				
	Regional water management plans				
	Guideline for setting municipal water/wastewater fees				
					Guideline, under the Pollution act: Municipal

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					permissions/requirements for use of sludge
		Green Loans from the Municipal Bank (Kommunalbanken)			

7.4 Mix of policy domains and rationales

In Norway, the Ministry of Climate and Environment is responsible for water quality, while the Ministry of Energy governs water quantity, water uses and management of water bodies, and the Norwegian Food Safety Authority oversees drinking water and water for irrigation. The overarching policy for use of freshwater resources is laid down in the Water Resources Act (Norwegian Ministry of Energy, 2000) whose main objectives are to promote sustainable development and maintain biological diversity and riverine ecosystems. Reflecting the environmental conditions noted above, water quality, rather than quantity, is in focus. The Water Resources Act frames a number of regulations that set conditions for the management of water and wastewater. Wastewater management is regulated under the Norwegian Pollution Regulation (Norwegian Ministry of Climate and Environment, 2004). Energy recovery through biogas production is promoted actively, however, more in relation to sustainable energy transition and the Renewable Energy Directive, than in terms of CE.

Treatment, storage, and use of sewage sludge in Norway is a shared responsibility. The current Regulation of Fertilisers of Organic Origin (2003) places the Norwegian Food Safety Authority as the central authority, and aims to ensure sufficient quality of organic fertilisers, while preventing pollution, health and hygienic challenges and facilitating utilisation of the products. Quality controlled sewage sludge has been used directly as soil improvement at Norwegian farms for decades, and 50-60% of the sludge from Norwegian wastewater facilities is already used this way, mainly for grain production (Norwegian Water, 2024).

The production and use of organic fertilisers based on sewage sludge falls under the Regulation of Fertilisers of Organic Origin (Norwegian Ministry of Health and Care Services, Ministry of Climate and Environment, Ministry of Agriculture, 2003) as well. A Guideline from the Norwegian Food Safety Authority (2023) stresses that the regulation is grounded in the Pollution Act, the Municipal Health Act, the Soil Act, and the Food Act, and that the development of rules and regulations in this domain is done in close collaboration between the relevant ministries. The Guideline further emphasizes that whereas the quality and use requirements are defined with a view to health and safety, it is important that the organic fertilisers are used in line with good farming practices, integrated in fertilisation plans. Regarding waste management, the national Waste Regulation (Norwegian Ministry of Climate and Environment, 2004) bans deposition of sewage waste that does not fulfil the requirements for organic fertilisers and specifies targets for recycling of household and garden waste, which can be conducive for soil improvement combining feedstock from municipal waste and wastewater treatment plants.

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7.5 Policy developments at different Levels

7.5.1 Transnational (EU) level

For the transnational level, our assessment has focused mostly on policies and regulations in the EU. However, the Protocol on Water and Health (UNECE/WHO, 1999), which aims to ensure sufficient drinking water and sanitation for all, should also be mentioned. Norway recently revised its national targets and will develop an inter-sectoral plan towards their realization. The targets state that all municipal wastewater (85% of the population) shall have chemical or biological treatment by 2030, and that larger settlements shall have wastewater treatment plants complying with the EU's requirements for secondary treatment (without any specified timeframe) (Norwegian Ministry of Health and Care, Ministry of Climate and Environment, 2024). The targets are seen as highly ambitious when it comes to rehabilitation of drainage and sewer networks and will influence how water and wastewater companies assess the sustainability of their services in future.

At the EU level, the Circular Economy Action Plan (European Commission, 2020), provides a sustainable product policy framework with important generic elements, e.g., the Ecodesign Directive, a European Dataspace for Smart Circular Application, ecolabelling and Product and Organisation Environmental Footprint methods, minimum mandatory green public procurement (GPP) criteria, promoting the use of digital technologies for tracking, tracing and mapping of resources; the Bioeconomy Action Plan, enabling industrial symbiosis, and an SME strategy. Regarding water and nutrients, promotion of water reuse, an Integrated Nutrient Management Plan stimulating the markets for recovered nutrients, and the recent revision of the Urban Wastewater Treatment Directive are key elements, influencing the scope for nutrient recovery in Norway.

According to the revised Urban Wastewater Treatment Directive, integrated urban wastewater management plans must be implemented. Urban wastewater collecting systems shall be extended to all agglomerations of 1 000 p.e. or more, and secondary treatment (i.e. the removal of biodegradable organic matter) must be implemented by 2035. Thresholds and timelines for tertiary treatment (i.e. the removal of nitrogen and phosphorus) and quaternary treatment (removing a broad spectrum of micropollutants) pertaining to larger plants (>150 000 p.e.), by 2039 and 2045, respectively. Moreover, an extended producer responsibility (EPR) for wastewater from production of cosmetics and pharmaceuticals, and requirement of energy neutrality by 2045 are included. The increased requirements will be challenging for the Norwegian water sector, considering the low level of compliance with the current requirements (noted above). The new requirements will require large investments but may also be seen to create a window of opportunity for implementing biological treatment and nutrient recovery in the coming years.

The EU Sewage Sludge Directive (The Council of the European Communities, 1986), regulates the use of sewage sludge in agriculture, with the aims to protect humans and the environment

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by ensuring that heavy metals in soil and sludge do not exceed set limits. A recent review concluded that the directive remains relevant, but that the set of pollutants regulated needs review, notably considering organic compounds, pathogens, pharmaceuticals, and microplastics (European Commission, 2023). How strict the updated limits will be, will influence the feasibility of production and social acceptance of using sludge-based products in agriculture.

The EU Action plan for Zero Pollution (European Commission, 2021), includes updated lists of water pollutants, as a proposed amendment of the Water Framework Directive (European Commission, 2022). The proposal aims to stimulate the reduction of emissions at source and benefit the potential for water reuse. Following the Circular Economy and Zero Pollution Action Plans, a revised Industrial Emissions Directive has also been proposed (European Commission, 2022). This includes stricter pollution limits, common cost-benefit assessment for derogations, and more transparency regarding discharge from larger industry facilities, as well as new, binding standards for resource use. This, again, will have implications for the wastewater treated at European wastewater treatment plants, and the volumes and qualities of sewage sludge generated.

Centrally for the case of nutrient recovery, the EU Fertilising Products Regulation (FPR) (European Union, 2019), aims to open the Single Market to organic and waste-based fertilisers, by providing requirements for safety, quality, and CE marking. It allows optional harmonisation, in that producers can choose whether to apply the new EU rules or comply with national rules under the principle of mutual recognition, to sell their products on the EU market. Importantly, sewage sludge is excluded from the list of eligible feedstocks for fertilisers or pyrolysis & gasification materials to be used in agriculture. This means that products such as the mineral-organic fertiliser produced by IVAR cannot be CE-marked and traded freely across the EU. While it could be sold in specific EU countries under the principle of mutual recognition, the EU Regulation is unconducive for the social acceptability and upscaling and export of fertilisers based on sewage sludge.

On the other hand, the EU Organic Farming Regulations (European Union, 2018) were recently modified to authorise “Recovered struvite and precipitated phosphate salts”, as defined in the FPR, with the specification that to be allowed for Certified Organic Farming, the recovered phosphates “must meet the requirements laid down in” the FPR. This development is conducive for the case of phosphorus recovery in form of struvite since it increases the potential market for struvite as a fertiliser product.

Furthermore, the EU Waste Framework Directive (European Union, 2008) targets a minimum recycling and recovery rate of 55% of municipal solid waste by 2025, increasing to 60% and 65% by 2030 and 2035. It also defines the concept end of waste with overarching criteria (Article 6), while introducing the "polluter pays principle" and the "extended producer responsibility". Since 2018, the directive states that bio-waste shall be either separated and recycled at source, or collected separately, to enable conversion into e.g., bioenergy, digestate or organic

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compost.[3] The EU Taxonomy for Sustainable Finance is not seen to influence Norwegian municipal water and waste companies directly but may affect the access to capital for upscaling green innovations in future. "Anaerobic digestion of sewage sludge" is recognised as an activity that may contribute substantially to climate change mitigation, and in the 2023 Annex "Phosphorus recovery from wastewater" is presented as an activity that can contribute substantially to the CE.

7.5.2 National level

Norway's National Strategy towards a Green, Circular Economy (Norwegian Government, 2021) is less concrete than the EU's action plan, in terms of targets and measures. It does, however, discuss the need for a national report on phosphorus, and a turnover requirement for phosphorus. Furthermore, most of the above-mentioned EU directives and regulations are implemented in Norwegian legislation. The national Bioeconomy Strategy (Norwegian Government, 2016), complements the National strategy towards a green circular economy. It emphasizes the need for coordination and collaboration across sectors and includes an action plan for bioeconomy research and innovation, as well as a dedicated grant scheme for bioeconomy and climate adaptation.

The national Water Regulation (Norwegian Ministry of Energy, Norwegian Ministry of Climate and the Environment, 2006) implementing the EU Water Framework Directive, specifies environmental standards and max. values for various substances in freshwater and seawater. It defines how water resources management is organised, and provides the basis for classification of normal, less sensitive, and sensitive water areas, as well as the timeframes and possibilities for extending set deadlines for achievement of the specified environmental goals.

The objective of the national Pollution Regulation, part 4, which implements the Urban Wastewater Treatment Directive in Norway, is to protect the environment against detrimental effects from wastewater discharge. It specifies requirements for the design of wastewater treatment plants, and regarding treatment, monitoring, and discharge for smaller and larger settlements. It also provides definitions of different types of wastewaters, e.g., stating that when the volume of sanitary water is less than 2000 p.e., and less than 5% of the total wastewater volume, the wastewater is not considered as municipal wastewater. Implementation of the revised Urban Wastewater Treatment Directive will thus imply a momentous change, in that many small wastewater treatment plants in Norway will face new treatment requirements. While this represents a window of opportunity for resource recovery, many actors tend to emphasize the challenges, and Norwegian Water has recommended chemical treatment as the best option in many cases.[4]

[3] By 2023, a targeted amendment, focusing on textiles waste, has been proposed by the Commission. The amendment is not relevant for the innovation cases in WIDER UPTAKE.

[4] E.g., at their national conference (Norsk Vann Fagtreff) 2023).

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The Act on municipal water and wastewater utilities (Norwegian Ministry of Climate and Environment, 2012) provides national rules regarding ownership and financing of water cycle services. The principle of full cost recovery is linked to "best available technologies", which may limit the room for development and implementation of radical, innovative technologies. On the other hand, the national Law on Public Procurement (Norwegian Ministry of Trade, Industry and Fisheries, 2016) emphasizes that public procurement shall contribute towards reduction of negative environmental impacts, and promote climate-friendly solutions, amongst other by considering Life Cycle Costs. The Regulation of Public Procurement (Ministry of Trade, Industry and Fisheries, 2017), further specifies that climate and environmental criteria shall be weighted minimum 30%, and where the procurer provides a set of criteria in a prioritised order, climate and environment shall be among the top three.[5] While these specifications can be favourable for biological treatment and nutrient recovery, actors note that economic cost-benefit analysis still tends to override other criteria.

The national Regulation of Fertilisers of Organic Origin implements the EU Sewage Sludge Directive and includes requirements for municipal wastewater treatment plants, as regards quality control and quality of sludge. Following a revision process that has lasted since 2009, proposals for two new regulations, one for trade in fertilisers of organic origin, and the other for its production and use, were launched in November 2023 and March 2024, respectively. According to the current version, all sludge to be used for farming must be hygienized and stabilized, thus Norwegian legislation is stricter than the EU on this point. Moreover, soil improvement mixtures may not contain more than 30% of organic feedstocks such as sewage sludge, sludge from aquaculture, and food waste. At the same time, the Norwegian Food Authority can make individual decisions or rules for specific materials, thus, actors such as Hias and Sirkula can apply for dispensation allowing use of soil improvement containing biomass from wastewater treatment also in fields where fruits, potatoes and vegetables are produced.

The prolonged revision process has caused uncertainty regarding future framework conditions. The proposed new Regulation of trade in fertilisers of organic origin provides specific requirements regarding content (e.g., of phosphorus and heavy metals), conditions for different uses, and labelling and marketing. It introduces tighter requirements for the spreading of phosphorus, and for the use of sludge from fisheries and aquaculture in fertilisers, while specifying that the products shall be of high agronomic quality and easy to apply. The proposed Regulation of use of fertilisers of organic origin (Norwegian Ministries of Agriculture, Health and Care, Climate and Environment, Trade, Industry and Fisheries, 2024) will apply across fertiliser types, and replace previous minimum required spreading areas with new limits in terms of phosphorus content.

Further related to the use of the recovered nutrients, the national Regulation of Organic Production and marking of Organic products (Norwegian Ministries of Agriculture, and of Trade, Industry and Fisheries, 2022), implements EU Organic Farming Regulations (Regulation (EU)

[5] Recently amended and in force from January 2024.

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2018/848) in Norway. Thus, use of recovered struvite as fertiliser for organic farming is permitted in Norway since 25.09.2023. However, there are no specific incentives for farmers: The Regulation of support for agricultural production (Ministry of Agriculture, 2015), provides subsidies to contribute to an active and sustainable agriculture sector, and support for the purchase of recommended fertilisers, however use of recovered resources and CE does not figure as a topic therein.

The national Waste Regulation implements the EU Waste Framework Directive. Thus, all household waste is separated before collection since 2023, and all enterprises generating household type, garden, or agricultural plastic waste are required to sort their waste accordingly. Moreover, the waste shall be subject to material recycling (Norwegian Ministry of Climate and Environment, 2022). These changes provide increased opportunities for the production of organic fertilizers, combining resource streams from municipal waste management and wastewater treatment. Also of relevance for the production and use of organic fertilizers and soil products, the National Climate Plan 2030 states that the Government will assess the possibility of imposing a special tax on mineral fertilizers, and of imposing a tax on the harvesting of peat, while working towards its phase-out, either by banning harvesting of peat completely, or by not allowing actors to break new fields (Norwegian Ministry of Climate and Environment, 2021). The Norwegian Environment Agency (2020) has proposed a plan for such out phasing.

7.5.3 Regional/local

Both the county of Rogaland and the county of Innlandet, where the Hamar case is located, have dedicated bioeconomy strategies. The one for Innlandet was recently revised, towards a strategy where industrial symbioses, linking traditional biobased industry with new types of production in local bio-hubs, are in focus. Although none of the strategies place special emphasis on water or wastewater, there is a drive for bioeconomy initiatives and important synergies, e.g. IVAR and key actors in the agriculture and agro-food industry, together with the regional energy company Lyse and Air Liquide Skagerak, have signed an intention agreement for a new, largescale biogas and fertiliser plant, and Hias, through Hias How2O, has been active in a regional bioeconomy cluster in Innlandet.

The County Governor specifies the discharge requirements for industry as well as wastewater treatment plants. Their practice, whether restrictive or more liberal, may thus influence the quality of wastewater sent to the wastewater treatment plants. Following the polluter pays principle, municipalities have the option to calculate extra costs due to treatment of polluted wastewater from industry and include this in the water and wastewater fees charged from the respective industries.

According to the principle of full cost recovery, water utilities must apply best available technologies and cannot accumulate profits from their services. Once an activity reaches an income of 20%, it must be singled out as a separate entity. While this may disincentivise radical

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innovation, the rules for public procurement encourage use of environmental criteria, and the specification of requirements may also be used to favour innovative, environment-friendly solutions. The experience with so-called innovative procurement is variable: whereas a National Programme for Supplier Development (LUP) has been set up, the actors have mixed experiences with Innovative Procurement, as developing a market can be more challenging than anticipated.

7.6 Consistency

Regarding consistency between existing policy elements, we find that the national goals developed under the Protocol on Water and Health (Norwegian Ministry of Health and Care, Ministry of Climate and Environment, 2024) are aligned with the ambitions for transition towards CE, as more wastewater will be treated at a higher level, providing increased opportunity for resource recovery. The EU Circular Economy Action Plan included review of the regulatory frameworks for wastewater treatment and sewage sludge, as well as an Integrated Nutrient Management Plan, to stimulate use of recovered resources. Nutrient recovery from wastewater and sewage sludge, specifically, is not highlighted in the Farm to Fork strategy. However, two of the targets in Farm to Fork are to reduce nutrient losses to the environment by 50% and increase organic farming to 25 % of the EU's agricultural land by 2030, which may increase the demand for struvite. The EU's overarching water strategy, A blueprint to safeguard Europe's water resources (2012), is rather old. Besides enforcing the Water Framework Directive, improving compliance rates on wastewater treatment through long-term investment planning (including EU funds and EIB loans) is one of its objectives. Although it highlights the opportunity to take action at EU level to increase water reuse, it is not explicit about nutrient recovery.

The revision of the Urban Wastewater Treatment Directive does indeed aim for increased alignment with the Green Deal objectives, by reducing GHG emissions from the wastewater sector, and making it more circular through enhanced recovery of nitrogen and phosphorus as well as water reuse (European Commission, 2022, p. 3). Thus, it is associated with a high degree of consistency with overarching goals in sustainability transition, and related EU policies. However, the increased treatment requirements are debated in Norway, as they will be based on p.e., without distinguishing between geographical and ecological contexts (except for "outermost regions"). According to the Norwegian Association of Local and Regional Authorities (2023), the minimum requirements in the Urban Wastewater Treatment Directive must be flexible as to allow national authorities to impose stricter requirements where needed, and to ensure that the financial and environmental costs of the requirements are proportional with the environmental gains achieved.

The goals and instruments of the EU Zero Pollution Action Plan (European Commission, 2021) are consistent with the Water Framework Directive and the revised Urban Wastewater Treatment Directive. The revised Industrial Emissions Directive complements the Urban Wastewater Treatment Directive, aiming to combat water scarcity. Environmental

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performance targets are strongly encouraged also for resource efficiency, energy efficiency and raw material use. The new directive will in many cases imply that the wastewater received from industry and livestock (pigs and poultry) installations will be less polluted. However, it allows for more flexibility than the Urban Wastewater Treatment Directive, stating the need to avoid disproportionately higher costs compared to the environmental benefits.

The double aim of the Sewage Sludge Directive, of protection against heavy metals and increasing the use of sewage sludge in agriculture, must be carefully balanced. The review of the directive (European Commission, 2023) highlights the need for flexibility, considering the synergies and trade-offs associated with different sludge management alternatives in different localities. Still, the current variability of practices, where some EU countries allow use of untreated sludge in farming under specified conditions and others do not allow this at all, is associated with uncertainty regarding future limits and how these in turn may influence the acceptability of sludge-based fertilisers.

The goal of the EU Regulation on Fertiliser Products (European Union, 2019) is also consistent with the wider policies promoting CE and organic fertilizers. As noted in section 7.4.1, the regulation excludes sewage sludge from the list of eligible feedstocks. This reflects the Member Countries' different practices and attitudes towards sewage sludge and is not conducive for co-composting. In this sense, it may be argued that it conflicts with the goals of circularity and resource recovery. On the other hand, a delegated act adds STRUvite, Biochar and ASh-based products (STRUBIAS), to the list of new material component categories, and provides a framework for conformity assessment for such products (European Commission, 2021). The modification of the EU Organic Farming Regulations (2018) to authorize the use of struvite in Certified Organic Farming helped increase consistency and alignment across sectors.

Furthermore, the targets of the Waste Framework Directive (European Union, 2008), are conducive for co-composting, considering the potential synergies between recycling and resource recovery from municipal waste and wastewater treatment. The inclusion of anaerobic digestion of sewage sludge and phosphorus recovery of wastewater under the EU Taxonomy is also well aligned and may be conducive in a longer-term perspective.

As noted above (section 7.4.1), Norway's strategy towards a Green, Circular Economy (Norwegian Government, 2021) emphasizes the benefits of resource recovery from wastewater. As regards the water and wastewater sector, the key EU Directives are implemented in Norwegian legislation. Thus, there is consistency across these levels. However, until recently, the current wastewater treatment requirements were not enforced very strictly, and there is a certain tension regarding the requirements in the revised Urban Wastewater Treatment Directive relating to geographical conditions and existing infrastructure backlogs. Moreover, the principle of full cost recovery and application of "best available technologies" (Norwegian Ministry of Climate and Environment, 2012), while constituting a stable financing system, may limit the scope for utilities to develop, test, and implement radical innovations.

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When it comes to production and trade of products including recovered nutrients, the EU Regulation on Fertiliser Products (European Commission, 2021) will be implemented in Norway, with some adjustments. The Norwegian EEC position paper last updated 03.04.2023, notes that the EU regulation differs significantly from the current national regulations, and claims that in practice, the principle of free trade based on mutual recognition has not worked. The strictest requirements pertain to compost and digestate from biogas production, where assessment by an independent conformity control committee is required. However, component material category 11, which was added late in the revision process, opens up for by-products classified as such under the EU Waste Framework Directive, provided they satisfy the REACH requirements, and as noted above, struvite may be approved.

The revised EU regulation does not regulate the *use* of organic fertilisers. The current Norwegian Regulation of Fertilisers of Organic Origin, however, contains rules for their use, based on heavy metal content. In this sense, the Norwegian regulation is stricter, as it specifies requirements and limit values based on the content, rather than the source of feedstock. Since the EU rules for CE-marking[6] do not make it obligatory to provide information on heavy metal content, this complicates the rules for use, i.e. there are certain conflicts between the EU and national regulations per now.

The proposal for a revised version of the national Regulation of Fertilisers of Organic Origin aims to reduce the current increase in P levels in large parts of Norway's farmland, while facilitating CE. Farms may either follow the fixed P limits or do 'spreading in balance' with yield absorbsion. This will provide flexibility for farmers, but not favour clean sludge-based products with high levels of plant-available P. The shortening of the period when spreading and storing sewage sludge is allowed, will create challenges for many utilities, e.g., require new/increased storage facilities (Norwegian Water, 2024). According to Norwegian Water, the proposal lacks foundation in a proper impact assessment and suggest extending the proposed time window for sludge spreading (ibid.). However, this could also provide an impetus, for production of new organic fertiliser or soil improvement products. Considering use of fertilisers and novel soil improvement products with content from sewage treatment, there are so far few incentives. As noted above, the Norwegian Regulation of support for agricultural production does not promote use of organic fertilizers, nor those specifically based on sewage sludge. While green public procurement may be used to pull demand and build acceptance, e.g., for use in public parks and construction work, this has up to now mainly been applied to products and services of a different nature. The key policy elements and relationships highlighted in this section are illustrated below (Figure 7).

[6] CE here stands for Conformité Européenne, and CE-marking serves as a confirmation that the product has been assessed and found to meet EU safety, health and environmental protection requirements.

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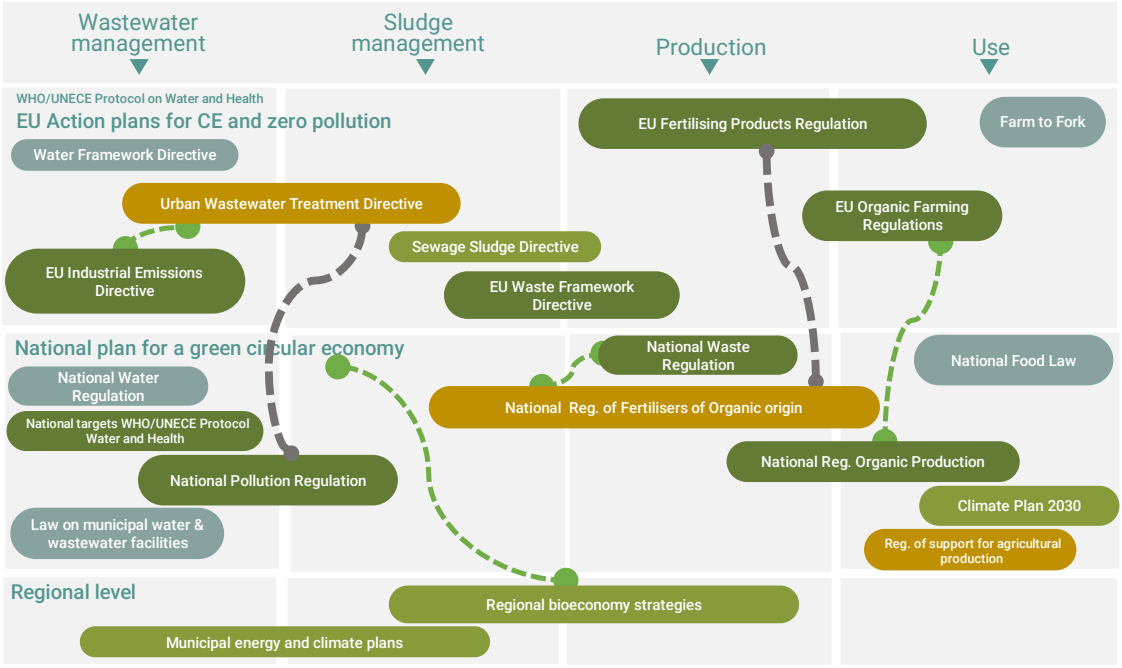


Figure 7: Key policy elements and relationships identified for the Norwegian case.

In the figure, green boxes illustrate elements promoting the circular water solutions in question (the darker colour the more active promotion), the grey boxes signify elements that are relevant, without actively promoting, and orange boxes illustrate elements that are critical or central to the case. Green dotted lines connect element that strongly support each other, grey dotted lines illustrate tension, and red arrows show a conflicting relationship (not all illustrations are used in all figures).

7.7 Comprehensiveness

Considering comprehensiveness, or the degree to which all necessary policy objectives and measures to address barriers to uptake are covered in the current policy mix, some shortcomings can be identified. First, we noted above that the EU's overarching water policy document per now (European Commission, 2012) does not directly address nutrient recovery from wastewater treatment. The Declaration for an EU Blue Deal (EESC, 2023) argues for a much stronger water policy, where water is not considered mainly in terms of environmental policy, but as an independent strategic priority. The declaration states that incentives should be put in place for a transition to a circular water economy, however the text focuses mostly on reuse and recycling of water, to reduce water poverty. The European Water Resilience Initiative, paused in February 2024, likewise emphasizes the need to ensure water availability both for 'traditional' (drinking water, agriculture, etc.) and new water uses (hydrogen, chip plants, etc.), while reducing pollution and restoring biodiversity. When the European Council since added water resilience to the strategic agenda for 2024-2029, President van der Leyen mentioned the need for a CE approach,[7] but how this will be reflected in the coming Water Resilience Strategy remains to be seen.

[7] https://www.waternewseurope.com/von-der-leyen-puts-water-resilience-strategy-back-on-the-table/

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The EU CEAP, as discussed above, comes with a broad mix of regulatory, economic, and informational instruments, including support for research and innovation through e.g., Horizon 2020 and Water for All. However, the Integrated Nutrient Management Action Plan (INMAP) is not yet in place. As it will address the use side, by "ensuring more sustainable application of nutrients and stimulating the markets for recovered nutrients", this policy element may foster acceptance and demand, also for products with recovered nutrients from wastewater treatment. The EU's Biodiversity, Farm to Fork strategies and Zero Pollution Action Plan aim for at least 20% reduction of the use of fertilisers. Moreover, the JRC Knowledge for Integrated Nutrient Management Action Plan report (Grizzetti et al., 2023) recommends that the EU should reduce its mineral fertiliser input of nitrogen and phosphorous by approx. 60%. The European Environmental Bureau (EEB) has argued that nutrient recovery from organic waste streams can contribute significantly to replacing synthetic fertilisers, hence, it is crucial that the INMAP provides clear direction and addresses EU end-of-life criteria and markets for secondary raw materials.[8] Further on this point, EurEau has proposed that compulsory requirements to blend a certain ratio of recovered P and N should be introduced for all mineral fertiliser in the EU – in parallel with the blend-in requirements for biofuel in conventional transport fuels (EurEau, 2022).

The revised UWWTD is also associated with a mix of measures, i.e. both progressive reinforcement of different treatments, improving the monitoring of contaminants, and better implementation of the polluter pays principle, to contribute to the financing of enhanced wastewater treatment. Whereas the revised UWWTD relates very explicitly to reuse and was preceded by a specific regulation providing guidelines and requirements for this (Regulation (EU) 2020/741, on minimum requirements for water reuse), parallel rules and recommendations for nutrient recovery have not been provided. However, the provisional agreement on the revised UWWTD (European Parliament, 2024) includes the following note from the evaluation:

> In order to ensure the proper and safe recovery of nutrients, including the critical substance phosphorus, from sludge, a minimum combined reuse and recycling rate should be set at Union level. Member States should be able to choose whether to reuse or recycle, or both, urban wastewater or sludge, or both, in order to recover phosphorus. The combined minimum reuse and recycling rate should take into account the phosphorus content of sludge, which can vary from one area to another. It should also take into account the level of saturation of each national market (…). (European Parliament, 2024, note (42), no page numbers).

Furthermore, the FPR is associated with requirements and informational instruments: maximum levels of contaminants and pathogens; minimum content of nutrients and other

[8] https://eeb.org/wp-content/uploads/2023/09/Letter-to-COM-re-INMAP_Sept-2023-FINAL.pdf

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relevant characteristics; conditions for labelling; and prescription for conformity testing (European Union, 2019). It is also associated with delegated acts with more specific criteria and requirements, but sewage sludge is not included as an eligible material. The modification of the EU Organic Farming Regulations to authorising recovered struvite, on the other hand, may contribute to increased awareness and acceptance of these products.

At the national level, the push towards enhanced wastewater treatment was up to recently limited, in the sense that it was possible for utilities to get exemptions or discharge requirements adjusted to the local conditions. However, stricter enforcement and the new requirements associated with the revised UWWTD is strengthening the drive towards wider implementation of more advanced treatment solutions. While the principle of full cost recovery provides a stable funding mechanism, the high investments required raise concerns about the social impacts of increasing water and wastewater fees. A grant scheme for development and testing of innovative solutions by water utilities was established in 2021, however the funding is limited and emphasises water supply rather than wastewater management. Some grant funding is also available for the establishment of biogas production, but there are no economic instruments specifically targeting innovation in wastewater treatment or nutrient recovery.[9]

Regarding sludge management, there are guidelines for the spreading of hygienized sludge but no information-type instruments or regulatory instruments (e.g., standards) for co-composting or struvite production. The proposed new regulations for production, trade and use of organic fertilisers help reduce uncertainty regarding future framework conditions and place new restrictions on the direct use of sewage sludge for soil improvement. In some cases, this could stimulate uptake of the solutions demonstrated in WIDER UPTAKE (though some consulted stakeholders argued that it might generate more interest in sludge pyrolysis). At any rate, the national report on phosphorous and turnover requirement for phosphorous, mentioned in the National Strategy towards a Green, Circular Economy would provide an important impetus, but has yet to materialise.

While there are recent policy developments aiming to disrupt the harvesting of peat, the public authorities in Norway do not provide specific incentives for the use of organic fertilisers or struvite at present.

7.8 Summary characterisation of policy mix

The policy mix relating to the case of nutrient recovery in Norway centres around the following main objectives: limiting pollution and protecting human health, minimising the environmental footprint, and ensuring resource efficiency in water cycle services, and contributing to sustainable soil management.

[9] It should be noted that the Government's proposed state budget for 2025 suggests removing the VAT on water fees, as to ease the economic burden on citizens.

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In terms of instrument types, regulation-type instruments (regulations and requirements) are most prominent. We also note that economic-type instruments, notably more stringent application of the polluter pays principle and defining an extended producer responsibility for some industries, are used to some extent. However, full cost recovery remains as the key funding mechanism, and the policy mix does not include any dedicated economic incentives for nutrient recovery. There is also a lack of informational instruments. The promised EU INMAP and national report on phosphorous presented in Norway's the National plan for a circular and greener economy have not materialised, and there are no specific regulation-type (standards) or information-type instruments (e.g., guidelines) for struvite production or fertilisers based on wastewater treatment residues.

Considering the respective action situations, we find a strong emphasis on fostering change in wastewater treatment and, to a lesser extent, in sludge management. Despite the targets to restrict use of conventional fertilisers and peat-based products, less has so far been done to stimulate the use of organic fertilizers and soil improvement among farmers, greenkeepers, and the citizens in general.

The policy mix contains both creative and destructive instruments (Kivimaa and Kern, 2016). The destructive instruments tend towards the incremental, in that they first aim to increase treatment levels and change existing systems and practices. The creative measures include strategic measures (e.g., establishing goals for wastewater treatment and energy neutrality) and regulation-type instruments aimed at establishing market niches (e.g., by approving struvite for ecological farming). The implications of these findings in terms of pathway realisation and policy recommendations are discussed in chapter 10.

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8 Policy mix for biocomposites (NL)

8.1 National context

In the Netherlands, water is high on the agenda. 55% of the territory is below sea level, and the country has a strong track record in water management. Still, water quality and the resilience of freshwater ecosystems are pressing issues (OECD, 2014), The climate is getting wetter and dryer simultaneously, causing both increased flooding challenges and periods of water shortage (Dutch Directorate General for Public Works and Water Management, 2019). As the Netherlands is a member state, EU directives have direct influence, and the Water Framework Directive (WFD) is integrated into the National Water Plan. The Dutch Government aims for a 50% reduction in waste by 2030, and a waste-free economy in 2050. According to the Circular Economy Implementation Programme, the Rijkswaterstaat (Ministry of Infrastructure and Water Management) is to make all its operations circular by 2030.[10]

In the Netherlands, 21 Regional Water Authorities govern the wastewater and drinking water sector. In the greater Amsterdam region, where the demonstration case is located, Waternet is responsible for both wastewater treatment, surface water management, and drinking water treatment. Waternet is governed by the Regional Public Water Authority Amstel, Gooi en Vecht. Building and construction work falls under the Ministry of Housing and Spatial Planning, and the Environment Buildings Decree of the Netherlands (Ministry of the Interior and Kingdom Relations, 2020).

8.2 Solution in focus

The solution demonstrated in the Dutch case is production of biocomposite materials. The targeted use is in riverbank protection panels, which may substitute panels made of tropical hardwood. The biocomposite materials are developed by the research and consulting company NPSP and utilize raw materials obtained from various sources, including wastewater treatment, the drinking water production process, and biomass collected from surface water management conducted by Waternet. A notable component is calcite, a byproduct of the water softening process in drinking water treatment, which acts as a filler in the biocomposite. Additionally, cellulose is recovered from wastewater treatment, specifically recovered toilet paper, to be incorporated as a fibre in the biocomposites. Water reeds and dike grass sourced from Waternet are also key elements in this innovative material.

As WIDER UPTAKE has evolved, NPSP has continued to enhance their biocomposite materials, resulting in four distinct biocomposite types. These variants feature varying percentages of biomaterial inputs, ranging from Nabasco 8010, which incorporates only water reeds, to the 100% biobased Nabasco 8040. The latter includes grass fibres, resin, and filler, with the two latter being sourced from agricultural waste. In addition to application of biocomposites in canal riverbank protection panels, the material has also been used for façade claddings, road and canal signs, and interior design pieces.

[10] https://www.government.nl/topics/circular-economy/circular-dutch-economy-by-2050

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Considering the categories of bioresources and set of actions involved, the circular value chain emerging in this case can be said to have five steps: Harvesting of reeds as part of surface water management, utilisation of calcite from the drinking water treatment process, recovery of cellulose from wastewater management, design and production of the novel biocomposite materials, and eventually their use, for riverbank protection and other applications, mainly in the building and construction sector. The different steps may be considered as a partly overlapping network of action situations, involving different actors, institutions, and policy domains. The circularity is illustrated below (Figure 8).

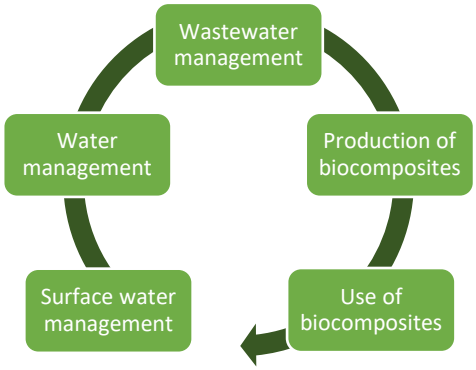


Figure 8: Circularity and action situations considered for the Dutch case.

8.3 Policy documents considered

To assess the policy mix for the Dutch demonstration case, we identified a series of relevant policy documents and instruments in collaboration with the project partners at TU Delft and NPSP. Further documents were identified through strategic searches, and subsequently double-checked with NPSP. The assessed documents are listed in Table 7. Instruments are denoted by soft colours while policies and strategic documents are denoted by strong colours. The columns relate to the different action situations in question. The rows are ordered according to administrative level (transnational (EU), national, regional). Furthermore, they visualise which policy elements that only target one or few of the linked action situations, and which ones that are more generic, by the length of the coloured row and the number of columns it cuts across.

Table 6: Overview of policy documents assessed for the Dutch demonstration case.

	Surface water management	Drinking water treatment	Wastewater treatment	Production of biocomposite materials	Use of biocomposite materials
Transnational	Blueprint to safeguard Europe's water resources				
	Water Framework Directive (WFD)				
			Urban wastewater		

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			treatment directive		
		Drinking water directive			
			Sewage sludge Directive		
				EU REACH regulation	
	EU Action Plan for the Circular Economy				
			EU updated Bioeconomy strategy (2018)		
				EU Construction products regulation	
				EN 15804:2012+A2:2019 Sustainability of construction works	
	EU Action Plan for Zero Pollution				
	Waste Framework Directive				
				EU Industrial Emissions Directive (IED)	
	EU Taxonomy for Sustainable Finance				
	Directive 2014/25/EU procurement in water, energy, transport..)				
					EU Public procurement directive
					Proposal for a new Ecodesign Directive
National	Transition the Dutch Economy to zero waste by 2050				
	"Point on the Circular Water 2050 horizon"				
	Water in the Circular Economy (WiCE) joint research programme				
	National Water Plan 2022-2027 (with preview towards 2050)				
	Dutch water act (Water Wet)				
	Regional Water Authorities Act				
	NWB Bank				
	Dutch Water Authorities association				
	STOWA (for applied research in water governance)				
	Energy and Raw Materials Factory (ERMF)				
	Environment and Planning Act				
	Environmental Management Act				
	NWMP			National Waste Management Plan	
					Dutch Public Procurement Act
					Public procurement decree
					National Plan on Sustainable Public Procurement for 2021-2025
				Building Decree 2012	
				Transition agenda for construction	
Regional	Amsterdam Circular Strategy 2023-2026				

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	Regional water by-law				Regional water by-law
	Water Authority Tax				
			Municipalities Act		
			Municipal sewerage charges		

8.4 Mix of policy domains and rationales

While the national government has the overarching responsibility for managing the natural water resources, the relevant tasks are largely in the hands of the Regional Water Authorities, which enjoy financial independence as a result of their own tax system (Havekes et al., 2017). As regards surface water, flood protection is a key issue involving major economic interests as well as human safety. Maintaining dykes and canals, and making sure that resilient water resources are available, are key concerns. This includes preventing water surpluses as well as shortages and protecting both groundwater and surface water from pollution.

Drinking water supply is not included in the definition of water management (ibid). It is handled by separate entities, including nine private companies with public part-ownership, and the Amsterdam Waternet, as a non-profit foundation, and regulated under the Drinking Water Act. Here, health and safety are the main concerns. It should be noted, however, that the institutions and rationale of this part of the water governance system are implicated in the studied case only to the extent that they influence the availability of calcite.

Water quality management and wastewater treatment is on the hands of the regional water authorities. However, sewerage is the responsibility of municipalities and regulated under the Environmental Protection Act. The municipalities can fund this through municipal sewerage charges, implemented through by-laws under the Municipalities Act. The wastewater treatment is thus financed through the same regional tax as the surface water and groundwater management, while sewerage is financed through the local, municipal taxes. This means that wastewater treatment is financed independently of national politics, and without having to compete with other government expenditures. Moreover, a dedicated bank, NWB Bank (Nederlandse Waterschapsbank) has been established to help finance local water management projects. As we shall see below, the rationale in wastewater management is strongly influenced by CE and the conception of wastewater treatment plants as resource factories.

The production of biocomposite materials, as process manufacturing, is an industrial activity, falling under EU Regulation no. 1272/2008 on classification, labelling and packaging of substances and mixtures, and the REACH regulation, which aims to "ensure a high level of protection of human health and the environment, including the promotion of alternative methods for assessment of hazards of substances, as well as the free circulation of substances

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on the internal market while enhancing competitiveness and innovation" (European Commission, Reg. 1278/2008, p.1). The goal of minimising pollution is pronounced also in the Industrial Emissions Directive and CEAP.

Many of NPSP's biocomposite products are made for uses classified as building and construction. Here, the central legislation at the EU level is the Construction Products Regulation (European Commission, 2011), which provides harmonized rules to ensure the proper functioning of the internal market for construction products. The seven Basic Requirements of this regulation include the requirement Sustainable use of natural resources. For public works, construction activities also fall under the EU procurement law, which amongst other aims to widen the uptake of innovative, green and social procurement (European Commission, 2017).[11] As the construction sector is one of eight sectors associated with high potential for circularity, e.g. in EU's Circular Economy Action Plan (2020), this domain has seen a stronger focus on CE than e.g., process manufacturing. Thus, the different policy domains are associated with different, but related goals and rationales, and vary in the degree to which they call for circularity and sustainability transition.

8.5 Policy developments at different levels

8.5.1 Transnational level (EU)

As shown in Table 5, policy developments at three levels of governance are considered. At the EU level, the overarching water strategy (A Blueprint to Safeguard Europe's Water Resources) dates back to 2012. As noted above, CE is not a topic in the strategy, which was drawn before this concept entered the public agenda. The Water Framework Directive is relevant in that it regulates the collection, transport, treatment, and discharge of wastewater in Netherlands. Of more impact is the revised Urban Wastewater Directive, which is conducive, in that it will extend the obligation to set up urban wastewater collecting systems to all agglomerations with 1000 inhabitants or more, impose stricter standards for nutrient removal, require locally integrated urban wastewater management plans, and subject urban wastewater to quaternary treatment. As Member States will be required to improve and maintain access to sanitation for all, the revised Urban Wastewater Treatment Directive may also increase the amount of biomass potentially available from wastewater treatment. The Sewage Sludge Directive (SSD) has less influence on the uptake of biocomposites, as its main aim is to encourage the use of sewage sludge in agriculture while preventing negative impacts.

The Circular Economy Action Plan (2020) states an intent to support the circular bio-based sector through the Bioeconomy Action Plan, provision of a new SME strategy, and promoting various certification and tracking schemes. As noted above, it does not focus particularly on water or wastewater. However, the potential for CE in the building and construction sector is emphasized, and will be further addressed, e.g., through revised recycled content and material recovery targets, integrating life-cycle assessment in public procurement and the sustainable

[11] https://single-market-economy.ec.europa.eu/single-market/public-procurement_en

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finance framework, and a new Strategy for a Sustainable Built Environment. By October 2024 the latter is not yet in place, but the aim is that it will increase material efficiency and reduce climate impacts, while promoting circularity throughout the life cycle of buildings.[12] At the same time, the CEAP emphasizes that circularity shall be enhanced in a toxic-free environment, amongst other implying a shift towards safe-by-design chemicals, which could lead to increased documentation and monitoring requirements.

As noted above, the EU Waste Framework Directive (2008/98/EC) sets ambitious targets for recycling and recovery. It also defines the end-of-waste criteria that allow certain waste streams to be considered as "recovered materials" approved for use in new products. Moreover, it promotes the implementation of Extended Producer Responsibility (EPR) schemes. The Directive is under revision, and the current proposal for a revised version (COM(2023) 420 final) seeks to halve food waste at the retail and consumer level by 2030, and also extends the current EPR to the textiles sector. However, there is no mentioning of change in end-of-waste criteria. So far, recovered materials from wastewater treatment, such as cellulose or reeds from surface water management, do not have end-of-waste status, [13]and as noted above, this constitutes a barrier to upscaling.

The Bioeconomy strategy and action plan (2018) aims to strengthen and scale up the biobased sectors, amongst other through a €100 million circular bioeconomy thematic investment platform, development of standards, launching pilot actions for bioeconomy in urban, rural, and coastal areas, and facilitating new sustainable biorefineries. Also stressed is the need for improved environmental performance information.

The EU zero pollution vision for 2050 is for air, water, and soil pollution to be reduced to levels no longer harmful to health and natural ecosystems. The Zero pollution action plan contains targets for improving water quality, by reducing waste, plastic litter, and microplastics released into the environment, and reducing waste generation, e.g., reducing municipal waste by 50%. However, it is not specific for wastewater. The Industrial Emissions Directive (IED) is the main EU instrument regulating pollutant emissions from industrial installations. IED does not regulate wastewater treatment plants. However, its implementation can influence the wastewater that enters the plants, and in turn the resources that can be recovered. The IED is also relevant for the production at NPSP – the company must comply with the requirements, e.g. in terms of minimizing leaching.

The REACH Regulation (EC 1907/2006) makes industry responsible for the management of risks from chemicals and to provide safety information on produced substances, through registration in the central database of the European Chemicals Agency (ECHA). Materials that

[12]https://www.europarl.europa.eu/legislative-train/theme-a-european-green-deal/file-strategy-for-a-sustainable-built-environment .
[13] https://www.eureau.org/documents/diverse/6080-factsheet-on-products-from-waste-water-fibres-and-polymers/file

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are considered as waste under the EU Waste Framework Directive (2008/98/EC, WFD) are not considered as substances, mixtures or articles in the scope of REACH. However, the REACH Regulation applies to by-products or materials that have ceased to be waste. These are materials that meet the end-of-waste criteria set by the Waste Framework Directive. All forms of recovery are considered as processes under REACH when resulting in generation of one or several substances, or in a mixture or article that has ceased to be waste (IMPEL, 2023). This is not a problem in itself but involves a number of obligations and documentation requirements that can be challenging for small, emerging actors.

The Construction Products Regulation is under revision, as a follow-up of the European Green Deal. The proposal for a new regulation includes new rules on how to express the environmental, climate and safety performance of construction products, as well as the establishment of environmental, climate, functional and safety product requirements (European Parliament, 2022). Furthermore, EN 15804:2012+A2:2019 Sustainability of construction works, provides core product category rules (PCR) for Type III environmental declarations for any construction product and service. It sets the rules for life cycle assessment (LCA) and reporting of environmental and health information not covered by LCA, and the conditions under which construction products can be compared based on the information provided by environmental product declarations (EPDs). While in many ways positive, the increased documentation requirements add costs, and some of the environmental advantages of riverbank protection made of biocomposite material, compared to tropical hardwood or e.g. recycled plastics, are not visible within its scope.

The Public procurement directive (2014/24) specifies criteria including overall cost effectiveness, technical quality, environmental and social aspects, trading and delivery conditions, etc. Moreover, public authorities are encouraged to divide up large contracts, to facilitate the participation of small companies. Directive 2014/25 - Procurement by entities operating in the water, energy, transport, and postal services sectors is based on the same principles. Thus, there are no EU level regulations that specifically require emphasis on environmental criteria. However, the communication Public procurement for a better environment (COM(2008) 400 final) encourages specification of green public procurement targets and criteria. More recently, the EC has developed voluntary GPP criteria for several product groups, and the CEAP proposes use of minimum mandatory GPP criteria in sectoral legislation, as well as ecolabelling. So far, voluntary GPP criteria have not been defined for water management or riverbank protection panels.[14] Furthermore, the proposed Ecodesign for Sustainable Products Regulation (COM(2022) 142 final) will apply to all types of products, providing a range of environmental requirements, together with required use of Digital Product Passports. The Taxonomy Regulation ((EU) 2020/852) aims to give clarity on which economic activities are sustainable, but the consulted stakeholders did not see that it has any direct influence on the uptake of biocomposites presently.

[14] GPP Criteria and Requirements - European Commission (europa.eu)

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8.5.2 National level

The national strategy A Circular Economy in the Netherlands by 2050 (Government of the Netherlands, 2016) sets the ambition to use 50% less primary abiotic resources by 2030, and for the Dutch economy to be fully circular by 2050. The generic strategy has been followed by more specific transition agendas for five sectors: Plastics, Consumer goods, Manufacturing, Building & Construction, and Biomass and Food. The National Circular Economy Programme 2023-2030 provides a range of new measures, both in the areas of pricing (better inclusion of environmental damage into prices), standard-setting and stimulus (taxes, levies, and focusing on circularity in existing subsidy schemes) more generally, and in relation to the prioritised sectors. For the Circular materials plan, the national authorities aim to better distinguish between high-grade and low-grade recycling and will include this in the minimum standards.

The part on Building and Construction does not contain specific new measures. One of the challenges noted is that "public clients currently are making insufficient use of their position as a buyer to focus on circularity" (Dutch Ministry of Infrastructure and Water Management, 2023; p.79). The strategic focus points therefore include enhanced used of the Environmental Cost Indicator (ECI), a data strategy for the circular construction economy, obtaining insight into how circularity and spatial planning are interconnected, and contributing to a proper market in 2030 (aligning supply and demand) for high-value reuse and high-grade recycling. Interestingly, the focus points for Biomass and Food include the development of a new value chain for bio-based construction materials, to strengthen the demand and supply of raw material flows that will help realise the transition targets for agriculture. This is highly relevant for innovations of the kind considered here, where uncertainty regarding raw material availability can be a barrier. By the end of 2023, a National Approach to Biobased Building (NABB) was launched by the central government, to promote non-wood biobased materials, based on e.g., flax, fibre hemp, elephant grass or straw. NABB includes substantial financing as well as soft instruments to support market and value chain development. While fibres and polymers from water and wastewater treatment are not emphasized, the strategy includes the quite specific target that by 2030, > 30% of new riverbank protections shall be made of biocomposite or willow wood (Government of the Netherlands, 2023).

While water is less in focus in the national CE programme, the National Water Plan 2022-2027 includes "a competitive, sustainable and circular delta" as one of its main objectives (Government of the Netherlands, 2022). The plan stresses the need for integrated management, across water, energy, housing, and agriculture, and for the central government to work together with other tiers of government, civil society organisations, sectoral parties, drinking water companies and other businesses, and the general public. The plan is grounded in the Water Framework Directive and the Dutch Water Act, which is further elaborated in the Water Degree, the National Water Regulation, and provincial and regional water authority by-laws.

The Water Act (Dutch Ministry for Transport, Public Works and Water Management, 2010) provides the administrative-legal, organisational and financial instruments for implementation

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of the national water policy. The Regional Water Authorities (RWA) Act, defines the authority of the Regional Water Authorities and its limits, as presented in the preceding section. The individual authorities collaborate in the Dutch Association of Regional Water Authorities, to promote their interest at a national and international level, and develop joint visions and viewpoints (Havekes et al., 2017). They have also established other cooperative organizations, including STOWA (Stiching Toegepast Oderzoek Waterbeheer), as their joint foundation for applied research in water governance.

Together, the Dutch Water Authorities and STOWA, and a number of knowledge institutions have established a collaborative network organization called Energy and Resources Factory (ERMF).[15] ERMF aims to enhance CE in the form of recovery of biogas, phosphate and cellulose, as well as bioplastics, phosphate, and bio-ALE, and has been important to ensure continued research and development, as well as cross-stakeholder learning, innovation and management improvements (van Leewuen et al., 2018). Aquaminerals BV is a joint enterprise originally established by drinking companies, which now includes multiple regional water authorities, trading and working to develop value chains for various material streams generated in their treatment processes.

Another joint programme, 'Water in the Circular Economy' (WiCE), is an initiative of the Dutch and Flemish drinking water utilities and KWR Water Research. In WiCE, the water utilities and other actors in the sector conduct joint research aiming towards CE, climate adaptation and sustainable energy transition. Under WiCE, a CE strategy for the Dutch water sector has been developed. The "Point on the Circular Water 2050 horizon", as it is called, has multiple objectives that may be conducive for biocomposite materials based on raw material from water and wastewater treatment, e.g., a fully circular/renewable and safe use of feedstock; 100% reuse of components, feedstock and raw materials; a climate- and environment-positive water cycle; and a minimal footprint.

The Dutch Environmental Management Act (Government of the Netherlands, 2024)[16] lays out an integrated approach to environmental management. The main tools are environmental plans, quality criteria for emissions and discharges, environmental impact assessment, and an Act on general provisions for environmental law. Communal wastewater is subject to Chapter 10 of the Environmental Management Act, and the Policy Document on Water Management.[17] The Activities Decree (Ministry of Infrastructure and the Environment, 2024) lays out environmental regulations for companies, and applies amongst other to work on materials. The Decree sets goals for each environmental aspect, such as soil, waist, air, water, and safety, while the related Activities Regulations provide the means for compliance.

15 https://www.efgf.nl/english
16 Reference is a link to a brief summary of this Act, in English language.
17 https://rwsenvironment.eu/subjects/from-waste-resources/national-activities/national-waste/

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Recovered resources (such as cellulose) and other residues (such as reeds) fall under the National Waste Management Plan (Ministry of Infrastructure and the Environment, 2024), which specifies end-of-waste criteria in accordance with the EU Waste Framework Directive. Here, it should be noted that the Dutch authorities have been a frontrunner, e.g., due to efforts from the regional water authorities and Aquaminerals, struvite was recognized as a raw material in the Netherlands in 2021. More recently, a court ruling on the legality of utilizing recovered grass and reed clippings in new products concluded that – subject to conditions – such use of clippings concerns 'continued use' and not a 'removal' (i.e. waste). Users must verify that they meet the criteria of a traceable origin, and a supply chain dedicated to an application with sufficiently high resource efficiency (higher than waste). Moreover, the ruling is a case law, not valid for export, which means that cross-border activity remains restricted.[18] A new law for spatial development, the Environment and Planning Act, entered into force from 1. January 2024. It applies to all changes in the physical environment, and aims to provide a one-stop shop, simplifying permitting procedures and enabling more integrated land use planning. It is therefore relevant, e.g., for the sourcing of reeds, and the placement of riverbank protection panels (Ministry of the Interior and Kingdom Relations, 2021).

Building Decree 2012 provides requirements related to safety, health, usability, energy efficiency, and the environment, and includes material applications in relation to fire safety. It sets a limit value for environmental performance (environmental impact per m² of gross floor area per year, including climate change impact and depletion of raw materials). The Building Decree is expected to be amended soon, with new regulations to further stimulate circularity. The transition agenda Circular economy in the Dutch construction sector (Dutch Ministry of Infrastructure and the Environment, 2015), notes that other bio-based materials than wood will mostly be for design of targeted products. They should not hinder circular use of materials, nor compete with food production or nature conservation. A Netherlands Technical Agreement (NTA 8080-1:2015) specifying sustainability criteria for biomass and the entire chain, is recommended for formulation of circular design and assessment criteria. Thus, the transition agenda provides support, while also raising a word of caution as regards advanced, biobased materials.

Further on the use side, the National Plan on Sustainable Public Procurement for 2021-2025, aims to boost the application of SPP by government, partly overlapping with the government strategy 'Procuring with Impact', and Better Procurement, of the Dutch Ministry of Economic Affairs. To maximise impact, buyer groups have been set up in certain sectors, and international coordination is strongly encouraged. A centre of expertise and online SPP criteria, self-evaluation, and monitoring tools are also provided. Otherwise, the EU public procurement directive is implemented through the Public Procurement Act (Aanbestedingswet 2012), which applies to all public contracts.

18 https://circulairterreinbeheer.nl/rechtsoordeel-bermgras/

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8.5.3 Subnational level

As noted above, the Regional Water Authorities have a crucial role in the transition to a circular water economy in the Netherlands. As many of their strategies and instruments are developed and applied in collaboration, by the national association of Dutch Regional Water Authorities, they were presented in the previous section. In the region of the studied case, Amsterdam, Waternet manages the entire urban water cycle, including drinking water, water safety, surface water and wastewater transport, and treatment. This integrated approach, together with close ties with the municipality and Regional Water Authority (Amstel, Gooi en Vecht), is considered conducive for CE innovation (van Leeuwen et al., 2018). As an autonomous, non-profit organisation, The Regional Water Authority has carried out multiple CE projects, in cooperation with different actors, which enabled cost savings and operational efficiency (Van der Hoek et al. 2016).

In Amsterdam Circular Strategy 2023-2026, a central aim is to substitute fossil materials with biomaterials in buildings, and there are ambitions to become 100% circular in procurement by 2030. By 2026, the Regional Water Authority and the City of Amsterdam will define a roadmap for making Amsterdam's water management 50% circular by 2030, and 100% circular by 2050 (City of Amsterdam 2022, p. 40). The strategy includes use of ECI in public procurement, and from 2025, " true pricing".[19] This is a voluntary system, which is like an extended version of the CO2 price, where the market price of products is taken to include the negative impacts on people and planet, of production, transport, usage and waste, in order to create a level playing field for circular businesses. In terms of funding, €17.5 million are available for the current period. Development of local legislation to stimulate circularity, knowledge sharing with other regions and countries, and dedication of capacities to attract financing from central government and the EU are other key steps.

8.6 Consistency

In terms of consistency, there is alignment between many of the policy elements considered for the Dutch case. While the overarching water policy document for the EU, A Blueprint to safeguard Europe's water resources, is relatively old, the Water Framework Directive provides the key objectives for water management in the EU. The CEAP does not emphasize water management as such. However, the goals of eliminating waste and decoupling economic growth from resource use resonate well with the WFD objectives. The revised UWWTD is also synergetic, in that it aims to stimulate sustainable reuse and recycling. However, it does not provide any goals or instruments specifically targeting recovery of materials that may be used in biocomposite materials. The goals of the Drinking Water Directive are well aligned with the WFD and UWWTD, but do not influence the case of biocomposite production strongly. Likewise, the SSD does not apply to this case specifically.

[19] https://trueprice.org/

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Of more relevance is the EU's updated Bioeconomy Strategy (2018), which works hand in hand with the CEAP. With the aims to manage natural resources sustainably while reducing dependence on non-renewable, unsustainable resources, it is also a driver for innovative materials based on bioresources.

With the Waste Framework Directive, which is under revision, there are apparent synergies in the sense that waste reduction, EPR, and recycling for higher-value materials and products are emphasized, in line with the objectives in the CEAP and the UWWTD. However, there is also a certain tension regarding end-of-waste criteria. Lack of end-of-waste criteria for raw materials such as reed clippings and recovered cellulose is conflicting with the aim to make these resources available for new products.

When it comes to use in building and construction, we also find a level of alignment between objectives, as the proposal for a new Construction Products Regulation aims to promote sustainable, circular products, and the Public procurement for a better environment communication (COM(2008) 400 final) encourages green procurement. Likewise, the proposed Ecodesign for Sustainable Products Regulation (COM (2022) 142 final), and the Taxonomy Regulation may be conducive in a long-term perspective.

The national policy developments promoting CE in Netherlands have partly preceded those at EU level and are characterized by high ambitions. E.g., the National Water Plan 2022-2027 aims specifically for circularity, and provides multiple instruments related to water and wastewater management target this aim. Regarding waste management, we note that a national court ruling permits the use of reed clippings in biocomposite materials despite the current lack of end-of-waste criteria. However, this is not valid for trade across borders in the EU.

For the application of novel biocomposites in building and construction, the current policy frameworks are under revision, both at the EU and national level. These processes are aligned towards the same goals, of increased circularity and sustainability. When it comes to public procurement, there seems to be a stronger drive at national level, but no conflict as such. Although it is consistent with the policy objectives, the emphasis on LCA means that some of the environmental benefits of using advanced biocomposite materials (e.g., instead of tropical hardwood) fall out of sight. In the studied case, there is also a high degree of consistency with policies and instruments at the regional/local level. The Regional Water Authority and the City of Amsterdam both have high ambitions regarding circular water economy and have implemented various instruments towards this end. The key policy elements and relationships identified for the Dutch demonstration case are illustrated below (Figure

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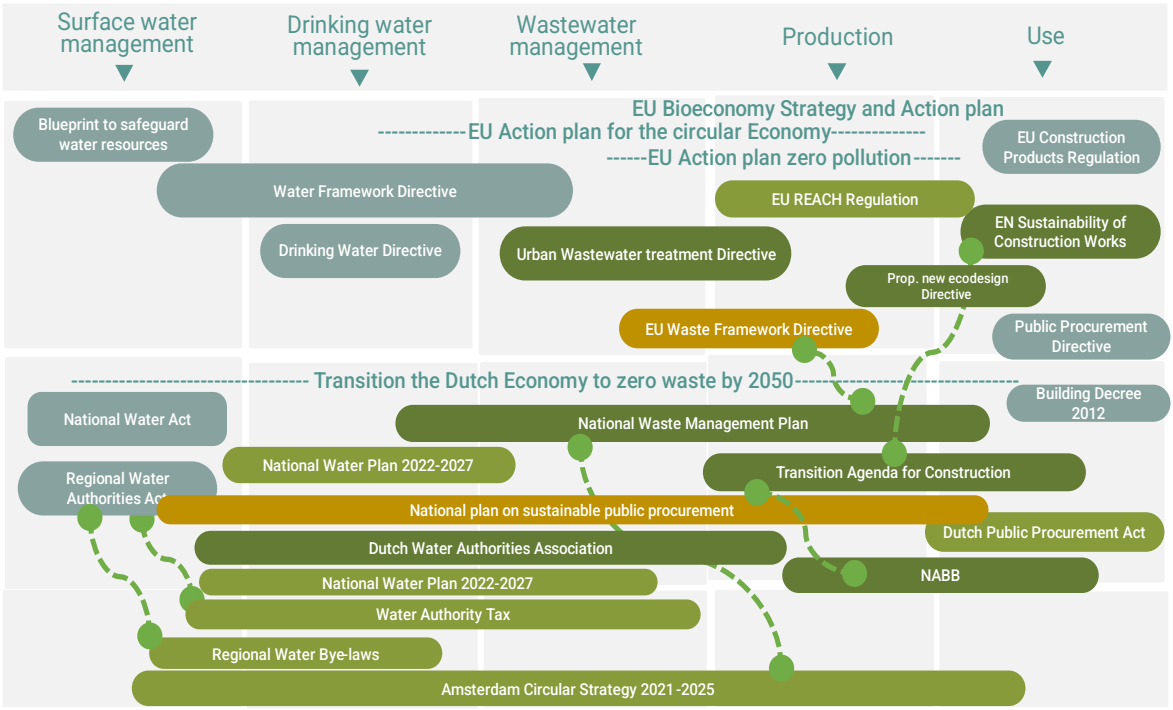


Figure 9: Key policy elements and relationships identified for the Dutch case.

In the figure, green boxes illustrate elements promoting the circular water solutions in question (the darker colour the more active promotion), the grey boxes signify elements that are relevant, without actively promoting, and orange boxes illustrate elements that are critical or central to the case. Green dotted lines connect element that strongly support each other, grey dotted lines illustrate tension, and red arrows show a conflicting relationship (not all illustrations are used in all figures).

8.7 Comprehensiveness

The assessment of the Dutch case does not reveal any major gaps or imbalances in terms of comprehensiveness. Considering surface water and drinking water management, we have noted that A Blueprint to safeguard Europe's water resources is old and that the WFD is scheduled for revision. However, the National Water Plan and CE strategy developed for the Dutch water sector set a clear direction.

When it comes to wastewater management, the revised UWWTD relates more explicitly to water reuse and nutrients than recovery of other materials, such as those in NPSP's biocomposites. Still, one of its aims is to enhance circularity, and in addition to specifying more ambitious treatment requirements, the revised UWWTD also emphasizes integrated planning and advanced quality monitoring and defines an Extended Producer Responsibility for wastewater. While the Water Act defines the national frameworks, the Dutch Association of Regional Water Authorities, in collaboration with relevant knowledge institutions, has

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established key instruments such as STOWA, the ERMF, and WiCE for knowledge development and experience sharing and Aquaminerals as a platform for market formation.

Considering production, the EU Bioeconomy action plan (2018), is accompanied by a range of economic instruments, in terms of R&D and innovation funding, as well as informational instruments. Moreover, the national CE strategy includes a focus point on the development of a value chain for new, bio-based construction materials, as one of several measures to enable transition in Dutch agriculture. However, the studied case illustrates that for some feedstocks, lack of end-of-waste criteria is a challenge. As noted above, ongoing and planned revision of policies and regulations for the building and construction sector are likely to increase the demand for circular products. Detailed criteria for bio-based products other than wood are yet to be developed. At the national and regional/local level, SPP is strongly promoted, both via regulative and soft instruments. Although extensive documentation requirements add costs, this is a development that may support the uptake of innovative bio-based materials.

8.8 Summary characterisation of policy mix

The policy mix considered for the case of biocomposite materials containing resources from water and wastewater management spans across several objectives, from protecting the environment and human health, via ensuring flood protection and sustainable agriculture, enabling circular bioeconomy, and reducing environmental footprints in the water and building and construction sectors.

Our assessment finds a relatively high degree of consistency in the policy mix. Although the overarching water policy of the EU needs to be updated, there are no conflicts between the policy developments for wastewater related to the CEAP and the current WFD, nor between these and the policies and instruments promoting the bioeconomy as part of the sustainability transition in the EU. While the relevant policies and regulations for building and constructions are under revision, the transition agenda for this sector at the national level has a strong focus on CE, where biocomposites are seen as relevant for design of targeted products. At the national and regional/local level, we also find a set of instruments promoting SPP, which may be conducive for the focal innovation in this case.

In terms of instrument types, a broad mix is observed. While the revised UWWTD tightens the treatment requirements and the IED brings more stringent rules, EPR and increased emphasis on “polluter pays” are included, and the Bioeconomy plan comes with a range of relevant economic and information instruments. There is an emphasis on development of criteria and advanced monitoring and assessment methods, which may contribute to the uptake of sustainable solutions, though these may also incur additional costs.

In terms of the distinction between creative and destructive policy elements (Kivimaa and Kern, 2016), there is a certain variance. In the assessed EU policies and instruments related to material recovery from water and wastewater management, we find more incremental and destructive, than creative elements. However, at the national level, there are more creative elements, such as the NWB Bank, and the multiple instruments provided by the Association of

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Regional Water Authorities, which both address knowledge development and diffusion (e.g., STOWA, EMRF, WiCE), aim to address market formation (e.g., Aquaminerals), and influence the direction of search (Point of the Circular Water 2050 Horizon). For the production and application of bio-based products there is more focus on creative measures both at the EU, national and regional/local level, e.g., in terms of establishing goals for the waste and construction sectors, resource mobilisation, and efforts to pull customer-side demand through SPP.

These findings resonate with statements from the actors in WIDER UPTAKE. Whereas a lack of end-of-waste criteria is a barrier to trade across borders in the EU, the above-mentioned court ruling solved this for the national level. Thus, there are no major policy or regulatory challenges at the present stage. It is noted, however, that SPP is not consistently applied, and the use of ECI in public tenders does not take into account CO2 sequestration or biodiversity aspects (which would add value in this case). As noted above, the strict documentation requirements for novel bio-based products, (e.g., LCA) can also be a challenge for small innovating firms.

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9 Policy mix for urban reuse (Czech Republic)

9.1 National context

In 2022, a total of 9.19 million people in the Czech Republic (from the total of about 10.5 million people) lived in houses connected to the sewage system. 453 million m^3 of wastewater (without precipitations) was released into the sewage systems. Of this amount, 97.7 % was treated (excluding rainwater). An average citizen consumed about 90 litres of water per day in 2022, which is moderate in European context.

The goal of the Czech Republic's water management policy is to foster the sustainable management of national water resources. As an EU country, the EU Water Framework Directive and other EU level policies provide the main regulatory framework conditions for the national water management policy. In addition, Czech Republic is part of several bilateral and multilateral agreements, treaties and conventions with other countries regarding, e.g., protection of Danube, Elbe and Oder rivers that cross national borders. At the national level, the Water Protection Department of the Ministry of the Environment is a key water management authority. Climate policy in the country is defined in the Climate Protection Policy of the Czech Republic from 2017. Key ministries for wastewater in the country are Ministry of Agriculture which has an overall role in water supply and sewage, and Ministry of Environment which e.g. regulates discharged wastewater quality. Moreover, counties and municipalities govern the sewage systems within their jurisdictions through specialized water authorities. Hydrologically, the Czech Republic is divided into five river basins, which are managed by state river basin companies. Three of these river basins have the status of international river basins (Elbe, Morava, Odra).

9.2 Solution in focus

The solution explored in the Czech case is water reuse for irrigation of urban greeneries in the city of Prague. The business case is thus to sell reclaimed water from wastewater treatment plant on Cisarsky island for irrigation in nearby public green areas. Green areas require irrigation during summer and are currently usually watered with drinking water from the water system, or with water from (e.g.) the river Vltava. A multi-year drought affecting the Czech Republic in 2014–2019 and in 2022–2023 put pressure on the water supply system and brought attention to the use of treated wastewater as an alternative water resource for irrigation. The drought period happened to coincide with the building of the new treatment plant on Cisarsky island. The location of the treatment plant nearby several urban green areas creates possibilities for several possible use sites for the reclaimed water with short transport distances (see Figure 10).

The new water line of the plant (finished in 2019) is a mechanical-biological wastewater treatment plant with chemical phosphorus precipitation, located underground on the island. Treated water is currently not reused but released to Vltava river. In an early phase of possible

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reuse, it has been planned that about 20 m³ of treated water would be transported for irrigation per day. The wastewater is disinfected either with only ultraviolet treatment, which already exists in the new treatment plant, or also with ultrafiltration treatment (membranes), which would require a new investment in the plant. It is expected that the possible additional ultrafiltration treatment would decrease the amounts of undesired substances in the reclaimed water, including micro-pollutants.

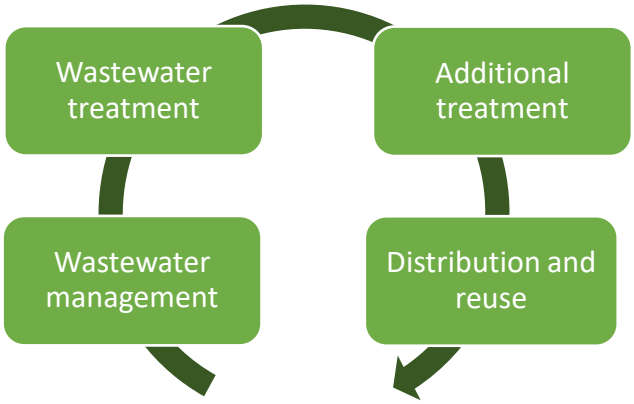


Figure 10: Circularity and action situations in focus in the Czech case

9.3 Policy documents considered

The documents that have been considered in the Czech case are listed in Table 8. Instruments are denoted by soft colours while policies and strategic documents are denoted by strong colours. The columns relate to the different action situations in question. The rows are ordered according to administrative level (transnational (EU), national, regional). Furthermore, they visualise which policy elements that only target one or few of the linked action situations, and which ones that are more generic, by the length of the coloured row and the number of columns it cuts across.

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Table 7: Overview of policy documents assessed for the Czech demonstration case.

	Wastewater management	Wastewater treatment	Additional treatment	Transport and use
Transnational	A Blueprint to safeguard Europe's water resources (COM(2012)673)			
	Water Framework Directive (WFD)			
	UWWTD			
			EU Regulation 2020/741 minimum requirements for water reuse	
			Guidelines to support the application of Regulation 2020/741	
	EU Industrial Emissions Directive (IED)			
	EU Circular Economy Action Plan			
	EU Zero Pollution Action Plan			
			EU Taxonomy for Sustainable Finance	
National		Water Act (254/2001 Coll.)		
		Surface water and wastewater regulation No. 401/2015 Coll.		
	Act No. 258/2000 Coll. Act on the Protection of Public Health and on Amendments to Certain Related Acts			
	Ministry of Health Bulletin 16/2023: Methodological recommendations for hygienic requirements for drinking water			
			Circular Czechia 2040 strategy	
		New Green Savings Programme		
	Environment for the Life funding (Program Prostředí pro život)			
Regional			Circular Prague 2030 strategy	
	Prague Innovation funding (Praha Inovacni)			

9.4 Policy developments at different levels

9.4.1 Transnational (EU) level

EU regulations and policies set the framework for the wastewater sector and CE. The Water Framework Directive sets the main water protection law in Europe and has stated an encouragement for increased water reuse. The Urban Wastewater Treatment Directive (UWWTD) regulates the management and treatment of wastewater from urban settlements and certain industrial sectors, while the Industrial Emissions Directive regulates wastewater discharges from industrial production processes. The UWWTD was revised in 2022 with an aim of, amongst other things, increasing circularity in the wastewater sector. The revised text of the UWWTD was adopted by the European Parliament in 2024.

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Several of the EU's environmental strategies and plans have also sought to promote circularity in the wastewater sector. EU Circular Economy Action Plan outlines an objective of promoting safe and cost-effective water reuse through non-regulatory means. The EU's Zero Pollution Action Plan seeks to remove nutrients and pollutants that can be harmful for health and ecosystems, including in the case of wastewater reuse. Meanwhile, the EU Taxonomy for Sustainable Finance recognizes activities related to building and refurbishing water reuse facilities that replaces natural water and drinking water use for other than human consumption use, qualifying them as environmentally sustainable in the Taxonomy.

EU Regulation 2020/741 defines minimum requirements for water quality, monitoring and risk management in order to ensure safe use of reclaimed water, aiming to ensure safety for use in agricultural irrigation. The regulation seeks to facilitate the implementation of water reuse where appropriate and cost-efficient by forming a framework which EU member states can use to foster water reuse.

9.4.2 National level

In the Czech Republic, the Water Act governs water and wastewater domains. Currently, the Water Act considers the use of treated wastewater (water reuse) in irrigation as discharge to groundwater, which requires a special permit from the authorities. This is considered to effectively disallow water reuse for irrigation in the country. The Water Act does state that in drought situations the authorities can ban the use of potable water for (e.g.) irrigation of grass areas. Surface water and wastewater regulation No. 401/2015 Col. sets the indicators and values of permissible pollution in surface water and wastewater and outlines the requirements for receiving a permit to discharge wastewater into surface waters and to sewers.

In the absence of specific national regulations and parameters for wastewater reuse, alternative regulations such as those surrounding "service (utility) water" may become pertinent. The category of service water is described and defined in two key documents: Act No. 258/2000 Coll. Act on the Protection of Public Health and on Amendments to Certain Related Acts and Ministry of Health Bulletin 16/2023: Methodological recommendations for hygienic requirements for drinking water.

Circular Czechia 2040 strategy, outlining plans and goals for increased circularity in Czech Republic, outlines an ambition to maximize water saving through, e.g., sustainable water management. The strategy also outlines that the state will support water reuse and recycling projects in industry, and support research into new technologies in wastewater management. In addition, the New Green Savings Programme offers financial support for rainwater retention and wastewater recovery, but only for family houses and apartment buildings, and not for centralized wastewater treatment facilities. Moreover, Ministry of Environment has the Environment for Life programme which offers funding for specific technologies, sometimes including wastewater-related technology, every 2–3 years.

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9.4.3 Regional/local level

In the city of Prague, Circular Prague 2030 Strategy mentions the use of rainwater, grey water and wastewater as one way of promoting CE. The city also uses part of its income to fund various project promoting innovations under the program Praha Inovacni, which can be relevant for promotion of wastewater. Finally, District 7 in the city has indicated that it could be interested in using wastewater in irrigation of its green areas, but by mid-March 2024, this issue had not yet been formally decided.

9.5 Mix of policy domains and rationales

Unsurprisingly, the policy landscape is more mature in the traditional wastewater management and treatment parts than in the production and use parts of the circular value chain. Wastewater management and treatment are regulated and guided with long-established EU and national regulations. Protection of human health and natural environment are key objectives for wastewater management and treatment legislation at the EU level. The key aim of the Water Framework Directive is to maintain and improve the aquatic environment within the EU, while the UWWTD aims for environmental protection and effective wastewater handling. The revised UWWTD proposed to promote reuse and resource recovery, in line with the EU Circular Economy Action Plan. A Blueprint to Safeguard Europe's Water Resources seeks to strengthen the general aims of protection of Europe's water resources, while including the promotion of reuse as one of its objectives. The EU's Industrial Emissions Directive is focused on prevention and control of pollution from industry, but also encourages circularity.

Regulations relating to additional treatment, distribution and reuse of reclaimed water are more incomplete. The EU's Regulation 2020/741 sets the key parameters for the required features of reused water to ensure safety. This regulation is motivated by responding to the challenges of water scarcity and deterioration of water quality by seeking to promote water reuse that limits extraction of water from primary water bodies. The regulation establishes the necessary requirements for using reclaimed water for agricultural irrigation and other irrigation purposes, while maintaining safety in regard to human and animal health and environmental protection. By early 2024, EU Regulation 2020/741 had not yet been translated into national policy in the Czech Republic. EU Circular Economy Action Plan aims for transformational change towards more sustainable and circular production and consumption patterns. EU Zero Pollution Action Plan seeks to reduce the pollution to non-harmful levels in order to protect human health and environment. Importantly, EU's Taxonomy for Sustainable Finance recognizes the production segment of wastewater reuse value chain, thus expanding opportunities for financing of building refurbishing and construction of facilities that allow wastewater reuse. The Taxonomy does not apply to the previous steps of wastewater management or treatment.

The general picture is the same in the Czech Republic. The Water Act, with its scope on wastewater management and treatment, has the aim of protecting surface water and groundwater bodies, provide conditions for economic use of water resources and their

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conservation, avoid water scarcity, and preserve and improve water quality. The key rationale is thus to handle wastewater in a way that protects human health and environment. In other words, it is not within the logic of the regulation to consider broader sustainability or other societal challenges beyond the core considerations of wastewater management and treatment, at least under normal conditions (i.e., when there is no acute drought).

Circular Czechia 2040 has a key goal of achieving "less waste and more value" for the country, promoting CE to create environmental, economic and social benefits. Similarly, the Circular Prague 2030 strategy aims to reduce the environmental and carbon footprint of the city. Both strategies, among several other topics, highlight the opportunities related to water reuse, and are thus focused on the production and use segments of the value chain. Indeed, the ambition of promoting water reuse through these strategies is motivated by and anchored in achieving broader sustainability benefits. Funding available for projects is challenge-led in rationale. New Green Savings Programme aims to reduce energy and water consumption, improve quality of living, reduce greenhouse gases, and so on, while Environment for the Life funding aims for improving health and environmental protection, but also to promote sustainable use of resources and improving life quality. Their scope is thus on going beyond traditional wastewater management and handling, seeking to promote segments such as wastewater reuse.

9.6 Consistency

In terms of policy mixes, we can identify two key themes in the policies having an influence on the water reuse in Czech Republic. First, policies and instruments which emphasise wastewater management and treatment. For example, the goals of EU policies such as WFD, UWWTD and EU Zero Pollution Action Plan have been environmental protection, human health, and effective handling of wastewater. The national level policies, especially the Water Act, are also aligned with such goals. These policies and instruments are well-established and mature in terms of regulatory development, and supportive of typical extract-use-discharge water management. This group of policies is well-developed and internally consistent between the EU and national levels.

Second, we can identify policies that focus on water reuse (and circularity more generally). For example, EU Regulation 2020/741 and EU's Circular Economy Action Plan address sustainability challenges such as water and resource scarcity, while also ensuring safety in terms of human health and environment. The revised UWWTD introduces goals of reduced pollution, energy use and greenhouse gas emissions, but also discusses circular economy principles. The European Commission's A blueprint to safeguard Europe's water resources emphasises topics relevant for both themes, such as land use, pollution reduction, efficiency and efficiency of water resources, and lists water reuse as a means to achieve such goals.

In the Czech Republic, the circular economy is emphasised in the Circular Czechia 2040 strategy as well as the city of Prague's own Circular Economy strategy. As for instruments, generic economic-type instruments are implemented at the national and regional levels. The policies

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and instruments focussing on water reuse, or the circular economy more generally are less developed than those relating to long-established principles in water management and treatment. Simultaneously, ambitions to promote a circular economy are consistent across the EU and national levels, though the EU has come further than the Czech Republic in terms of supporting its strategies with relevant instruments.

As shown in Figure 11 , which illustrates some of the key policy elements and relationships here discussed, the key inconsistency in the Czech policy mix relates to EU Regulation 2020/741, which defines key parameters for water reuse to ensure human health and environmental protection. National authorities in the Czech Republic do not intend to implement the regulation. Rather, water reuse is regulated through Act No. 258/2000 Coll and Ministry of Health Bulletin 16/2023, which imposes strict rules on the reuse of reclaimed water. Hence, the regulation-type instrument at the EU level stands in tension with the same type of instruments at the national level. While this barrier may be addressed by re-categorising reclaimed water as utility water, it points to an inconsistency between the EU and national levels.

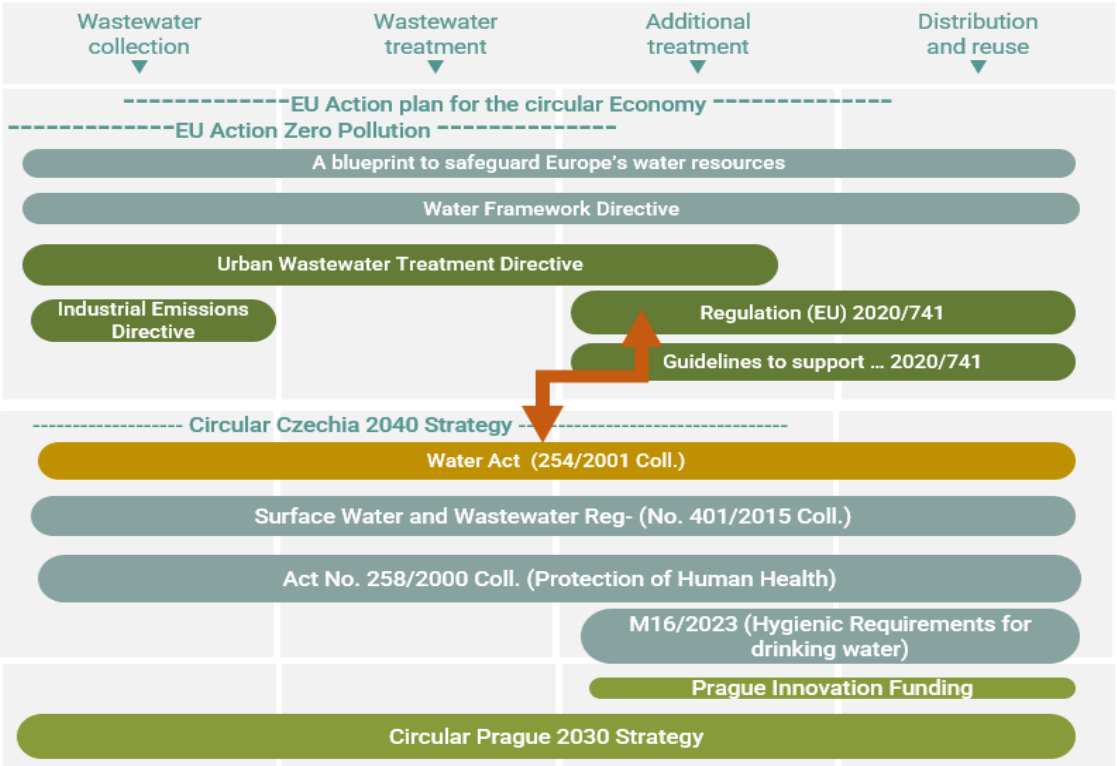


Figure 11: Key policy elements and relationships identified for the Czech case.

In the figure, green boxes illustrate elements promoting the circular water solutions in question (the darker colour, the more active promotion), the grey boxes signify elements that are relevant, without actively promoting, and orange boxes illustrate elements that are critical or central to the case. Green dotted lines connect element that strongly support each other, grey

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dotted lines illustrate tension, and red arrows show a conflicting relationship (not all illustrations are used in all figures).

9.7 Comprehensiveness

Reuse of wastewater in Czech Republic is affected by the general regulations and other policies governing wastewater management and treatment at both EU and national level. However, water reuse has traditionally not been a key objective of these policies in either level, even though at EU-level, some of the key regulations now include circularity goals. Wastewater reuse is however not an objective or a topic in national wastewater regulations under regular conditions (e.g., when there is no severe droughts). Moreover, the Czech Ministry of Agriculture has indicated that it does not intend to apply the EU Regulation 2020/741 into national legislation (Wanner, 2023; Wanner et al., 2023).

Instead, the national policies that have mentioned wastewater reuse are generic CE policies which lack the necessary specificity and leverage to effectively promote water reuse in Czech Republic. Moreover, there is no strategy document by national or Prague authorities that promotes water reuse specifically. Relevant national and city-levels strategies are generic in nature, mentioning water reuse as one form of solution to increase circularity in the country but do not address its specific bottlenecks and hindrances, such as regulatory barriers. Moreover, there is no concrete goal for water reuse in Czech Republic, for example, how much water reuse should be in place by a specific year. In other words, the key strategies lack dedicated attention to water reuse, and there are no instruments in place to promote water reuse.

While wastewater reuse has only partial support from policies in general, this is particularly the case for specific regulatory and economic instruments with an aim to foster water reuse at the national level. At the EU level, the recent EU Taxonomy lists water reuse as a sustainable activity, which may have an effect in future. However, there are no concrete goals for water reuse in EU. Though the EU Regulation 2020/741 provides key parameters for safe water reuse, this regulation has not been implemented in the Czech Republic. There is thus no concrete legislation that supports water reuse in the Czech Republic, and that would follow up on developments at the EU level. In fact, researchers in WIDER UPTAKE have pointed out that the use of reclaimed water for irrigation of non-agricultural green areas (such as public parks) is not explicitly banned by the current regulation. This *lack* of legislation may allow for water reuse following the legal principle of *praeter legem*, meaning if something is not explicitly regulated, it is *not illegal* (Wanner, 2023). This may change if a regulation that bans the action is introduced at a later period. In addition, application of "service (utility) water" regulation, governed by Ministry of Health, may provide an alternative regulatory framework for water reuse in Czech Republic (see Figure 11).

The policy landscape at national level is still dominated by policies with a focus on the conventional approach to wastewater management, while EU regulations are moving towards

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including circular economy principles in key regulations. In practice, there is no concrete pressure to promote water reuse in Czech Republic, which aligns with the national government's seeming disinterest in water reuse.

9.8 Summary characterisation of policy mix

Overall, the current policy mix in the Czech Republic is not conducive for a transition towards increased water reuse for irrigation. There are no specific instruments which incentivise water reuse. National and city-level CE plans list water reuse as a way to increase circularity, but there are no specific goals or instruments relating to water reuse. The available economic-type instruments that might promote reuse are generic instruments. According to a report by the OECD (2021), the above-mentioned issues apply more generally to promoting CE in the Czech Republic. There are few systematic and comprehensive policies cutting across sectors, and limited guidance with concrete measures to implement CE goals, for example in terms of economic instruments (funding, R&D) with specific links to CE. The policies promoting CE in the wastewater sector thus lack credibility in Czech Republic.

The delay in implementing EU regulation 2020/741 in the Czech Republic hinders the building of a clear regulatory framework for water reuse. Policymaking is thus not actively seeking to realize water reuse in the country at a regulatory level. This situation means that also other policy instruments lack the power to realize water reuse. Establishing water reuse as a practice will first require changes to the regulation-type instruments. Removing barriers or re-categorising reclaimed water as utility water would be a key steppingstone for promoting water reuse. However, as the Czech national authorities currently express little interest in water reuse (except during droughts), the dismantling or circumvention of this barrier will only be a first step. The low comprehensiveness of the policy mix for water reuse suggests that further instruments must be established in order to *promote* water reuse, as opposed to simply allowing it.

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10 Policy mix influence on transition pathways

10.1 Linking policy mixes and pathways

As noted above, policy developments and changes in policies are closely intertwined with other changes in socio-technical systems (Lindberg et al., 2019). As we have seen, policies can both be creative, i.e. support sustainable innovations, and destructive, in the sense of constraining existing technologies and practices that are deemed unsustainable. The resulting developments in the socio-technical system will in turn lead to further changes in the policy mix. These interactions can lend shape to different pathways, or patterns of change characterising the development and uptake of the focal solution, and their contribution to transition of the wider system (Geels and Schot, 2007). As shown in chapter 1, four key dimensions or conditions can be seen to influence the feasibility of transition pathways. These include political feasibility, as well as the maturity of the solutions, their level of integration with existing systems and infrastructure, and their social acceptability (Turnheim and Nykvist, 2019).

In this chapter, we therefore discuss how the policy mix characteristics found in the previous chapters are seen to influence the further uptake of the solutions demonstrated in WIDER UPTAKE. Our focus is on how the assessed policy mixes influence the feasibility of the solutions, and how they interact with the other three conditions shaping their development as pathways towards a circular water economy. Following one sub-chapter dedicated to each solution, we provide a more general discussion of the opportunities and challenges identified and outline a set of overarching implications.

10.2 Policies and pathway for water reuse in Sicily

The policy mix in the Sicilian case is consistent across levels. Policies and instruments at the EU, national, and regional levels which were previously out of sync have recently been harmonised. Water reuse is promoted within certain parameters. However, it is worth noting that generally, the policies and instruments targeting reuse specify the preconditions for water reuse to happen, rather than actively promoting reuse as a practice. This is further reflected in the policy mix's level of comprehensiveness. Though the policy mix is consistent, it is not especially comprehensive. There is a strong emphasis on water treatment over reuse, which is natural given that extract-use-discharge approaches to water management have long been prevalent.

Instruments regulating water reuse have come some way both in the EU and at the national level, but there are few specific instruments promoting reuse as a practice (e.g., in terms of organisation, funding, operations and maintenance). The policies and instruments focussing on distribution of reclaimed water are mostly generic, and, with one exception, they cover distribution and use as part of broader policies focussing on water and/or circular economy. The sole exception is Corleone Municipality's temporary funding for distributing reclaimed water, which is highly specific, both in terms of what it funds and the duration of the support. The situation is similar for the use of reclaimed water. The EU has provided significant funding

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for refurbishing the wastewater treatment plants and pipe network in Sicily, and there are specific regulatory instruments that establish the minimum requirements for reuse. However, there are no policies or instruments that specifically *promote* water reuse practices. Rather, water reuse is usually discussed as one circular economy solution among many. The same applies to funding schemes: water reuse may be covered by many of these (e.g., EU funding schemes for agriculture or rural development), but their orientation is generic.

We now combine the results from the policy mix analysis with Turnheim & Nykvist's (2019) four dimensions, as to indicate extra-political conditions currently influencing the prospect of increased water reuse in Sicily and subsequently indicate elements in a transition pathway towards increased water reuse. In terms of *maturity*, the technology for treating water to the appropriate quality is readily available and, in some cases, already installed at Sicily's wastewater treatment plants. Though these are currently facing some technical and/or organisational problems, the uptake of water reuse does not hinge upon technology development. Rather, the challenge lies in establishing a viable business model for reclaimed water. For users to choose reclaimed water over fresh groundwater, the reclaimed water must be cheaper.

This also relates to *system integration*. The lack of dedicated infrastructure for reclaimed water increases transport costs, and while Corleone Municipality currently intends to provide temporary funding for transport costs, cover distribution costs through temporary funding, this does not secure a permanent supply of reclaimed water. In terms of *societal acceptability*, the consistency of the analysed policy mix indicates that water reuse is relatively uncontroversial in Sicily. This has been further substantiated through exchanges between Sicilian farmers and WIDER UPTAKE researchers. Furthermore, the increasing prevalence of droughts in Southern Italy has increased the acceptability of water reuse. This has also translated into *political feasibility*: national and regional authorities have responded to the droughts with policies that explicitly identify water reuse as one measure for countering droughts. Simultaneously, the kind of measures that are currently pursued at the national and regional levels do not necessarily translate into actual reuse. Other instruments are needed, and as always, the political feasibility will differ depending on the kind of policies and instruments.

Based on the preceding discussion, we suggest that in order to promote and foster water reuse, the policy mix must be made more comprehensive. A key challenge regarding water reuse in Sicily is establishing reclaimed water as a cheap(er) alternative to fresh groundwater. This suggests that systemic-type economic instruments – for example, subsidies and/or a restructuring of the national and/or regional water tariffs – may be necessary for promoting, rather than simply facilitating water reuse. In terms of viable business models, system integration is a further challenge, though there are some apparent – but not straightforward – strategies for reducing transport costs (e.g., rehabilitating the currently decommissioned infrastructure or, more generally, identifying and pursuing use-cases near treatment plants). Due to variations in water governance throughout Europe, there is a gap regarding systemic instruments of the economic and regulatory types is best addressed at the national and regional

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levels, though the EU may provide an additional push by developing more specific economic instruments for promoting water reuse.

In Sicily, the key challenges for promoting water reuse relate to infrastructure provision and customer-side demand, challenges which also impact the maturation of business models for reclaimed water. On this basis, we present the following recommendations:

- Focus on creative instruments. Except for the temporary municipal funding from Corleone, creative instruments are currently absent from the policy mix. The primary focus should be instruments for establishing the practice of reusing water.
- Costs may be redistributed with systemic-type, economic-purpose instruments (for example, water tariff reforms) to make reclaimed water less expensive than groundwater. This may fruitfully be combined with the identification of use-cases near wastewater treatment plants, as to avoid the costs associated with developing new infrastructures.

10.3 Policies and pathway for reuse in the Czech Republic

In the Czech Republic, the policy mix for wastewater management and treatment is consistent. The overarching principles for water treatment are well-established, and the policies and instruments ensuring proper water treatment are consistent between the EU, national, and regional levels. However, the results from the policy mix analysis point toward a key inconsistency regarding water reuse.

The EU has developed policies *and* instruments for promoting circular economy principles, including generic policies and instruments (e.g., the Circular Economy Action Plans) and policies and instrument specifically promoting water reuse (e.g., Regulation 2020/741). To some extent, circular economy objectives have entered national and regional governance in the Czech Republic, but usually in the form of policies – mainly generic strategies (e.g., Circular Czechia 2040, Circular Prague 2030) – rather than specific instruments for promoting water reuse. Furthermore, the Czech Republic has opted for not implementing EU's Regulation 2020/741 at the national level, thus introducing an inconsistency between the EU and national levels. Reuse is further complicated by the current categorisation of reclaimed water. However, researchers in WIDER UPTAKE believe this regulatory barrier may be circumvented by re-categorising reclaimed water as utility water.

Regarding comprehensiveness, the policy mix at the EU level is comprehensive in terms of the number of policies that discuss water reuse. In terms of instruments, there are a series of generic economic-type technology push instruments. The instruments specific to water treatment and reuse are predominantly regulations aimed at wastewater treatment plants. In the Czech Republic, the policy mix for water reuse at the national and regional levels is characterised by low comprehensiveness. Whereas the circular economy strategies at the

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national and regional levels discuss water reuse as a circular economy solution among many, there are no dedicated policies or instruments for water reuse. Further, there is the challenge regarding the categorisation of reclaimed water, which – in combination with the previously mentioned delay in implementing Regulation 2020/741 at the national level, means that there currently is a regulatory barrier for water reuse. In addition to this specific barrier, specific and systemic instruments – especially of the regulatory (market design) and economic (infrastructure) types – are notably absent at the national and regional levels.

Based on Turnheim & Nykvist's (2019) four dimensions, we now provide a brief assessment of the conditions that influence the uptake of water reuse in the Czech Republic, before combining these with insights from the policy mix analysis to indicate a possible transition pathway. In terms of *maturity*, the technology necessary for bringing treated wastewater within the required parameters is mature (and already under implementation at some plants). However, establishing a viable business model for reclaimed water relies upon the establishment of dedicated infrastructure (such as a pipeline) for reclaimed water and/or finding use-cases near treatment plants. This also indicates that *system integration* is a currently a challenge – but also a necessary requirement – for use-cases far away from a treatment plant.

In terms of *societal acceptability*, two aspects have become clear during WIDER UPTAKE. First, health authorities harbour some concerns regarding the safety of using reclaimed water for urban irrigation. These concerns persisted even when presented with analyses of the quality of the reclaimed water, showing that treated wastewater had low quantities of harmful substances. Second, the prospect of using reclaimed water for irrigation rose to attention during drought periods. As climate change is expected to exacerbate droughts, one might expect the societal acceptability of water reuse to increase in the years to come. In terms of *political feasibility*, the main challenge appears to be the concern regarding possible health hazards arising from water reuse. This challenge may be possible to circumvent through a re-classification of reclaimed water and/or through development of more knowledge regarding the safety of water reuse. The acceptability of using reclaimed water may also increase if droughts periods intensify.

The policy mix shows that there is considerable consistency in the policy mix for water management and treatment, with regulation-type instruments focussing on establishing demand-pull for specific solutions at treatment plants being predominant. The comprehensiveness appears to be sufficient for promoting proper management and treatment. For water reuse, the inconsistency between the EU and national levels remains a challenge. Beyond this, comprehensiveness in the part of the policy mix that applies to reuse is low: water reuse is mainly discussed in policies, with no dedicated instruments in place.

The key challenge for water reuse in the Czech Republic is the national authorities' relative disinterest in the practice. Barring drought periods, water reuse is considered irrelevant. Hence, the primary barrier for promoting water reuse lies in political (in)feasibility. However, the

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potentially increasing prevalence of drought periods suggest that having facilities for producing reclaimed water may be useful. With a view to future implementation, we offer the following recommendations for irrigation using reclaimed water:

- Prepare regulation-type instruments that establish quality requirements and acceptable areas of application for reclaimed water during drought periods. Such instruments may be supplemented with economic-type demand-purpose instruments to ensure that a certain number of wastewater treatment plants may produce a certain volume of reclaimed water of a certain quality during such emergencies.
- As transport costs are a key challenge for establishing a viable business case for reclaimed water, infrastructure for such water may be established when developing new areas, especially areas close to wastewater treatment plants. Elsewhere, public financial support schemes for water reuse infrastructure investments and operations may help to overcome business model challenges.
- Information-type instruments may raise awareness regarding the merits of reclaimed water, as well as reduce public scepticism toward such water. Moreover, further research and knowledge development regarding the safety of water reuse in Czech context may provide improved basis for fact-based policymaking.
- Public ownership in water utilities allows for setting goals for water reuse per year and prioritising reclaimed water in public procurement processes, both of which may help establish customer-demand and a supply of reclaimed water.

An alternative pathway may be water reuse for industrial applications rather than urban and/or agricultural irrigation. The situation is particularly favourable where an industrial plant operates its own wastewater treatment plant or where it jointly treats its wastewater with urban wastewater at its own plant. In this case, the management of the treated wastewater is not covered by the Water Act and the wastewater can be used in accordance with general safety and hygiene regulations. In the Czech Republic, there are already several successful water reuse project in the glass, rubber and automotive industries, and pulp and paper mills are also switching to water reuse.

10.4 Policies and pathways for nutrient recovery in Norway

The characteristics of the policy mix assessed for the two cases of nutrient recovery in Norway are seen to influence the feasibility of this transition pathway in multiple ways.

First, in terms of political feasibility, the analysis did not reveal any major policy or regulatory barriers for the demonstrated solutions at the present stage. Following the approval for use of struvite in organic/ecological farming in the EU, struvite has been approved for use as fertiliser in pure form in ecological farming in Norway, where it also may be used as ingredient in combination with other feedstocks, in organic soil improvement and fertilisers. Thus, the struvite from Hias will enter the market via NORGRO, which is a major supplier to Norwegian horticulture, gardening and agriculture. The soil improvement and organic fertilisers provided

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by Sirkula in Innlandet are already in the market, permitted under the present and revised regulations for production, trade and use of fertilisers of organic origin. Considering the high quality of the hygienised products and the soil conditions in the region, the local authorities have further granted an exemption, whereby the products can be used for vegetable production.

Minorga, the mineral-organic fertiliser produced at IVAR, has also been permitted in Norway, but is currently optimised for the market in Vietnam, where there is a better fit with local soil conditions and equipment. The export is allowed and profitable for Terramarine under the current market conditions. The EU FPR does pose a barrier to trade of Minorga or products such as Sirkula's with EU Member States, however this is not a target market.

The recent revision of the UWWTD implies that more wastewater treatment plants in Norway must establish secondary and tertiary treatment in the coming years, and the revised IED will influence the volume and quality of inflow to the plants in some areas. This constitutes a window of opportunity for nutrient recovery and other circular water solutions. Moreover, the proposed new regulations for production, trade and use of organic fertilizers in Norway can be conducive, as the time window and conditions for direct use of sewage sludge as soil improvement on Norwegian farms will be narrowed, whereas the same rules will apply to all sludge, from aquaculture, poultry and livestock, as well as sewage. This may make it more attractive to produce fertilizers combining feedstock from different sources, in areas where such opportunities exist. On the other hand, many utilities and wastewater treatment plants are small and distant from each other and from relevant other waste streams. While the implementation of the revised Waste Framework Directive in Norway will increase the availability of other types of biomass, it is likely that there will be increasing competition for these resources, e.g., for sustainable biofuels and high-value, low volume products.

Although the struvite harvesting facility at Hias is the first of its kind in Norway, the solutions demonstrated in the Norwegian cases are relatively mature. Both Hias and IVAR have benefitted from public research and development funding in earlier phases, e.g., through the Research Council of Norway and Innovation Norway. As noted above, there is a modest grant scheme available to reduce the risk for enterprises who embark on full scale piloting and testing of novel, environment friendly technologies. It is also possible to apply for green loans from the Municipal Bank.

However, there are no dedicated support schemes for nutrient recovery, as the principle of full cost recovery works as the key financing mechanism for water utilities. While some consider this a sound situation, others hold that the reliance on full cost recovery may limit upscaling of circular water solutions, as municipalities are reluctant to increase the water fees too much. Furthermore, the rules laid down in the Act on Municipal Water and Wastewater imply that there are no financial incentives for innovation: the utilities are not supposed to generate profit, and if an activity generates income beyond a certain limit, it must be singled out as a separate entity.

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Another factor to consider, as regards maturity, is the noted lack of end-of-waste criteria for the utilised biomass, and the lack of standards and guidelines for production of organic fertilisers and soil improvers. As noted above, the announced Integrated Nutrient Management Plan at EU level will be crucial to stimulate acceptance and market uptake of recovered nutrients, and the turnover requirement for P considered by Norwegian authorities may pull in the same direction.

In terms of system integration, we have already noted that the revised UWWTD provides a window of opportunity, as implementation of more advanced treatment solutions also may increase the scope for circularity, e.g., by selecting biological, rather than chemical treatment, providing space for additional process steps, and so on. The emphasis on integrated management and planning is also crucial, to identify and implement the most sustainable solutions for different contexts and localities. Nutrient recovery does not require infrastructure in form of pipework for distribution. However, the bulky nature of the components, and the need to combine different types of feedstocks for co-composted soil improvers, means that geographical proximity is a crucial factor for this kind of products. Following the Act on municipal water and wastewater utilities, most Norwegian utilities are owned by individual municipalities. Some regions have established inter-municipal companies, but most utilities are small, with limited capabilities. This can be a hindrance to integrated planning and coordination across regions, hence there may be reason to reassess the current sector structure in some cases.

As shown above (chapter 7), the policy mix assessed for the Norwegian case, includes more regulatory than financial or information instruments. At the same time, there is limited awareness of the values in water in Norway, both among business actors and the general public. According to the local stakeholders, there is a widespread attitude that sewage sludge should be given away for free. A survey carried out by Terramarine showed that farmers willingness to pay for mineral-organic fertilisers was limited, unless cheaper than the conventional alternatives. This suggests that more emphasis on informational instruments, illuminating the benefits of organic products, could have a positive effect. On the other hand, the widespread practice of sludge-spreading means that people generally are used to the idea of fertilising with sewage-based products, and there is less scepticism than observed e.g., for reuse of treated wastewater.

The planned updating of the limits for heavy metals and pollutants under the SSD may have an impact on future acceptance, as the variable practices and attitudes among EU Member States is creating uncertainty about the future acceptance and frame conditions for products based on sewage sludge, also in Norway. Furthermore, the national policy to limit harvesting of peat may be conducive, as so-called issue-linking can be positive for the uptake of radical innovations (Rouillard et al., 2019). On the other hand, there are no financial incentives for users. While GPP and the subsidies for farmers could be used to promote organic fertilisers and soil improvement products, this is not done today.

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In pathway terms, these findings suggest that the development for nutrient recovery in Norway is at a critical branching point. The revised UWWTD constitutes a window of opportunity, but how it will be implemented at the national level is a matter of debate. The proposals for new regulations for the production, trade and use of organic fertilisers in Norway will complicate the practice of direct spreading of sewage sludge. However, given the variation in volumes and opportunities for utilising recovered nutrients in different regions, other solutions (e.g., pyrolysis) are also seen as relevant. Upscaling will at any rate take time, since substantial investments are required to install the biological treatment processes the nutrient recovery solutions are based on. However, the updated national targets under the Protocol on Water and Health (UNECE/WHO, 1999) signal an ambition to accelerate the upgrading of wastewater treatment, which also can accelerate the development for nutrient recovery.

Among the policy implications of these developments, we would like to highlight the following:

- The coming implementation of the revised UWWTD constitutes a window of opportunity, where the wider environmental, economic, and social benefits of biological treatment and nutrient recovery must be carefully considered vis-à-vis narrow cost-benefit assessment of how to meet the specified requirements.
- The Integrated Nutrient Management Plan at EU level may provide further direction for nutrient recovery and speed up the development of instruments for market pull, which will be important also in the Norwegian context.
- Likewise, the National plan on phosphorus, proposed in Norway's strategy for developing a green, circular economy (Norwegian Government, 2021) should be put in place, as many utilities now are assessing what solutions to implement in order to meet the requirements of the revised UWWTD.
- Use of creative instruments is important also in this case, to build capacity and stimulate demand, e.g., by incentivising users, taking steps to deploy demand-pull measures such GPP, and considering a turnover requirement for P, as mentioned in the national CE strategy, and proposed by EurEau.

10.5 Policies and pathway for biocomposite in the Netherlands

In the chapter presenting the policy analysis related to production of biocomposites for riverbank protection in the Netherlands, we found a consistent and comprehensive policy mix. In this section we further discuss how the identified characteristics influence the feasibility of this solution as a pathway in the transition towards a circular water economy in the Netherlands.

Considering political feasibility, it can be noted that prevailing policies and regulations are not associated with any major barriers in this case. While the EU's overarching water policy, as stated in the Blueprint, has been focused on environment and health, with less attention paid to circularity, there are clearly stated CE ambitions for the water and wastewater sector in the

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Netherlands, with a drive to innovate, recover and reuse various materials. This reflects the overarching CE strategy in the country. Here, building and construction is a prioritised sector, and developing a market for high-value reuse and high-grade recycling by 2030, and a novel value chain for bio-based construction materials are among the focus areas.

SPP is applied as a key instrument to stimulate demand, and ECI has recently been introduced to strengthen the pull. While lack of end-of-life criteria hinders trading the biocomposites across borders within the EU, a case law defines the use of grass and reed clippings as sustainable 'continued use' at national level. The potential for upscaling is good, considering the stated ambitions in both sectors, and the synergy with the need for alternative value chains in in Dutch agriculture. However, the Transition agenda for building and construction sees non-wood mainly for design of targeted products.

Regarding the maturity of solutions, the biocomposites are already in the market for some applications, e.g., interior design, façade claddings, signposts. The riverbank protection panels are also near market ready, but high costs are still a challenge.

While environmental costing and assessment methods such as LCA can provide competitive advantage, the extensive documentation requirements for novel bio-based products are a challenge, as they add considerable costs for small actors dealing with small volumes. Moreover, some of the environmental benefits associated with the biocomposites as a replacement for tropical hardwood fall outside the scope of e.g., LCA.

To accelerate market development, the National Approach to Biobased Construction (NABB) was launched towards the end of 2023, providing substantial funds, and a target of (> 30% of new bank protections of biocomposite or willow wood) by 2030.

Coming to system integration and infrastructure, the production of biocomposite materials does not require any huge infrastructure investments. However, there is considerable uncertainty regarding the availability of biomass, e.g. the harvesting of reed clippings is strictly regulated, in terms of when and where it can be done to not disturb ecosystems. As to cellulose, only few wastewater treatment plants have the sieves that are needed for collection and recovery. Thus, the case underscores the need to consider how circularity and spatial planning are interconnected, as noted in the Dutch CE strategy (Government of the Netherlands, 2016), and the need for integrated water management, stressed in the revised UWWTD.

In terms of societal acceptability, biobased construction materials are not controversial in the Netherlands, unless they conflict with sustainable food production or could have negative impacts on biodiversity. However, willingness to pay is still limited today. Against this background, NABB includes knowledge sharing and education, and testing and certification, in addition to its effort to develop and scale biobased chains.

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In all, the identified policy mix seems quite conducive for this pathway, and if the current targets are achieved, the kind of biocomposite materials produced by NPSP will contribute significantly to the Dutch transition to CE by 2030. Considering the circularity targets for the sector, there are also strong motivations for the inclusion of materials from water management, but there are some uncertainties regarding the security of supply for such components, and multiple sources of biomass are therefore considered.

As implications for future policy, we note the following:

- A good balance between destructive and constructive policy elements should be maintained in coming years, as transition pathways take time to evolve.
- Lack of end-of-waste criteria constitute a barrier for cross-border trade in biocomposite components, which should be carefully addressed.
- Assessment methods such as LCA and ECI might well be supplemented with less resource-demanding approaches, which shed light on environmental, economic and social sustainability impacts at both local and global level.
- While several steps have been taken to strengthen GPP or SPP as instrument to generate demand, voluntary GPP criteria do not yet exist for water and wastewater-based products. Such criteria could have a positive signal effect.

10.6 Policies and pathways for reuse and biochar in Ghana

The policy mix analysed for the case in Ghana highlights policy and regulatory gaps for reuse of treated wastewater and utilisation of sewage sludge for biochar production. In terms of political feasibility, there are no conflicts or instruments that specifically hinders development and diffusion of the two solutions. The relevant policies at transnational level focus mainly on water supply and not on wastewater. At the national level, there are well developed frameworks for IWRM and the provision of water cycle services, but no policy elements that address circular water solutions. The recent CE Roadmap and Action Plan (MESTI, 2023) underscores environmental sanitation challenges, but does define specific targets or activities related to water and wastewater. However, key representatives of the relevant ministries have shown interest in the demonstrations and expressed willingness to support further exploration and development of circular water solutions that are suited to the Ghanaian context.

For water reuse, the lack of legal recognition and national standards is a key challenge. Regarding biochar, there is little coordination between the Ministry of Sanitation and Water Resources and the Ministry of Energy, and there are no specific goals or strategies promoting CE based on water resources. However, there is one potential synergy, as the National Energy Transition and Investment Plan (GoG, 2023) includes promotion of more energy-efficient charcoal stoves, and there is a stated willingness to promote the demonstrated biochar in parallel.

Considering maturity of options, we have already note that water reuse is a technologically mature solution, however the WIDER UPTAKE demonstration is the first case in Ghana. The

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demonstration shows that full-scale implementation of reuse for urban farming in Accra will be challenging, as there are no supportive instruments. Urban farmers have limited ability to pay, and treatment, distribution and monitoring would incur substantial costs. The production of biochar is also a mature technology. It is also near market-ready, as it is cost-competitive with regular charcoal, and a local retailer is willing to distribute via her existing channels for charcoal. In terms of regulation, the production will fall partly under existing regulations for regular charcoal, and partly under regulations for renewable energy and biofuel production. This means that a special permit is required, but according to the local partners, the main barrier currently is lack of certification and standards.

System integration and infrastructure is another major challenge for water reuse for urban farming, due to lack of a piped distribution system and congested traffic in Accra. Land ownership and land use rights are often unclear and disputed, and urban farmers tend to operate informally under unpredictable conditions. Addressing these issues requires coordinated action and long-term planning in collaboration with the Accra Metropolitan Assembly. As the central parts of the city already are densely built, SSGL has stated that reuse may be more sustainable for their facilities in Northern Ghana, where the climate is dryer and farmers are located near the facilities. The biochar production does not require pipework and as noted it may be integrated into existing channels for trade in charcoal. However, it is dependent on additional feedstock in form of sawdust from nearby wood industry.

Lastly, in terms of social acceptance, there is a certain scepticism regarding water reuse, due to hygiene and health concerns. This is less of a challenge for the biochar, but in both cases the current lack of recognition and standards is a critical factor in this respect.

Overall, both reuse and biochar for energy are seen as feasible pathways towards a circular water economy in Ghana. Biochar is near market ready, so if certification can be put in place and the solution also could be actively promoted as a fuel for more energy-efficient stoves, upscaling can take place in the near future. The pathway of reuse requires more policy support and may take longer to realize, due to the need for distribution and monitoring. Establishing a viable business model may require systemic instruments involving multiple ministries, which also will require careful assessment and take time to develop. Importantly, the wastewater utility itself considers different transition pathways for different localities, depending on the possible synergies and contributions to sustainability in each context.

Against this background, we highlight the following implications for policy and practice:

- Certification for the production and safe use of biochar from sewage sludge should be prioritised by the relevant authorities in Ghana, as it may unleash the market for a sustainable substitute to regular charcoal production, associated with deforestation and loss of biodiversity.
- The possibility of promoting the biochar in parallel with more energy-efficient charcoal stoves should be pursued and if possible realised, as it could be a win-win in the effort to introduce more sustainable energy solutions in Ghana.

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- Creative instruments, related to knowledge development and diffusion, goal setting, as well as resource mobilisation (e.g., financial, but also in terms of human and social capital, in collaboration with the country's development partners) will be crucial to address the identified gaps and enable a circular water economy in Ghana.

10.7 Implications and recommendations

10.7.1 Overarching insights

The preceding policy mix analyses and discussions regarding transition pathways are specific to the solutions demonstrated in each location. However, it is also possible to discuss more overarching insights regarding policies, instruments, and policy mixes for CE solutions based on water resources.

The first set of insights relates to consistency. When a sector has been organised according to the same principle for a long time, there is usually considerable consistency in the policy mix (e.g., the extract-use-discharge principle in water management). This consistency is particularly apparent when compared to the (lack of) consistency relating to newer sectoral principles (e.g., circular water economy). While old and new principles do not necessarily compete, they are at different stages of development, which influences their alignment: resource recovery from wastewater does not necessarily reduce the safety of wastewater management for public health, but the specific policies to support the implementation of safe and effective resource recovery may be lacking. Among the relevant policies at the EU level, many are either recently revised, under revision, or up for revision or updating. Therefore, it is challenging to assess the level of consistency.

We argue that for the reviewed policy strategies, the overall trend is towards increasing alignment. Considering the instrument mix, we find a relatively high level of consistency between existing instruments for several of the demonstrated solutions (e.g., water reuse for agriculture, struvite for organic farming, and biocomposite materials for application in the building and construction sector). For the use of recovered nutrients in innovative products based on co-composting, it may be argued that there is a certain inconsistency, in that nutrients recovered this way are not eligible as materials for CE-marked fertilisers in the EU.

Regarding consistency between the transnational and national level, we find a high level of consistency for water reuse in Sicily, for struvite in Norway, and also for the biocomposite materials in the Dutch case. Still, there are differences in how actively the specific solutions are promoted, and the degree to which there are specific instruments dedicated to their uptake. For water reuse, quite specific instruments are in place, both at EU level, nationally and locally in the Sicily case. For biocomposites in the Netherlands, the National approach to biobased construction (NABB) provides a broad mix of measures, but there are no dedicated instruments specifically aiming to promote struvite harvesting in Norway.

When it comes to water reuse in the Czech case, in contrast to the EU-level, which is quite clear in its efforts to promote water reuse, the national government has decided not to implement

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Regulation 2020/741 per now and does not promote water reuse actively. In the Norwegian case, there is an ongoing debate on how the revised UWWTD will be implemented, but there is full agreement on its overarching objectives. The FPR has been recommended for implementation by the National Food Authority, but with adjustments to ensure that EEA countries may keep their own limit values for certain substances, e.g., where Norwegian regulations are stricter. Moreover, we note that in the Dutch, Sicilian and Norwegian cases, there are supporting regional and local policies, which indeed have been important drivers.

The second set of insights relates to comprehensiveness. In some cases, a policy mix analysis reveals inconsistencies and/or tensions (e.g., Norway, the Czech Republic). However, in other cases, the policy mix is seemingly consistent without necessarily yielding the desired results, in terms of scale or time of implementation of the new solution (e.g., Sicily). This suggests that attention should also be paid to gaps or *absences* in the policy mix. Consistency is not in itself a sufficient condition. In the case of reuse in Sicily, the policy mix is highly consistent in terms of requirements for water reuse, but not sufficiently comprehensive to promote water reuse. Most of the reviewed instruments are generic, and the brunt of the instruments that specifically target water reuse focus on regulatory pull. In this specific case, the absence of systemic instruments (e.g., for market design, infrastructure for distribution, possible reform in water tariffs, or alternative business models to fund operations and maintenance of water reuse systems) is particularly notable, but the overarching point is that the absence of key types of instruments might render a consistent policy mix inert.

In the case of water reuse and biochar as an alternative energy solution in Ghana, no direct conflicts or tensions between policies and instruments were identified. However, this relates to the wider national context, where increasing the coverage of basic water supply and sanitation services is a main priority, and CE mainly has been linked to pressing challenges in terms of plastic and electronic waste. While the establishment of wastewater treatment plants in populous areas where there was none before offers a great opportunity for implementing innovative CE solutions, reuse is not recognised in the national legislative frameworks, and there are no policies or instruments specifically targeting resource recovery from wastewater. Thus, addressing the current policy and regulatory gaps is a crucial first step.

In terms of comprehensiveness, we further note that in the European cases there has so far been an emphasis on regulatory instruments, to pull demand for more advanced treatment and CE technologies on the side of water utilities. Economic and informational instruments are also applied, e.g., in Sicily where several facilities for water reuse have been established with EU support, and by way of R&D funding, guidelines, etc. We observe that less attention has been paid to the user side, and how to build acceptance and demand for the novel resources and products. To address the challenges and opportunities experienced in the WIDER UPTAKE cases, a stronger focus on value chain and market development is needed.

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10.7.2 Key implications of the study

Overall, the policy mix analyses revolve around three core topics: coordination, incentives, and the need to transition from a conventional linear principle towards a circular water economy perspective, in policy as well as in practice. Below, we discuss these in turn.

10.7.2.1 Coordinating circular water economy solutions

Although an increasing level of alignment is observed, our results show that policies and strategies are not always well aligned. This observation holds true for alignment between governance levels (vertical alignment) and between different governance sectors (horizontal alignment). In terms of vertical alignment, inconsistencies between levels may cause delays in the uptake of already mature technologies, and the development of viable business models around these. For example, the Czech Republic's decision to postpone implementation of EU Regulation 2020/741 constitutes a serious barrier to water reuse in Prague. In this specific case, the local CoP identified a strategy to circumvent the inconsistency, i.e., by defining the treated water as "utility water", rather than pursuing the creation of new national regulation for "reused water", as was done at the EU level. Another example is found in the Dutch case, where a national court ruling was made to permit the use of reed clippings in the novel biocomposites, if their sustainability is well documented, even though no end-of-life criteria for this feedstock exists.

Furthermore, the analyses show that horizontal misalignment may be particularly challenging for the uptake of CE solutions. Such solutions often cut across sectors, and thus are subject to sectoral policies that are configured to realise sector-specific goals rather than promote CE. This is exemplified in the Norwegian case, where the prolonged revision of the national Regulations for production, trade and use of organic fertilisers is seen to result from different sector interests and concerns in the three responsible ministries: Agriculture and Food, Climate and Environment, and Trade, Industry and Fisheries. However, challenges relating to horizontal alignment are not exclusively cross-sectoral, as exemplified in the Czech case.

Simultaneously, there is a need for context-sensitivity. This is illustrated in the Norwegian case, where IVAR in Rogaland has the option to inject biogas into a regional gas grid and combine their resources with biomass from livestock and poultry production in the area, albeit less demand for P. Hias has to find different uses for their biogas, but their process technology and the soil and form of agriculture in their region, combined with the high CE ambitions at Sirkula, has driven their CE innovation in a slightly different direction. The importance of context is also highlighted in the case in Ghana, where SSGL sees biochar as a sustainable pathway for their plant in Accra, due to high demand and nearby wood industry providing sawdust as additional feedstock but considers reuse as a long-term option that can be more suited for their plants in Northern Ghana, where systems can be built from scratch and farms in need of water are located closer to the facilities. This underscores the importance of regional coordination and integrated planning, as emphasized in the revised UWWTD. At the same time, it highlights the need for flexibility and adaptation to local circumstances.

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10.7.2.2 Incentivizing circular economy solutions

The results from our analyses suggest that there can be an (over)emphasis on regulatory instruments. For example, the policy mix in the Sicily case is characterised by regulatory pull instruments at all levels of governance. While these instruments instate new requirements for wastewater treatment and establish the conditions for using reclaimed water, they are not accompanied by instruments focussing on establishing demand for reclaimed and recovered resources. In short, (public and private) utilities must comply with requirements aiming to promote circular water solutions, but there are few specific creative instruments – i.e., instruments that seek to establish a demand for recovered resources such as reclaimed water, struvite, and organic fertilisers, rather than control the production and application of these products.

Establishing a market for reuse and resource recovery requires creative instruments in addition to destructive ones. While products manufactured from reclaimed and recovered resources often perform as well, and in some regards better than virgin products, the markets for such products are often less mature. Hence, products from reclaimed/recovered resources must be accompanied by instruments aiming to establish them as viable alternatives (e.g., public procurement, subsidies, purchase quotas, identifying niche applications).

While demand-pull instruments may require (e.g.) wastewater treatment plants to comply with new regulations, fostering a market for recovered resources requires instruments meant to stimulate demand for such products *and* have these products be incorporated in user/organisational practices. For resource recovery, the experience the Netherlands suggests that pooling resources and establishing joint platforms such as Aquaminerals can help to develop a market. GPP or SSP are also highly relevant instruments, which could be further developed and more systematically applied. Establishing a turnover requirement for P has also been proposed, both for the EU and at the national level in Norway. For water reuse, the development and implementation of systemic instruments is needed. Market design will be important, as will infrastructure provision (or, alternatively, identifying use-cases that do not require water to be transported far) and possibly water tariff reforms.

Regarding incentives for circular economy solutions in the water sector, the core implication of the policy mix analyses is that the mixes often miss creative instruments aimed at establishing a market for recovered resources. This gap appears to relate to policy objectives. The analysed policies and instruments are primarily concerned with – and fit to address – sectoral issues, though many of them (implicitly or explicitly) include goals that also relate to the circular economy. However, if the EU seeks to foster circular economy solutions in Europe's water sector – as indicated by the Call for a Blue Deal (EESC, 2024) – more instruments must be tailored to foster demand for reclaimed and recovered products, rather than the uptake of facilitating technologies and solutions at wastewater treatment plants.

10.7.2.3 Prioritising circular water economy in policymaking

The preceding points indicate that transitioning towards a circular water economy should be integrated more strongly in the relevant policy mixes – also at the level of objectives. The

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European Green Deal states that CE principles shall be integrated across all policy areas. However, as noted in the Declaration of a Blue Deal for Europe, water has for a large part been associated with environmental policy and considered less as a scarce and valuable resource (e.g., in the Green Deal). This is reflected in recent policy developments for the water and wastewater sector, where CE in most cases is mentioned, but comes across as a secondary objective. For example, the revised UWWTD emphasises reduction of pollution, and energy neutrality, then addressing water scarcity. Although the revised version relates very explicitly to water reuse and also addresses resource recovery, how to manage the resources and values in water optimally in different regional and industrial contexts is a less prominent theme, both here and in other relevant frameworks, such as the SSD. As a result, too little attention is paid to the rather comprehensive set of instruments that may be needed to establish viable business models and develop a demand for reuse water and novel products based on recovered resources.

Thus, we argue that when relevant policies and instruments are developed and revised in future, their relationship to the circular economy must be carefully considered. This requires a systemic perspective, where the interaction between resources, technologies and institutions is taken into account. A stronger focus on the values in water, implies a perspective where both environmental, economic and social benefits are taken into account, and where long-term planning considering alternative scenarios and capabilities to change and innovate are combined. This requires a combination of destructive and creative instruments, where knowledge development and diffusion, resource mobilisation, and room for innovation are essential elements, together with the emerging CE regulations and financial instruments required to enable the foreseen investments. Moreover, consistency and comprehensiveness of instruments in promoting circular economy is needed, both across governance levels and policy domains.

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11 Conclusion

This report has provided an analysis of the current policy mixes surrounding the solutions demonstrated in WIDER UPTAKE, as a basis for discussion of the feasibility and future development of the solutions as pathways to a circular water economy in Europe and beyond. Based on the extended policy mix concept developed within the field of socio-technical transition studies, we assessed the policy mixes surrounding the key solutions in each demonstration case, focusing on their level of consistency and comprehensiveness.

The analysis highlights the major policy developments and revisions that are ongoing at the EU level. Overall, we see an increasing level of consistency, between the strategies and policies for water and wastewater, the overarching frameworks for transition towards a more sustainable society, and transition strategies in the targeted use sectors. However, we also identify certain tensions between elements at the EU and national levels, and between objectives and embedded interests in different sector policies. The findings also highlight differences across the case countries. For example, the Netherlands has a quite consistent and comprehensive mix related to the sector structure and generally high CE ambitions in the country. Ghana, an emerging economy where the main focus has been extending safe water supply and basic sanitation to a fast-growing population, faces larger gaps in policy and regulation.

In terms of comprehensiveness, we further note an emphasis on the development and implementation of CE technologies, by way of technology-push and demand-pull instruments directed towards utilities. We argue that there is a need to pay more attention to the target users of the reclaimed water and products based on recovered resources, to build acceptance and generate the demand needed to establish viable value chains.

The preceding discussion provided a set of recommendations specifically for each case and national context. We conclude the report by highlighting three overarching challenges:

First, there is the need for increased coordination and collaboration – across levels, sectors, and regions. This underscores the need for integrated planning and creative policy instruments, e.g., networks, innovation platforms, guidelines, and/or training schemes, to build capacity and open for broader perspectives.

Second, CE solutions based on water resources must be further incentivised. This applies to utilities, who tend to lack financial incentives for reclamation and recovery, and to users, who are not currently incentivised to use reclaimed and recovered resources. The willingness to pay for products associated with wastewater or sludge is often limited at the outset, either because they are associated with contaminants and health risk or because they previously have been freely available.

Third, we argue that CE should be more prioritised in policymaking. By this we mean that circular water economy should be a main strategic priority, rather than a secondary objective.

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Future policy must pay more attention to the values in the water cycle and how they best can be realised to maximise the economic, social, *and* environmental benefits of sustainability. As part of this transition, it is tantamount to combine long-term planning towards resilience with capabilities to change and innovate in response to more immediate challenges and opportunities. This requires a comprehensive mix of instruments, including both creative ones, aiming to foster identify and support novel solutions, and destructive ones, aiming to change and do away with less sustainable systems and practices.

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Annex 1: Policy documents

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